

Jingdao Chen – Promotion Dossier

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Application Letter



**DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING**

665 George Perry Street
300 Butler Hall
Post Office Box 9637
Mississippi State, MS 39762
P 662.325.2756
cse.msstate.edu

Dear Dr. Andy Perkins,

I am writing to provide a Letter of Intent to submit my application for promotion to Associate Professor in the Department of Computer Science and Engineering (CSE) at Mississippi State University (MSU). It has been a great honor to serve in this faculty for the past several years. With a lineup of esteemed professors together with talented, hardworking students, this great institution is the perfect place to call home.

The convergence of robotics, automation, and Artificial Intelligence is a key breakthrough of the 21st-century and has the potential to radically change many aspects of industries of the future ranging from transportation, manufacturing, construction, and agriculture. My research combines 3D computer vision, digital twins, and machine learning to achieve robust, accurate, real-time capabilities for intelligent, autonomous agents to operate in challenging environments. Since starting my position at MSU, my research has resulted in **6 journal papers** and **16 conference papers**. I have also received best paper awards at EG-ICE 2018, ISARC 2016, and ISARC 2017.

I am grateful to have received grants from various institutions including NSF, IARPA, and NASA to support my research. The following shows a list of external funding received (**total funding \$3,538,960**) :

- NSF CAREER award IIS-2440583 for \$510,532, PI
- NSF Computer Science Research Initiation Initiative (CRII) award IIS-2153101 for \$175,000, PI
- NSF Secure and Trustworthy Cyberspace (SaTC) award DGE-2246920 for \$399,978, co-PI
- NSF Innovative Technology Experiences for Students & Teachers (ITEST) \$1,192,951, co-PI
- IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA) for \$160,499, PI
- NASA EPSCoR Rapid Response Research (R3) award for \$100,000, PI, \$88,500 to MSU
- USDA Agricultural Marketing Service (AMS) award for \$1,000,000, co-PI, \$446,354 to MSU

I am committed to advance the educational goals of MSU through my teaching and mentoring of MSU students. I have taught CSE 4633/6633 Artificial Intelligence and CSE 4643/6643 AI Robotics for 4 years and have also introduced two new classes, CSE 3683 AI Fundamentals and CSE 8990 Advanced AI Robotics. At MSU, I have served on the departmental Courses and Curricula Committee as well as the Graduate Studies Committee. To foster interest in AI and robotics, I hosted robotics lab tours through the Academic Insight and Engineering Discovery Day programs and advised the AI Club. I have continuously chaired workshops at the IEEE International Conference on Robotics and Automation (ICRA), the largest and most prestigious international conference in robotics. Additionally, as part of my service to the scientific community, I regularly serve on professional committees, conduct tutorials, review journal/conference papers, and have served on four NSF panels.

It has been a wonderful journey thus far and I am grateful and excited about continuing to learn and to grow within the MSU community.

Sincerely,

Jingdao Chen
Assistant Professor, Mississippi State University

Promotion and Tenure Form

Mississippi State University
Application for Promotion and/or Tenure

<i>Please check response(s) in both columns</i>	
TENURE:	PROMOTION:
☐ Mandatory tenure decision	☒ Promotion to Associate Professor
☒ Not applicable (early promotion or already possess tenure)	☐ Promotion to Full Professor
	☐ Not applicable (already at associate or full level)

Faculty members eligible for consideration for promotion or tenure must provide the department head or appropriate official with all pertinent available information by **October 1**. The department head or other appropriate official has the responsibility to assist the faculty member in preparing for tenure or promotion review.

Materials to be provided in the applicant's dossier include:

- *1. Cover letter from the candidate requesting promotion and/or tenure.
- *2. Completed University Promotion and Tenure application form (this cover page and attached pages) with appropriate responses and associated documentation. This must include a summary sheet of teaching evaluations.
- *3. Complete up-to-date vita.
- *4. Copy of initial offer letter and, if necessary, additional letter detailing significant changes.
- *5. Letters from external reviewers (to be added by department head). The department head should include a sample letter sent to external reviewers and biographical information about reviewers as appropriate.
- 6. All materials required by the academic unit's procedural guidelines.
- 7. All supporting documentation desired by the candidate.

*Only these items will be reviewed routinely above the college level. Items 1-7 must go to dept. head and dept. committee. Deans, college committees, and the Provost require items 1-5 but may also request items 6 and 7. Department heads and deans can use their discretion in sending forward any important information included in items 6 and 7. All department head, dean, and committee recommendations should be included in the package to the Provost.

Note: Please refer to the Faculty Handbook for information pertaining to the Promotion & Tenure process.

To apply and be considered for tenure requires that you be a citizen of the United States or be a permanent resident or have begun the permanent residence process (verification required) in order to be eligible for permanent employment in this country.

Are you a citizen or permanent resident of the United States: ☒ Yes ☐ No

If No, have you applied for permanent residency: ☐ Yes ☐ No (Date process initiated (if Yes): _____)

Name of Applicant: **Jingdao Chen**

Present rank: **Assistant Professor**

Date of appointment at current rank: **August 16, 2021**

College/School: **Bagley College of Engineering** Department: **Computer Science and Engineering**

Department Head: **Dr. Andy Perkins**

Preferred Mailing Address (Include City and Zip Code): **102 Snead Way, Starkville MS 39759**

Initial rank at MSU with date of appointment: **Assistant Professor, August 16, 2021**

Tenure track date of appointment: **August 16, 2021**

Years of transferred service (if applicable): **N/A**

Advanced Degrees with Dates: **PhD in Robotics 2021, MS in CS 2019, BS in EE 2015**

Salary Funding (%): E&G: 100% MSU Research Unit: _____ Extension: _____ Other: _____

All other information contained in the attached application is correct, to the best of my knowledge.

Date: 08/15/2025

Signed: _____



Faculty Member

I. Current semester responsibilities:

A. Current instruction

	<u>Course number</u>	<u>Title</u>	<u>Credit hours</u>	<u>Number of students</u>
1. Undergraduate:	CSE 3683	AI Fundamentals	3	58
	CSE 4643	AI Robotics	3	47
2. Graduate:	CSE 6643	AI Robotics	3	27
	CSE 9000	Research / Thesis	3	9
3: Advises:	Undergraduate <u> 5 </u>	Master's/Specialist <u> 6 </u>	Doctoral <u> 15 </u>	

B. Current or on-going research/creative/performance activities

- *Robust off-road navigation for mobile robots*: This research explores machine learning methods for robot navigation in rough terrain that combines perception from camera or LiDAR data with self-supervised learning from prior trajectories. This research is funded by a 2-year NSF CRII award (IIS- 2153101) and has resulted in 2 conference publications at SPIE 2023 and 2 conference publications at I3CE.
- *Robotic digital twin mapping and modeling of construction environments*: This research deals with performing 3D scanning with LiDAR-equipped robots and constructing 3D information models of a building environment. The 3D model can be used for digital twin simulations, floorplan generation, and detection of damaged elements. This research has resulted in 1 journal publication, 2 conference publications, and 2 workshop posters.
- *Terrain segmentation for planetary rovers*: In collaboration with NASA's Jet Propulsion Lab (JPL) at Caltech, this research uses self-supervised learning methods to perform terrain segmentation in planetary surface images and explores techniques such as contrastive learning and network distillation to make navigation more accurate and efficient for planetary rovers. This research is funded by a 1-year NASA award EPSCOR R3 award and has resulted in 1 journal publication at the Journal of Spacecraft and Rockets, 2 conference publications at IEEE AeroConf, 2 conference publications at ECCV, and 1 in-progress paper.
- *Artifact correction and mitigation for image data*: In collaboration with BlueHalo, Princeton University, and Stanford University, this research investigates generative adversarial networks and diffusion models to remove artifacts such as blurring, over/under-exposure, compression, weather effects and transient objects from image data. This research is funded by a 3.5-year award from IARPA. This research has resulted in 2 in-progress papers.
- *Security and ethical considerations in robotics*: This research aims to address security and ethical issues in artificial intelligent robot systems such as trustworthiness, privacy, data poisoning, model inversion, and potential legal issues. This research is funded by a 3-year NSF SaTC award (DGE-2246920) and has resulted in 2 journal publications and 3 conference publications.

C. Current service/administrative assignments

1. Public service and off-campus professional service activities:

- NSF Panel 2025
- Invited talk, IEEE Robotics and Automation Society Student Branch Chapter, July 2025
- Organizing committee, 4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments at ICRA 2025

2. Professional association service, as offices held, etc.:

- Associate Editor, Journal of Computing in Civil Engineering
- Reviewer, IEEE Transactions on Field Robotics

- Reviewer, Journal of Computing in Civil Engineering
- Reviewer, International Conference on Computing in Civil Engineering
- Reviewer, International Symposium on Automation and Robotics in Construction (ISARC)

3. University and departmental committee and administrative accomplishments:

- Department of Computer Science and Engineering, Courses and Curricula Committee
- Department of Computer Science and Engineering, Graduate Studies Committee
- Faculty advisor for AI Club, State-Space Robotics Team

D. Other

II. Activities since last promotion (or initial appointment for tenure):

A. Teaching

1. Evidence of quality of instruction (check items submitted):
(The faculty member should provide material describing his/her teaching activities and documentation supporting effectiveness. This material must include a summary statement of student evaluations and may include any of the following, or any other items deemed appropriate: peer evaluations (internal or external) __✓__, course syllabi and exams __✓__, student input in the form of letters, emails, faculty nominations, etc. ____, recordings of teaching sessions ____, graduate student theses and dissertations ____, other materials that demonstrate teaching effectiveness ____.)
2. Master's students: major professor __15__ ; minor professor ____.
Specialist students: major professor ____ ; minor professor ____.
Doctoral students: major professor __16__ ; minor professor ____.
3. Courses initiated or innovations instituted:
 - A new course *CSE 3683: AI Fundamentals* is introduced in Spring 2025.
 - A new course *CSE 8990: Advanced AI Robotics* is introduced in Spring 2023.
 - The content of *CSE 4633/6633 Artificial Intelligence* is updated to include modern topics in AI such as computer vision and natural language processing.
 - The content of *CSE 4643/6643 AI Robotics* is updated to include to include modern topics in Robotics such as reinforcement learning and deep learning for perception.
4. Other (academic advisement may be described here or as service):

Undergraduate students mentored:

- Austin Cosby, 2023, defended Honor's thesis in April 2023
- Yuexin Xu, 2022 – 2023, defended Honor's thesis in November 2023
- Neil Sanipara, 2024 – 2025, defended Honor's thesis in May 2025
- Justin Yee, 2022
- Wes Davis, 2023

- Nicholas Beers, 2023
- Amanuel Tesfaye, 2023
- Prabesh Khanal, 2023
- Reyki Garcia, 2023 – 2024
- Arjun Koinkar, 2023 - 2024
- Hunter Moore, 2024
- Aadhav Senthil, 2024
- Rahul Kumar, 2024
- Callie Simons, 2024
- Abhinav Dhakal, 2024
- Alaa Mohammad, 2024 - 2025
- Carter Holton, 2024 - 2025
- Leonel Giacobbe, 2024 - 2025
- Tanner Grantham, 2024 - 2025
- Isha Shresta, 2025

M.S. students mentored as major professor:

- Erez Meoded, defended M.S. thesis in October 2023
- Charles Moore, 2022 – 2023, defended M.S. thesis in October 2023
- Joshua Waldbeiser, 2022 – 2023, defended M.S. thesis in October 2023
- Sushant Gautam, 2023 – 2025, defended M.S. thesis in April 2025
- Malika Dutta, 2023 – 2025, defended M.S. thesis in March 2025
- Jacob Kutch, 2023 – 2024, defended M.S. thesis in October 2024
- Lucian Murdock, 2022, graduated May 2022
- Monika Jayakumar, 2024, graduated December 2024
- Suryateja Reddy Maddirala, 2024-2025, graduated May 2025
- Jaylen Hopson, 2025
- Charles Sailors, 2025
- Jacob Roberts, 2025
- Ethan Heverly, 2025
- Connor Bell, 2025
- Nina Miles, 2025

Ph.D. students mentored as major professor:

- Kevin Gao, 2023 – 2025, defended PhD thesis in June 2025
- Grant Greenwood, 2023 - 2025
- Jeffrey Rugg, 2023 - 2025
- Jinhee Yu, 2023 – 2025
- Arjun Koinkar, 2024 - 2025
- Swayamjit Saha, 2023 - 2025

- Munib Ahsan, 2024 - 2025
- Erez Meoded, 2024 – 2025
- Mark Trovinger, 2024 – 2025
- Shawn Frye, 2024 - 2025
- Charles Raines, 2024 - 2025
- Don Bennett, 2025
- Mark Shiffer, 2025
- John Steiner, 2025
- Benquan Wang, 2025
- Alec Zoeller 2025

B. Research, creative endeavor, or performances

1. Publications, performances or creative activities:

(For books, indicate date of publication and publisher; for articles, indicate refereed journals; for art shows, indicate judged competition; for musical shows, attach copies of programs; for reports, indicate those done for in-house use.)

Note: Italicized names are student advisees.

Journal Publications

- Yang, Y., Wijewardane, N., Slonecki, J., Wadl, P., Andreason, S., **Chen, J.**, Harvey, L. (2025). “Nondestructive detection of sweet potato leaf curl virus using 3D laser imaging combined with deep learning”. *Smart Agricultural Technology*, Volume 11, 2025, 101004, ISSN 2772-3755
- Armeni, I., **Chen, J.**, Feng, C., Jeong, I., Liang, C.J., Morin, K., Tsagarakis, N., Wang, X., and Zhang, L. (2024). “Construction Robotics and Automation [TC Spot Light]”. *IEEE Robotics and Automation Magazine*, vol. 31, no. 4, pp. 186-192, Dec. 2024
- Neupane, S., Mitra, S., Fernandez, I., *Saha, S.*, Mittal, S., **Chen, J.**, Pillai, N., Rahimi, S. (2024). “Security Considerations in AI-Robotics: A Survey of Current Methods, Challenges, and Opportunities” *IEEE Access*, vol. 12, pp. 22072-22097, 2024
- *Gao, K.*, Haverly, A., Mittal, S., Wu, J., **Chen, J.** (2024). “AI Ethics: A Bibliometric Analysis, Critical Issues, and Key Gaps” *International Journal of Business Analytics (IJBAN)*, 11(1), 1-19.
- Vincent, G., Ward, I., *Moore, C.*, **Chen, J.**, Pak, K., Yepremyan, A., Wilson, B., and Goh, E. (2023). CLOVER: Contrastive Learning for Onboard Vision-Enabled Robotics. *Journal of Spacecraft and Rockets*. December 2023
- **Chen, J.**, and Cho, Y. (2022). CrackEmbed: Point feature embedding for crack segmentation from disaster site point clouds with anomaly detection. *Advanced Engineering Informatics*, Volume 52, April 2022

Conference Publications

- *Yu, J.*, *Jayakumar, M.*, Chen, Y., Cho, Y., and **Chen, J.** (2025). “Self-supervised Learning with LiDAR-Camera Fusion for Construction Site Traversability Estimation”. *Proceedings of the ASCE 2025 International Conference on Computing in Civil Engineering (i3CE)*, New Orleans, LA, USA, May 11-14
- *Rugg, J.*, Kim, S., Cho, Y., **Chen, J.** (2025). “Segment Anything Model for Point Cloud Temporal Change Detection on Construction Sites”. *Proceedings of the ASCE 2025 International Conference on Computing in Civil Engineering (i3CE)*, New Orleans, LA, USA, May 11-14
- *Fernandez, I.*, Neupane, S., Chakraborty, T., Mitra, S., Mittal, S., Pillai, N., **Chen, J.**, and Rahimi, S. (2024). “A Survey on Privacy Attacks Against Digital Twin Systems in AI-Robotics”. *2024 IEEE 10th International Conference on Collaboration and Internet Computing (CIC)*, Washington, District of Columbia, United States, 28 - 30 October 2024

- Gao, K., Mittal, S., Wu, J., Du, H., **Chen, J.**, and Rahimi, S. (2024). "The AI Pentad, the CHARME2D Model, and an Assessment of Current-State AI Regulation". *10th IEEE International Conference on Sustainable Technology and Engineering (i-COSTE)*, Perth, Australia, 18th - 20th December, 2024
- Yu, J., Saha, S., Jayakumar, M., Gugssa, M., **Chen, J.**, and Wang, J. (2024). "LiDAR-based Traversability Estimation for Ground Robots on Construction Sites using Self-Supervised Learning". *Proceedings of the ASCE 2024 International Conference on Computing in Civil Engineering (i3CE)*, Pittsburgh, PA, USA, July 28-31
- Rugg, J., **Chen, J.**, Gugssa, M., and Wang, J. (2024). "Object-level Temporal Change Detection on Construction Sites with 3D Deep Learning Models". *Proceedings of the ASCE 2024 International Conference on Computing in Civil Engineering (i3CE)*, Pittsburgh, PA, USA, July 28-31
- Chang, S., **Chen, J.**, and Park, J. (2024). "Review of Metaverse Technologies to Broaden Accessibility in Civil and Construction Engineering Education." *Construction Research Congress 2024*, March 20-23, Des Moines, IA
- Gao, K., Haverly, A., Mittal, S., **Chen, J.** (2023). "A bibliometric view of AI Ethics development." *IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*, Fiji, December 4-6
- Garshabi, A., Wang, J., **Chen, J.**, and Ma, J. (2023). "Human-Robot Collaboration in the Construction Industry: A Mini-review." *IEEE International Conference on Robotics and Automation (ICRA) Workshop on Future of Construction*, London, U.K., June 2
- **Chen, J.**, Gugssa, M., Yee, J., Wang, J., Goodin, C., and Ram Das, A. (2023). "Framework for digital twin creation in off-road environments from LiDAR scans." *Proc. SPIE 12529, Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications*, 125290F (13 June 2023)
- Yu, J., **Chen, J.**, Dabbiru, L., and Goodin, C. (2023). "Analysis of LiDAR configurations on off-road semantic segmentation performance." *Proc. SPIE 12540, Autonomous Systems: Sensors, Processing, and Security for Ground, Air, Sea, and Space Vehicles and Infrastructure 2023*, 1254003 (13 June 2023)
- Goh, E., Ward, I. R., Vincent, G., Pak, K., **Chen, J.**, and Wilson, B. (2023). "Self-supervised Distillation for Computer Vision Onboard Planetary Robots." *IEEE Aerospace Conference*, Big Sky, MT, USA, March 4-11
- Ward, I. R., Moore, C., Pak, K., **Chen, J.**, and Goh, E. (2022). "Improving Contrastive Learning on Visually Homogeneous Mars Rover Images." *European Conference on Computer Vision (ECCV) Workshop on AI4Space*, Tel Aviv, Israel, Oct 23
- Vincent, G., Yepremyan, A., **Chen, J.**, and Goh, E. (2022). "Mixed-Domain Training Improves Multi-Mission Terrain Segmentation." *European Conference on Computer Vision (ECCV) Workshop on AI4Space*, Tel Aviv, Israel, Oct 23
- Kim, S., Yajima, Y., Park, J., **Chen, J.**, and Cho, Y. (2022). "A Hybrid Semantic-Geometric Approach for Clutter-Resistant Floorplan Generation from Building Point Clouds." *Proceedings of the 9th International Conference on Construction Engineering and Project Management (ICCEPM)*, Las Vegas, NV, USA, June 20-23
- Goh, E., **Chen, J.**, and Wilson, B. (2022). "Mars Terrain Segmentation with Less Labels". *IEEE Aerospace Conference*, Big Sky, MT, USA, March 5-12

Book Chapters

- **Chen, J.**, and Cho, Y. (2022). Rapid scan-to-building information modeling using robotics and artificial intelligence for construction applications. *Research Companion to Building Information Modeling*, March 2022
2. Professional papers read; indicate whether invited, refereed, or volunteered.
Cite organization, date, and title:
 - (*invited*) **Chen, J.**, "Constructing Maps for Construction Robots", IEEE RAS Student Branch Chapter at Federal Institute of Science and Technology (FISAT), Kerala, India. July 17, 2025.
 - (*invited*) **Chen, J.**, "Robot Perception in Unstructured Environments", University of Alabama Distinguished Colloquium Speaker. November 20, 2024.
 - (*refereed*) Jenkins, W., **Chen, J.**, "Panel on AI and ML in HPC", Mississippi High Performance Computing Conference. March 27, 2024.
 - (*invited*) **Chen, J.**, "Robust Scene Understanding for Robots in Unstructured Environments", IEEE Mississippi Seminar. April 15, 2022.
 3. Grants for research or study:

Proposals submitted since last promotion and total dollar amount: 22 ; \$ 14,880,105 *

** This includes large, multi-institute center proposals with significant funds subawarded to other institutions.*

- (successful) IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA). Scene Understanding, Reconstruction and Rendering with Varying Environment Altitude and Limited Imagery. PI. \$634,082. Submitted July 2022.
- (successful) NSF Computer and Information Science and Engineering Research Initiation Initiative (CRII). Semantic Vector Fields for Robot Navigation and Exploration in Unstructured Environments. PI. \$175,000. Submitted September 2021.
- (successful) NSF Secure and Trustworthy Cyberspace (SaTC). Inculcate a culture of preparedness against AI security threats to pervasive robotic systems. Co-PI. \$399,978. Submitted October 2022.
- (successful) NASA EPSCoR Rapid Response Research (R3). Incremental learning with knowledge distillation for autonomous rover terrain characterization. PI. \$100,000. Submitted January 2023.
- (successful) USDA Agricultural Marketing Service (AMS). Advancing Optical Technologies for Enhanced Quality Evaluation, Grading and Sorting of Sweet Potatoes. Co-PI. \$1,000,000. Submitted December 2021.
- (unsuccessful) Mississippi Soybean Promotion Board. Using a Machine Vision Approach to Weed Detection towards Precision Weed Management. Co-PI. \$58,137. Submitted December 2021.
- (unsuccessful) USDA National Institute of Food and Agriculture (NIFA). Autonomy, Precision, and Climate-Smart Technologies for the Thriving Farm of the Future. Co-PI. \$3,935,914. Submitted October 2021.
- (unsuccessful) Amazon Research Awards. Semantically-aware Online Route Optimization for Off-road Delivery Robots. PI. \$79,055. Submitted October 2021.
- (unsuccessful) Facebook. City-level Information Modeling with Semi-supervised Discovery of Object Classes. PI. \$66,257. Submitted August 2021.
- (unsuccessful) USDOT. Southeast Transportation Regional Institute for Equity and Advanced Mobility (STREAM). Co-PI. \$200,000. Submitted August 2022.
- (unsuccessful) NSF National Robotics Initiative (NRI). Crop Anomaly Detection and Management with Continually Learning Robot Swarms. PI. \$1,199,997. Submitted February 2022.
- (unsuccessful) NSF Human-Environment and Geographical Sciences (HEGS). Understanding Human Behavior Through 3D Spatial Environment. Co-PI. \$399,995. Submitted January 2022.
- (unsuccessful) NSF CyberTraining. EMBRACE: Engaging the Metaverse to Broaden Research Accessibility In Civil and Construction Engineering. PI. \$259,993. Submitted September 2023.
- (unsuccessful) USDA National Institute of Food and Agriculture (NIFA). Ag2: Developing High-Fidelity Agricultural Environments in Gazebo – A Framework Towards Agricultural Autonomy Digital Twin. Co-PI. \$299,997. Submitted October 2023.
- (successful) NSF Innovative Technology Experiences for Students and Teachers (ITEST). Creating Intelligent Solutions with Machine Learning and Computer Vision: Smart Computational Thinking as a New Pathway to AI Careers for Diverse Students (Smart CT). Co-PI. \$1,192,951. Submitted August 2023.
- (unsuccessful) USDOT Rural Autonomous Vehicle (RAV). Mobility of People in Rural and Tribal Communities. Co-PI. \$1,748,945. Submitted March 2024.
- (unsuccessful) NSF PFI: AI-Enabled Scanning and Modeling Systems for Accelerated Digital Twin Creation. PI. \$1,000,000. Submitted May 2024.
- (unsuccessful) NASA. Versatile and Resilient Space Explorations by Collaborative Multi-Robot Team with Humans in the Loop: Situational Awareness, Communication, and Decision-Making. Co-PI. \$750,000. Submitted September 2023.
- (successful) NSF CAREER: Constructing Schematic Maps for Construction Robots in Highly Dynamic Environments. PI. \$510,532. Submitted July 2024.
- (unsuccessful) USDA NIFA. Monitoring Greenhouse Gases in Agricultural Systems Using UAVs: Integrating Sensing and Modeling Techniques. PI. Submitted September 2024. \$157,292.
- (pending) DoD DURIP: Development of an Integrative Testing Facility and its Digital Twin for Off-Road Autonomy. Co-PI. Submitted April 2025. \$411,980.
- (pending) USDA NIFA. Like-Real-Farm Simulator (Lrf Sim): Developing High-Fidelity Virtual Farm Environments In Gazebo. Co-PI. Submitted October 2024. \$300,000

Proposals funded (cite source, title of project, role [PI, etc.], \$ amount, dates): **\$3,538,960**

- **Jingdao Chen** (Role: PI). NSF CAREER: Constructing Schematic Maps for Construction Robots in Highly Dynamic Environments. \$510,532. June 2025 – May 2030.
- Yan Sun, Lisa Thomas, Zhiqian Chen, **Jingdao Chen** (Role: Co-PI). NSF Innovative Technology Experiences for Students and Teachers (ITEST). Co-PI. \$1,192,951. September 2024 – August 2027.
- **Jingdao Chen** (Role: PI). IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA). Scene Understanding, Reconstruction and Rendering with Varying Environment Altitude and Limited Imagery. \$160,499. June 2023 – June 2024.
- **Jingdao Chen** (Role: PI). NSF Computer and Information Science and Engineering Research Initiation Initiative (CRII). Semantic Vector Fields for Robot Navigation and Exploration in Unstructured Environments. \$175,000. April 2022 – April 2024.
- Sudip Mittal, **Jingdao Chen** (Role: Co-PI). NSF Secure and Trustworthy Cyberspace (SaTC). Inculcate a culture of preparedness against AI security threats to pervasive robotic systems. \$399,978. June 2023 – May 2026.
- **Jingdao Chen** (Role: PI). NASA EPSCoR Rapid Response Research (R3). Incremental learning with knowledge distillation for autonomous rover terrain characterization. \$100,000. September 2023 – August 2024.
- Yuzhen Lu, Nuwan Wijewardane, **Jingdao Chen** (Role: Co-PI). USDA Agricultural Marketing Service (AMS). Advancing Optical Technologies for Enhanced Quality Evaluation, Grading and Sorting of Sweet Potatoes. \$1,000,000. March 2022 – March 2025.

4. Other:

C. Service

1. Public service, as lectures, short courses, workshops (with dates, organizations, places):

- NSF panel, 2021, 2023, 2024, 2025
- Invited talk, IEEE Robotics and Automation Society Student Branch Chapter, July 2025
- Panel on AI and ML in HPC, Mississippi High Performance Computing Conference, March 2024
- Invited talk, Mississippi State University CSE Seminar, February 2022
- Invited talk, IEEE Mississippi Seminar, April 2022
- Workshop on Deep Learning Tools for Understanding and Modeling the Built Environment at I3CE 2024
- Organizing committee, 4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments at ICRA 2025
- Organizing committee, 3rd Workshop on Future of Construction: Lifelong Learning Robots in Changing Construction Sites at ICRA 2024
- Organizing committee, 2nd Workshop on Future of Construction: Robot Perception, Mapping, Navigation, Control in Unstructured and Cluttered Environments at ICRA 2023
- Organizing committee, Workshop on Future of Construction: Build Faster, Better, Safer - Together with Robots at ICRA 2022
- Virtual Scan-to-BIM Bootcamp with Seoul National University, 2022
- Tutorial on Scan-to-BIM using Python and Open3D at ICCEPM 2022
- Session Chair, International Conference on Construction Engineering and Project Management (ICCEPM) 2022
- Session Chair, International Conference on Computing in Civil Engineering (i3CE) 2021

2. Professional association service (offices held, journals edited, etc.):

- Associate Editor, Journal of Computing in Civil Engineering
- Reviewer, Automation in Construction
- Reviewer, Journal of Computing in Civil Engineering
- Reviewer, Advanced Engineering Informatics
- Reviewer, International Symposium on Automation and Robotics in Construction (ISARC)

- Reviewer, International Conference on Computing in Civil Engineering (i3CE)
- Reviewer, IEEE Transactions on Robotics
- Reviewer, IEEE Transactions on Field Robotics
- Reviewer, IEEE Robotics and Automation Letters
- Reviewer, IEEE International Conference on Robotics and Automation
- Reviewer, IEEE International Conference on Intelligent Robots and Systems
- Reviewer, IEEE International Conference on Automation Science and Engineering
- Reviewer, IEEE Access

3. University service (committees, administrative accomplishments, etc.):

- Department of Computer Science and Engineering, Courses and Curricula Committee
- Department of Computer Science and Engineering, Graduate Studies Committee
- Faculty advisor for AI Club, State-Space Robotics Team

4. Other (academic advisement may be described here or as teaching):

III. Awards and distinctions (title, date, organization):

IV. Memberships in learned and professional societies. Society, dates of membership, offices held:

- IEEE RAS Technical Committee on Construction Robotics, Co-Chair
- IEEE Robotics and Automation Society, 2018-2025, member

V. Previous academic ranks, institutions, dates:

- Assistant Professor, Mississippi State University, August 2021 - present

VI. Non-academic positions held prior to appointment at MSU: N/A

VII. Summary listing of all required and supporting documentation (items 6 and 7 on cover of application form). This listing should be less than one page in length.

- Table of Contents
- Application Letter
- Promotion and Tenure Form
- Initial Offer Letter
- Curriculum Vitae
- Teaching Accomplishments
- Research Accomplishments
- Service Accomplishments
- Selected Publications

Department Head's Recommendation for Promotion or Tenure
(Cite the following information and sign.)

1. Name of candidate: _____ Present rank: _____
2. Recommended for promotion to the rank of: _____
(Or not recommended): _____
3. Recommended for tenure: Yes/No/NA

Assessment and evaluation by department head: strong points that warrant promotion should be listed, with documentation wherever possible; stress such items as teaching and advising of students, research accomplishments, and university and community service. Please avoid platitudes or general, subjective opinions. It would be useful, too, to comment upon the quality of personal relationships of the candidates with peers, superiors, and any who may report to him/her, as well as upon his/her professional performance. Finally, consider the candidate in relation to what you picture as the ideal candidate for this recommended position, rather than in relation to other members of your department. (Where "demonstrated excellence" is called for, please supply some evidence, e.g., student evaluations, reviews of publications, letters of commendation.) Attach relevant departmental committee recommendations.

Date _____ Signed: _____
Department Head

Dean's Recommendation for Promotion or Tenure
(Cite the following information and sign.)

1. Name of candidate: _____ Present rank: _____
2. Recommended for promotion to the rank of: _____
(Or not recommended): _____
3. Recommended for tenure: Yes/No/NA

Recommendation: Use materials provided by candidate and department head, as appropriate, but please indicate your evaluation of the candidate's performance to date and prospects for the future. Avoid general, subjective opinions; stress obvious strong points, and indicate where further development may be expected. Attach relevant college/school committee recommendations.

Date _____ Signed: _____
Dean

Initial Offer Letter

Letter of Recommendation: Department Head

Letter of Recommendation: Department Promotion & Tenure Committee

Committee Report Form

Curriculum Vitae

Jingdao Chen

Butler Hall 314

Mississippi State, MS 39762

chenjingdao@cse.msstate.edu jingdao.github.io U.S. Permanent Resident

Appointments

Mississippi State University

Tenure-track Assistant Professor, Computer Science and Engineering

Starkville, MS

Aug 2021 - present

Education

Georgia Institute of Technology

Ph. D. in Robotics

Research Areas: 3D perception, deep learning, construction robotics, digital twins

Atlanta, GA

May 2021

Georgia Institute of Technology

M.S. in Computer Science

Atlanta, GA

Dec 2019

Washington University in St Louis

B. S. in Electrical Engineering

Summa Cum Laude

St. Louis, MO

May 2015

Work Experience

Georgia Institute of Technology, RICAL Lab

Graduate Research Assistant with Dr. Yong K. Cho

Atlanta, GA

Jun 2015 – May 2021

Meta, Inc. (formerly Facebook, Inc.)

AR/VR Research Intern with Dr. Daniel Huber

Menlo Park, CA

May 2020 – Aug 2020

Wheego Technologies, Inc.

Research Intern, Autonomous Vehicle Project

Atlanta, GA

May 2017 – Jul 2017

Hunter Engineering Company

Co-op, Wheel Balancer Project

St. Louis, MO

Jan 2014 – Jul 2014

Washington University, Electrical and Systems Engineering

Research Assistant, Autonomous Quadcopter Project

St. Louis, MO

Sep 2013 – Dec 2013

Teaching Experience

CSE 3683 – AI Fundamentals

Primary Instructor

Mississippi State University

Fall 2025, Spring 2025

CSE 4633 / 6633 – Artificial Intelligence

Primary Instructor

Mississippi State University

Spring 2024, Spring 2023

CSE 4643 / 6643 – AI Robotics

Primary Instructor

Mississippi State University

Fall 2021-2024

CSE 8990 – Special Topics in CS – Advanced AI Robotics

Primary Instructor

Mississippi State University

Spring 2022

Office of Institutional Diversity and Inclusion

Diversity Education Certificate

Mississippi State University

Spring 2023

CEE 8813 – Automation in Construction

Co-developer and Instructor

Georgia Institute of Technology

Fall 2018, Fall 2019, Spring 2021

Center for Teaching and Learning

Tech to Teaching Certificate

Georgia Institute of Technology

Spring 2019 – Fall 2019

Mentoring Experience

- Jinhee Yu, Grant Greenwood, Kevin Gao, Jeffrey Rugg, Munib Ahsan, Mark Trovinger, Mark Shiffer, Charles Raines, Shawn Frye, Mississippi State University, CS PhD, 2023 - current
- Charles Moore, Lucian Murdock, Erez Meoded, Joshua Waldbieser, Jacob Kutch, Sushant Gautam, Malika Dutta, Mississippi State University, CS Master's, 2022 - current

- Yuexin Xu, Justin Yi, Prabesh Khanal, Austin Cosby, Reyki Garcia, Arjun Koinkar, Rahul Kumar, Aadhav Senthil, Alaa Mohammad, Carter Holton, Leonel Giacobbe, Tanner Grantham, Isha Shresta, Neil Sanipara, Mississippi State University, CS Undergraduate Research, 2022 - current
- Shiqin Zeng, Georgia Tech, CSE/CEE Master's thesis, 2019 - 2020
- Mark Kahoush, John Yi, Georgia Tech, CS Undergraduate Research, 2019 - 2021

Honors / Awards

- Best Paper Award (co-author), "SLAM-driven Intelligent Autonomous Mobile Robot Navigation for Construction Applications." (2018) European Group for Intelligent Computing in Engineering (EG-ICE), 25th Workshop, Lausanne, Switzerland
- Best Poster Award: 2nd runner-up, "Semantic Parsing of 3D Point Clouds for Construction Progress Estimation." (2018) Construction Research Congress (CRC), New Orleans, LA.
- Best Poster Award, "Autonomous Mobile Robot Navigation and Scan Planning for 3D Mapping of Infrastructure" (2018). Georgia Institute of Technology Career, Research, and Innovation Development Conference (CRIDC), Atlanta GA.
- Best Paper Award (co-author), "Building element recognition with thermal-mapped point clouds." (2017) International Symposium on Automation and Robotics in Construction (ISARC), Taipei, Taiwan.
- Best Paper Award (co-author), "Target-Free Automatic Registration of Point Clouds." (2016) International Symposium on Automation and Robotics in Construction (ISARC), Auburn, AL.
- David Levy ESE Award for Design Excellence (2015). Washington University in St. Louis.

Research Grants and Gifts

NSF CAREER - \$510,532	PI
<i>Constructing Schematic Maps for Construction Robots in Highly Dynamic Environments</i>	2025 - 2030
NSF Innovative Technology Experiences for Students and Teachers (ITEST) - \$1,192,951	co-PI
<i>Learning to create Intelligent Solutions with Machine Learning and Computer Vision: A Pathway to AI Careers for Diverse High School Students</i>	2024 - 2027
IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA) - \$160,499	PI
<i>Scene Understanding, Reconstruction and Rendering with Varying Environment Altitude and Limited Imagery</i>	2023 - 2024
NSF Secure and Trustworthy Cyberspace (SaTC) - \$399,978	co-PI
<i>Inculcate a culture of preparedness against AI security threats to pervasive robotic systems</i>	2023 - 2026
NASA EPSCoR Rapid Response Research (R3) - \$100,000	PI
<i>Incremental learning with knowledge distillation for autonomous rover terrain characterization</i>	2023 - 2024
NSF Computer and Information Science and Engineering Research Initiation Initiative (CRII) - \$175,000	PI
<i>Semantic Vector Fields for Robot Navigation and Exploration in Unstructured Environments</i>	2022 - 2024
Google Cloud Computing Research Credits	2022 - 2023
XSEDE Cloud Computing Startup Allocation	2022 - 2023

Professional / Service Activities

Associate Editor

- Journal of Computing in Civil Engineering
- Guest Editor: Special Collection on Construction Robotics

Conference Chair

- Session Chair for International Conference on Computing in Civil Engineering (I3CE) 2024
- Session Chair for International Conference on Construction Engineering and Project Management (ICCEPM) 2022
- Session Chair for International Conference on Computing in Civil Engineering (I3CE) 2021

Organizer

- 4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments at ICRA 2025
- Workshop on Deep Learning Tools for Understanding and Modeling the Built Environment at I3CE 2024
- 3rd Workshop on Future of Construction: Lifelong Learning Robots in Changing Construction Sites at ICRA 2024
- 2nd Workshop on Future of Construction: Robot Perception, Mapping, Navigation, Control in Unstructured and Cluttered Environments at ICRA 2023
- Virtual Scan-to-BIM Bootcamp with Seoul National University, 2022
- Tutorial on Scan-to-BIM using Python and Open3D at ICCEPM 2022, presenter

- Workshop on Future of Construction: Build Faster, Better, Safer - Together with Robots at ICRA 2022
- 2019 Southeast Robotics Symposium, organizing committee

Invited Talks

- *Constructing Maps for Construction Robots*
 - IEEE RAS Student Branch Chapter, July 2025
- *Robot Perception in Unstructured Environments*
 - University of Alabama Distinguished Colloquium Speaker, November 2024
- *Panel on AI and ML in HPC*
 - Mississippi High Performance Computing Conference, March 2024
- *Robust Scene Understanding for Robots in Unstructured Environments*
 - Mississippi State University CSE Seminar, February 2022
 - IEEE Mississippi Seminar, April 2022

Reviewer

- Automation in Construction
- Journal of Computing in Civil Engineering
- Advanced Engineering Informatics
- International Symposium on Automation and Robotics in Construction
- International Conference on Computing in Civil Engineering (i3CE)
- IEEE Transactions on Robotics
- IEEE Transactions on Field Robotics
- IEEE Robotics and Automation Letters
- IEEE International Conference on Robotics and Automation
- IEEE International Conference on Intelligent Robots and Systems
- IEEE International Conference on Automation Science and Engineering
- IEEE Access

Panelist

- National Science Foundation (NSF) Panelist, 2021, 2023, 2024, 2025

Professional Memberships

- IEEE RAS Technical Committee on Construction Robotics, Co-Chair
- IEEE Robotics and Automation Society, Member

Departmental Service

- Courses and Curricula Committee
- Graduate Studies Committee
- Faculty advisor for AI Club, State-Space Robotics Team

Outreach

- Mississippi State University: Intro to Engineering Summer Camp, Engineering Demo Day, Academic Insight
- Georgia Tech : Girl Code Project, LEGO Mindstorms competition, National Robotics Week

Journal Publications

- [J1] Yang, Y., Wijewardane, N., Slonecki, J., Wadl, P., Andreason, S., **Chen, J.**, Harvey, L. (2025). "Nondestructive detection of sweet potato leaf curl virus using 3D laser imaging combined with deep learning". Smart Agricultural Technology, Volume 11, 2025, 101004, ISSN 2772-3755,
- [J2] Armeni, I., Chen, J., Feng, C., Jeong, I., Liang, C.J., Morin, K., Tsagarakis, N., Wang, X., and Zhang, L. (2024). "Construction Robotics and Automation [TC Spot Light]". IEEE Robotics and Automation Magazine, vol. 31, no. 4, pp. 186-192, Dec. 2024
- [J3] Neupane, S., Mitra, S., Fernandez, I., Saha, S., Mittal, S., **Chen, J.**, Pillai, N., Rahimi, S. (2024). "Security Considerations in AI-Robotics: A Survey of Current Methods, Challenges, and Opportunities" IEEE Access, vol. 12, pp. 22072-22097, 2024
- [J4] Gao, K., Haverly, A., Mittal, S., Wu, J., **Chen, J.** (2024). "AI Ethics: A Bibliometric Analysis, Critical Issues, and Key Gaps" International Journal of Business Analytics (IJBAN), 11(1), 1-19.
- [J5] Vincent, G., Ward, I., Moore, C., **Chen, J.**, Pak, K., Yepremyan, A., Wilson, B., and Goh, E. (2023). "CLOVER: Contrastive Learning for Onboard Vision-Enabled Robotics" Journal of Spacecraft and Rockets. December 2023.

- [J6] **Chen, J.**, and Cho, Y. (2022). "CrackEmbed: Point feature embedding for crack segmentation from disaster site point clouds with anomaly detection" *Advanced Engineering Informatics*, Volume 52, April 2022
- [J7] **Chen, J.**, Kira, Z., and Cho, Y. (2021). "LRGNet: Learnable Region Growing for Class-Agnostic Point Cloud Segmentation." *IEEE Robotics and Automation Letters*, 6(2), pp. 2799-2806. Accepted for oral presentation at the IEEE International Conference on Robotics and Automation (ICRA).
- [J8] Price, L., **Chen, J.**, Park, J.S. and Cho, Y. (2021). "Multisensor-driven Real-time Crane Monitoring System for Blind Lift Operations: Lessons Learned from a Case Study." *Automation in Construction*, Volume 124, April 2021, 103552
- [J9] **Chen, J.**, Yi, J., Kahoush, M., Cho, E. and Cho, Y. (2020). "Point Cloud Scene Completion of Obstructed Building Facades with Generative Adversarial Inpainting." *MDPI Sensors*, 20(18), 5029
- [J10] Zeng, S., **Chen, J.**, and Cho, Y. (2020). "User exemplar-based building element retrieval from raw point clouds using deep point-level features." *Automation in Construction*, Volume 114, June 2020, 103159
- [J11] Park, J.S., **Chen, J.**, Cho, Y., Kang, D., and Son, B. (2020). "CNN-Based Person Detection Using Infrared Images for Night-Time Intrusion Warning Systems." *MDPI Sensors*, 20(1), 34
- [J12] **Chen, J.**, Kira, Z. and Cho, Y. (2019). "Multi-view Incremental Segmentation of 3D Point Clouds for Mobile Robots." *IEEE Robotics and Automation Letters*. 4(2), pp. 1240-1246.
- [J13] **Chen, J.**, Kira, Z. and Cho, Y. (2019). "Deep Learning Approach to Point Cloud Scene Understanding for Automated Scan to 3D Reconstruction." *ASCE Journal of Computing in Civil Engineering*, 33(4).
- [J14] Fang, Y., **Chen, J.**, Cho, Y., and Kim, K.N. (2018). "Vision-based Load Sway Monitoring to Improve Crane Safety in Blind Lifts." *Journal of Structural Integrity and Maintenance*, 3(4), Pages 233-242.
- [J15] Kim, P., **Chen, J.**, and Cho, Y. (2018). "SLAM-driven robotic mapping and registration of 3D point clouds." *Automation in Construction*, Volume 89, Pages 38-48.
- [J16] Kim, P., **Chen, J.**, and Cho, Y. (2017). "Automated Point Clouds Registration using Visual and Planar Features for Construction Environments." *ASCE Journal of Computing in Civil Engineering*, Volume 32, March 2018
- [J17] **Chen, J.**, Fang, Y., and Cho, Y. (2017). "Performance Evaluation of 3D Descriptors for Object Recognition in Construction Applications." *Automation in Construction*, Volume 86, Pages 44-52, February 2018
- [J18] Kim, P., **Chen, J.**, and Cho, Y. (2017). "Robotic sensing and object recognition from thermal-mapped point clouds." *International Journal of Intelligent Robotics and Applications*. September 2017, Volume 1, Issue 3, Pages 243-254
- [J19] **Chen, J.**, Fang, Y., and Cho, Y. (2017). "Real-Time 3D Crane Workspace Update Using a Hybrid Visualization Approach." *ASCE Journal of Computing in Civil Engineering*, Vol. 31. Issue 5
- [J20] Park, J.W., **Chen, J.**, and Cho, Y. (2017). "Self-Corrective Knowledge-based Hybrid Tracking System Using BIM and Multimodal Sensors." *Advanced Engineering Informatics*, Volume 32, Issue C, April 2017, Pages 126-138
- [J21] **Chen, J.**, Fang, Y., Cho, Y., and Kim, C. (2016). "Principal Axes Descriptor (PAD) for Automated Construction Equipment Classification from Point Clouds." *ASCE Journal of Computing in Civil Engineering*, Volume 31, Issue 2
- [J22] Fang, Y., Cho, Y., and **Chen, J.** (2016). "A Framework for Real-time Pro-active Safety Assistance for Mobile Crane Lifting Operations." *Automation in Construction*, Volume 72, Part 3, December 2016, Pages 367-379

Conference Publications

- [C1] Yu, J., Jayakumar, M., Chen, Y., Cho, Y., and **Chen, J.** (2025). "Self-supervised Learning with LiDAR-Camera Fusion for Construction Site Traversability Estimation". *Proceedings of the ASCE 2025 International Conference on Computing in Civil Engineering (i3CE)*, New Orleans, LA, USA, May 11-14
- [C2] Rugg, J., Kim, S., Cho, Y., **Chen, J.** (2025). "Segment Anything Model for Point Cloud Temporal Change Detection on Construction Sites". *Proceedings of the ASCE 2025 International Conference on Computing in Civil Engineering (i3CE)*, New Orleans, LA, USA, May 11-14

- [C3] Fernandez, I., Neupane, S., Chakraborty, T., Mitra, S., Mittal, S., Pillai, N., **Chen, J.**, and Rahimi, S. (2024). "A Survey on Privacy Attacks Against Digital Twin Systems in AI-Robotics". 2024 IEEE 10th International Conference on Collaboration and Internet Computing (CIC), Washington, District of Columbia, United States, 28 - 30 October 2024
- [C4] Gao, K., Mittal, S., Wu, J., Du, H., **Chen, J.**, and Rahimi, S. (2024). "The AI Pentad, the CHARME2D Model, and an Assessment of Current-State AI Regulation". 10th IEEE International Conference on Sustainable Technology and Engineering (i-COSTE), Perth, Australia, 18th - 20th December, 2024
- [C5] Yu, J., Saha, S., Jayakumar, M., Gugssa, M., **Chen, J.**, and Wang, J. (2024). "LiDAR-based Traversability Estimation for Ground Robots on Construction Sites using Self-Supervised Learning". Proceedings of the ASCE 2024 International Conference on Computing in Civil Engineering (i3CE), Pittsburgh, PA, USA, July 28-31
- [C6] Rugg, J., **Chen, J.**, Gugssa, M., and Wang, J. (2024). "Object-level Temporal Change Detection on Construction Sites with 3D Deep Learning Models". Proceedings of the ASCE 2024 International Conference on Computing in Civil Engineering (i3CE), Pittsburgh, PA, USA, July 28-31
- [C7] Chang, S., **Chen, J.**, and Park, J. (2024). "Review of Metaverse Technologies to Broaden Accessibility in Civil and Construction Engineering Education." Construction Research Congress 2024, March 20-23, Des Moines, IA
- [C8] Gao, K., Haverly, A., Mittal, S., and **Chen, J.** (2023). "A bibliometric view of AI Ethics development." IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), Fiji, December 4-6
- [C9] Garshabi, A., Wang, J., **Chen, J.**, and Ma, J. (2023). "Human-Robot Collaboration in the Construction Industry: A Mini-review." IEEE International Conference on Robotics and Automation (ICRA) Workshop on Future of Construction, London, U.K., June 2
- [C10] **Chen, J.**, Gugssa, M., Yee, J., Wang, J., Goodin, C., and Ram Das, A. (2023). "Framework for digital twin creation in off-road environments from LiDAR scans." Proc. SPIE 12529, Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications, 125290F (13 June 2023)
- [C11] Yu, J., **Chen, J.**, Dabbiru, L., and Goodin, C. (2023). "Analysis of LiDAR configurations on off-road semantic segmentation performance." Proc. SPIE 12540, Autonomous Systems: Sensors, Processing, and Security for Ground, Air, Sea, and Space Vehicles and Infrastructure 2023, 1254003 (13 June 2023)
- [C12] Goh, E., Ward, I. R., Vincent, G., Pak, K., **Chen, J.**, and Wilson, B. (2023). "Self-supervised Distillation for Computer Vision Onboard Planetary Robots." IEEE Aerospace Conference, Big Sky, MT, USA, March 4-11
- [C13] Ward, I. R., Moore, C., Pak, K., **Chen, J.**, and Goh, E. (2022). "Improving Contrastive Learning on Visually Homogeneous Mars Rover Images." European Conference on Computer Vision (ECCV) Workshop on AI4Space, Tel Aviv, Israel, Oct 23
- [C14] Vincent, G., Yepremyan, A., **Chen, J.**, and Goh, E. (2022). "Mixed-Domain Training Improves Multi-Mission Terrain Segmentation." European Conference on Computer Vision (ECCV) Workshop on AI4Space, Tel Aviv, Israel, Oct 23
- [C15] Kim, S., Yajima, Y., Park, J., **Chen, J.**, and Cho, Y. (2022). "A Hybrid Semantic-Geometric Approach for Clutter-Resistant Floorplan Generation from Building Point Clouds." Proceedings of the 9th International Conference on Construction Engineering and Project Management (ICCEPM), Las Vegas, NV, USA, June 20-23
- [C16] Goh, E., **Chen, J.**, and Wilson, B. (2022). "Mars Terrain Segmentation with Less Labels". IEEE Aerospace Conference, Big Sky, MT, USA, March 5-12
- [C17] Yajima, Y., Kim, S., **Chen, J.**, Cho, Y. (2021). "Fast Online Incremental Segmentation of 3D Point Clouds from Disaster Sites." Proceedings of the 38th International Symposium on Automation and Robotics in Construction (ISARC), Dubai, UAE, November 2-5
- [C18] Kahoush, M., Yajima, Y., Kim, S., **Chen, J.**, Park, J., Kangisser, S., Irizarry, J., Cho, Y. (2021). "Analysis of Flight Parameters on UAV Semantic Segmentation Performance for Highway Infrastructure Monitoring." Proceedings of the ASCE 2021 International Conference on Computing in Civil Engineering (i3CE), Orlando, FL, USA, September 12-14
- [C19] Yajima, Y., Kahoush, M., Kim, S., **Chen, J.**, Park, J., Kangisser, S., Irizarry, J., Cho, Y. (2021). "AI-driven 3D Point Cloud-Based Highway Infrastructure Monitoring System using UAV." Proceedings of the ASCE 2021 International Conference on Computing in Civil Engineering (i3CE), Orlando, FL, USA, September 12-14

- [C20] **Chen, J.**, Kim, P., Sun, D.I., Han, C.S., Ahn, Y.H., Ueda, J., and Cho, Y. (2020). "Workspace Modeling: Visualization and Pose Estimation of Teleoperated Construction Equipment from Point Clouds." Proceedings of the 37th International Symposium on Automation and Robotics in Construction (ISARC), October 27-29
- [C21] Price, L., **Chen, J.**, and Cho, Y. (2020). "Dynamic Crane Workspace Update for Collision Avoidance during Blind Lift Operations." Proceedings of the 18th International Conference on Computing in Civil and Building Engineering (ICCCBE) 2020, August 18-20, São Paulo, Brazil
- [C22] **Chen, J.**, and Cho, Y. (2020). "Unsupervised Crack Segmentation from Disaster Site Point Clouds using Point Feature Clustering." Proceedings of the 27th International Workshop on Intelligent Computing in Engineering (EG-ICE) 2020, July 1-4, Berlin, Germany
- [C23] Park, J., **Chen, J.**, and Cho, Y. (2020). "Point Cloud Information Modeling (PCIM): an Innovative Framework for as-is Information Modeling of Construction Sites". ASCE Construction Research Congress (CRC) 2020, March 9-10, Tempe, AZ
- [C24] **Chen, J.** and Cho, Y. (2019). "Exemplar-based Building Element Retrieval from Point Clouds." International Conference on Smart Infrastructure and Construction (ICSIC), Churchill College, Cambridge, UK, July 8-9
- [C25] **Chen, J.** and Cho, Y. (2019). "Detection of Damaged Infrastructure on Disaster Sites using Mobile Robots." IEEE 2019 16th International Conference on Ubiquitous Robots (UR), Jeju, Korea, June 24-27
- [C26] **Chen, J.**, Kim, K.N., Cho, Y., Lee, J., Kim, B., Ahn, Y., and Kang, J. (2019). "Nuclear Power Plant Disaster Site Simulation using Rigid Body Physics." Proceedings of the ASCE 2019 International Conference on Computing in Civil Engineering (i3CE), Atlanta, GA, USA, June 17-19
- [C27] Kim, K.N., **Chen, J.**, and Cho, Y. (2019). "Evaluation of Machine Learning Algorithms for Worker's Motion Recognition using Motion Sensors." Proceedings of the ASCE 2019 International Conference on Computing in Civil Engineering (i3CE), Atlanta, GA, USA, June 17-19
- [C28] **Chen, J.**, Cho, Y., and Ueda, J. (2018). "Sampled-Point Network for Classification of Deformed Building Element Point Clouds". Proceedings of the 2018 IEEE Conference on Robotics and Automation (ICRA).
- [C29] **Chen, J.**, Kim, P., Cho, Y., and Ueda, J. (2018). "Object-sensitive potential fields for mobile robot navigation and mapping in indoor environments." Proceedings of the 2018 IEEE 15th International Conference on Ubiquitous Robots (UR), Honolulu, HI, USA, June 26-30.
- [C30] Fang, Y., **Chen, J.**, Cho, Y., Zhang, S., and Perez, E. (2018). "Enhance Blind Lift Safety by Real-Time Sensing and Visualization." Proceedings of the 18th International Conference on Construction Applications of Virtual Reality (CONVR2018), Auckland, New Zealand, Nov 22-23
- [C31] Kim, P., **Chen, J.**, Kim, J., and Cho, Y. (2018). "SLAM-Driven Intelligent Autonomous Mobile Robot Navigation for Construction Applications." Best Paper Award, Proceedings of Workshop of the European Group for Intelligent Computing in Engineering, EG-ICE. pp. 254-269, Lausanne, Switzerland,
- [C32] **Chen, J.** and Cho, Y. (2018). "Point-to-point Comparison Method for Automated Scan-vs-BIM Deviation Detection." Proceedings of 17th International Conference on Computing in Civil and Building Engineering, Tampere, Finland
- [C33] Kim, P., **Chen, J.**, Cho, Y. (2018). "Autonomous Mobile Robot Localization and Mapping for Unknown Construction Environments." ASCE Construction Research Congress (CRC) 2018, pp.147-156, April 2-4, New Orleans, LA
- [C34] **Chen, J.**, Cho, Y., and Kim, K. (2018). "Region Proposal Mechanism for Building Element Recognition for Advanced Scan-to-BIM Process." ASCE Construction Research Congress 2018, April 2 - 4, New Orleans, LA
- [C35] Kim, P., Cho, Y. and **Chen, J.** (2017). "Building element recognition with thermal-mapped point clouds." Proceedings of the 34rd International Symposium on Automation and Robotics in Construction (ISARC), Taipei, Taiwan, June 28-July 1
- [C36] **Chen, J.**, Fang, Y., and Cho, Y. (2017). "Mobile Asset Tracking for Dynamic 3D Crane Workspace Generation in Real Time." Proceedings of the 2017 International Workshop on Computing for Civil Engineering (IWCCE), Seattle, WA, USA, June 25-27

[C37] **Chen, J.**, Fang, Y., and Cho, Y. (2017). "Unsupervised Recognition of Volumetric Structural Components from Building Point Clouds." Proceedings of the 2017 International Workshop on Computing for Civil Engineering (IWCCE), Seattle, WA, USA, June 25-27,

[C38] Kim, P., Cho, Y., and **Chen, J.** (2016). " Automatic Registration of Laser Scanned Color Point Clouds Based on Common Feature Extraction." 16th International Conference on Construction Applications of Virtual Reality (CONVR), Hong Kong, Dec. 11-13

[C39] **Chen, J.**, Fang, Y., and Cho, Y. (2016). "Automated Equipment Recognition and Classification from Scattered Point Clouds for Construction Management." International Symposium on Automation and Robotics in Construction (ISARC), Auburn, AL, July 18-21, 2016

[C40] **Chen, J.**, Cho, Y. (2016). "Real-time 3D Mobile Mapping for the Built Environment". International Symposium on Automation and Robotics in Construction (ISARC), Auburn, AL, July 18-21, 2016

[C41] Fang, Y., **Chen, J.**, Cho, Y., and Zhang, P. (2016). "A Point Cloud-Vision Hybrid Approach for 3D Location Tracking of Mobile Construction Assets." International Symposium on Automation and Robotics in Construction (ISARC), Auburn, AL, July 18-21, 2016

[C42] Kim, P., Cho, Y., **Chen, J.** (2016). "Target-Free Automatic Registration of Point Clouds." International Symposium on Automation and Robotics in Construction (ISARC), Auburn, AL, July 18-21, 2016

Workshop Presentations

Chen, J., Park, J., Yajima, Y., Kim, S. (2021). GTS2B. 1st Workshop and Challenge on Computer Vision in the Built Environment. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2021

Book Chapters

[B1] **Chen, J.**, and Cho, Y. (2021). "Rapid Scan-to-BIM using Robotics and Artificial Intelligence for Construction Applications." Research Companion on Building Information Modeling, March 2022.

Teaching Accomplishments

Contents

- Highlights
- Teaching Philosophy
- Student Feedback
- Curriculum Development
- Student Mentoring
- Peer Evaluations
- Sample Syllabus for CSE 8990 Advanced AI Robotics

Highlights

- Overall positive student feedback and encouraging peer evaluations.
 - Developed two new courses (CSE 3683 AI Fundamentals and CSE 8990 Advanced AI Robotics) and updated existing courses CSE 4633/6633 Artificial Intelligence and CSE 4643/6643 AI Robotics.
 - Mentored 15 MS and 16 PhD students as major professor.
-

In this section of my dossier, I demonstrate my commitment to teaching and mentoring of students. Specifically, I describe my teaching philosophy; present overall scores from my student teaching evaluations, along with samples of student feedback; describe my efforts to develop and teach new courses at MSU; and elaborate on my involvement as an advisor/mentor.

Teaching Philosophy

Communicate effectively. Effective communication is key to keeping a class organized. At the start of each semester, I try to provide the students with a rough timeline and schedule of assignment deadlines on Canvas so that they can plan ahead and allocate their time accordingly. For each programming assignment, I also provide a written document with detailed instructions and also perform a "demo" of what a completed assignment should look like in class. These strategies have been effective in communicating expectations with students and preventing confusion when working on assignments.

Encourage creativity. AI is a rapidly-changing field with many novel developments, ideas, and models being released every day. This could offer new opportunities as well as present new challenges for students who are studying AI. In my AI classes, I try to encourage students to first understand the fundamental concepts in AI, but also to think outside the box and come up with different approaches to solve AI problems beyond the "textbook" algorithm given in class. To motivate this, I offer "open-ended extra credit questions" in each homework assignment where students can implement any visualization or algorithmic improvements in their code (with proper justification) and receive extra credit points for their efforts. In the field of AI, where there are still many open challenges and unsolved problems, this encourages students to be willing to experiment and try out their ideas, however big or small those ideas are. As a result of these "open-ended extra credit questions", I received several creative solutions to homework assignments every semester, and these have also greatly stimulated my thinking on how to solve different problems in AI.

Have in-class programming assignments. In computer science and engineering classes, connecting the theoretical concepts with actual code and using it to solve real-world problems can be challenging. My approach to this issue is to put the concepts learnt in class into practice through a series of in-class programming assignments throughout the semester. The assignments deal with practical problems in AI (e.g. path planning for delivery robots, building a chatbot, classifying images etc.) and students have the opportunity to follow along the coding process and build those AI applications over the course of a class period. Students are encouraged to bring their laptops and have the opportunity to work together with classmates and ask questions. The grade for in-class programming assignments is given based on participation, allowing students to experiment with code in a stress-free manner. In general, students have responded positively to these in-class programming assignments (see feedback comments below) and I now incorporate these assignments into all my classes.

Give students equal opportunities to succeed. A vital part of creating an inclusive classroom is to be able to understand and cater to diverse needs in the classroom. This could include transfer students and first-generation college students who may not have as much access to proper preparation for college classes, or international students who may have issues with language and cultural barriers, or students who have disabilities or other medical needs. I believe that fundamentally, every student has the capability to succeed, and part of our role as educators is to support students on the path to success. I have observed several examples where even when a student is struggling at the start of the semester, the student can still catch up and perform well at the end of the semester. My approach is to not give up on poorly-performing students but give students additional leeway with absences, allow students access to video recordings, and schedule extra office hours when applicable.

Facilitate online learning. The CSE department has a large population of online MS and online PhD students, thus the classes that I have taught tend to have many online students. AI is also an attractive field for people working in the industry across science and engineering, who want to pursue graduate studies and grow their career through incorporating AI techniques into their current workplace. In my teaching, I try to facilitate online learning and cater to the needs of these distance students. I have had to make several course modifications including creating lecture recordings, hosting virtual office hours, and converting in-person exams to online quizzes. However, facilitating online learning has an overall positive effect because online students are usually more mature and are able to use their industry experience to contribute to interesting projects and in-class discussions. I also allow on-campus students to collaborate with online students, helping them build a broader network and establish connections to industry.

Student Feedback

At the end of each semester, students may provide anonymous feedback in the form of specific scored questions, as well as written free-form comments. Here, I discuss my

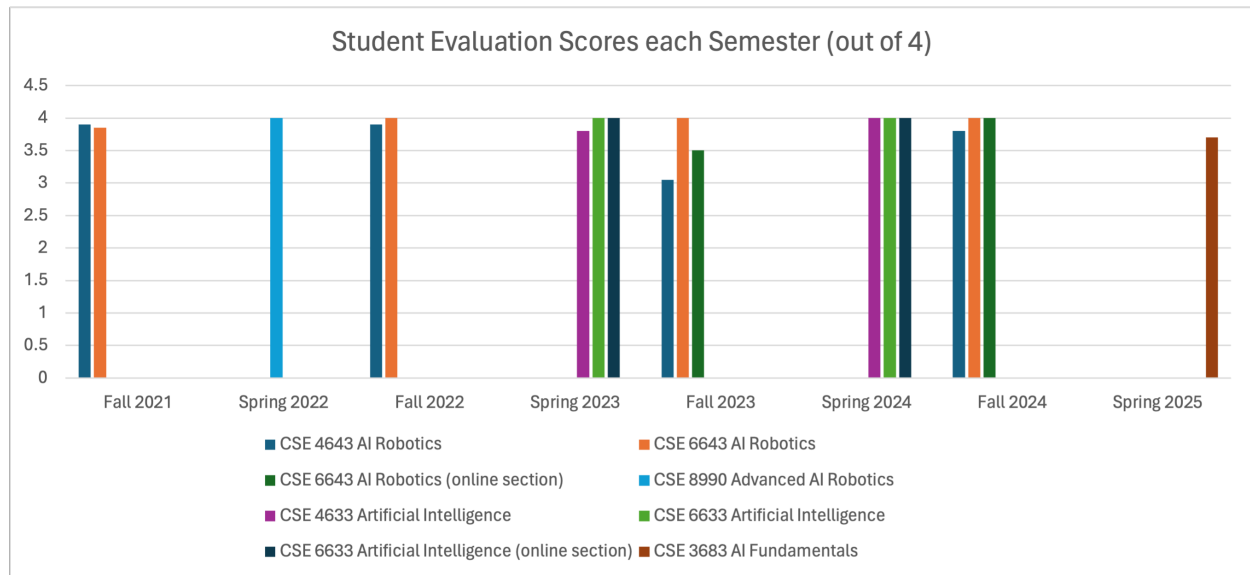


Figure 1: A summary of my student evaluation scores by semester.

numerical scores, as well as how I incorporate student feedback into my teaching efforts.

Summary of Numerical Student Feedback. Table 1 illustrates my average student evaluation scores for each class section I have taught. The scores are obtained from the institution-wide class climate surveys which are conducted every semester. The surveys consist of multiple questions about the students' experience in the course which the students can rate using a number from 1 (Strongly Disagree) to 4 (Strongly Agree). The survey reports the median student response for each question, and Table 1 further computes the average response over all questions. The scores show that my teaching quality has been consistently high, with scores ranging between 3.0 and 4.0 every semester.

Summary of Free-Form Student Feedback. In addition to numerical scores, students may also provide anonymous free-form written feedback. Below, I've included a sampling of this feedback. Under *Positive Student Feedback*, students indicated that enjoyed the design of the projects and homework assignments and especially appreciated the in-class programming assignments for getting hands-on practice and helping to understand abstract topics in AI.

Under *Students' Feedback that Motivated Course Changes*, I listed some comments from students which I incorporated into future iterations of the class. For example, in AI Robotics, students expressed concerns that they do not have adequate preparation for using frameworks such as numpy and ROS. To address this issue, I added tutorials for Python, numpy, and ROS with practice questions at the start of the semester so that students have a chance to get familiarized with those frameworks. In Artificial Intelligence, some students pointed out that there was too much code copying during the in-class assignments which hindered learning. Thus, for future in-class assignments, I included more template code and focused the in-class discussion on different algorithmic choices

Table 1: Summary of student evaluation scores by course

Course	Title	Semester	Evaluation Score
CSE 4643	AI Robotics	Fall 2021	3.9/4.0
CSE 6643	AI Robotics	Fall 2021	3.85/4.0
CSE 8990	Advanced AI Robotics	Spring 2022	4.0/4.0
CSE 4643	AI Robotics	Fall 2022	3.9/4.0
CSE 6643	AI Robotics	Fall 2022	4.0/4.0
CSE 4633	Artificial Intelligence	Spring 2023	3.8/4.0
CSE 6633	Artificial Intelligence	Spring 2023	4.0/4.0
CSE 6633	Artificial Intelligence (online section)	Spring 2023	4.0/4.0
CSE 4643	AI Robotics	Fall 2023	3.05/4.0
CSE 6643	AI Robotics	Fall 2023	4.0/4.0
CSE 6643	AI Robotics (online section)	Fall 2023	3.5/4.0
CSE 4633	Artificial Intelligence	Spring 2024	4.0/4.0
CSE 6633	Artificial Intelligence	Spring 2024	4.0/4.0
CSE 6633	Artificial Intelligence (online section)	Spring 2024	4.0/4.0
CSE 4643	AI Robotics	Fall 2024	3.8/4.0
CSE 6643	AI Robotics	Fall 2024	4.0/4.0
CSE 6643	AI Robotics (online section)	Fall 2024	4.0/4.0
CSE 3683	AI Fundamentals	Spring 2025	3.7/4.0

and comparison of results. Finally, some students indicated that certain lectures were lacking in student interaction. To address this, I increased the frequency of in-class activities and group discussions.

Curriculum Development

During my time at MSU, I have constantly worked to improved the curriculum of my CSE classes to ensure that students are well-prepared for a career in AI and robotics, whether in academia or industry. In my 3 years teaching CSE 4633/6633 Artificial Intelligence and CSE 4643/6643 AI Robotics, I have put in significant efforts to revise and update the class material before each semester based on student feedback and also incorporate recent developments from AI research. For example, the content of Artificial Intelligence was updated to include modern topics in AI such as generative AI and large language models, and the content of AI Robotics was updated to include modern topics in robotics such as reinforcement learning and deep learning for robotics. In my 3rd year teaching AI Robotics, I have also submitted a course modification to the University Committee on Courses and Curricula to create an online section for the course, with modifications including creating lecture recordings, hosting virtual office hours, and the conversion of in-person exams to online quizzes. The creation of an online section was vital to cater to the needs of online MS and PhD students who want to specialize in the Artificial Intelligence concentration area.

Positive Student Feedback

“Instructor is always willing to work with students and he is good at explaining problems and its solution. He is real helpful. He wants students to work and know the materials. Great Instructor.” –AI Robotics. Fall 2021

“The coding assignments were very practical and clear. Likely the best coding projects I have been given in school. The paper review presentations were also productive and interesting.” –Advanced AI Robotics. Spring 2022

“Dr. Chen responds quickly to emails, and is more than happy for you to drop by his office when you need help on an assignment. Having an instructor who cares and wants you to succeed, who works with you and helps you learn is absolutely fantastic. When the class struggled on Lab 2, he redesigned the lab document instructions for future labs to help with clarification, without taking away from the rigor. It is difficult to put into words how much I appreciate that as a student, when the instructor is putting in the work, I find it very motivational.” –AI Robotics. Fall 2022

“Every project and homework in this class was well-designed and enjoyable. Dr. Chen did an amazing job of assigning us things that were not only instructional but engaging and interesting to interact with. The battlesnake competition was some of the most fun I’ve had in a class and was also extremely instructional.” –Artificial Intelligence. Spring 2023

“I really liked the organization, the hands-on activities via Jupyter notebooks, and the quality of the lectures.” –AI Fundamentals. Spring 2025

In addition, I have also introduced a new special topics course, CSE 8990: Advanced AI Robotics, which covers autonomous driving, Simultaneous Localization and Mapping, deep reinforcement learning, self-supervised learning, and deep learning for perception. The programming assignments and research paper reading lists were designed from scratch based on my own research experience in robotics. This class was taught in Spring 2022 and a sample syllabus is attached towards the end of this section.

In Spring 2025, I developed a new undergraduate course, CSE 3683 AI Fundamentals. The course is designed to be an introductory class and precursor to CSE 4633 Artificial Intelligence. With the rapid pace of recent developments in AI, the new course plays an important role for the university to adequately prepare students for an AI future and cater to the newly introduced Bachelor’s degree in AI.

Student Mentoring

Undergraduate student mentoring. I regularly mentor undergraduate students as part

Students' Feedback That Motivated Course Changes

"The pace was slow at times, though this is not a totally bad thing. More ROS examples (even if they are not assignments) may be helpful." –Advanced AI Robotics. Spring 2022

"When looking at numpy for the first time was a little difficult. Maybe some more resources when learning this library." –AI Robotics. Fall 2022

"The only thing I can think of is this: On the in class assignments, the fact that I have to rush to copy the code that is being typed keeps me from actually listening to the explanation and learning. I had to go back later and actually analyze the code, since I was busy copying the code while the explanation was being given." –Artificial Intelligence. Spring 2023

"Overall, I thoroughly enjoyed the class but the lectures should be more interactive to keep the attention of the students." –AI Robotics. Fall 2023

"I wish group projects were better organized." –Artificial Intelligence. Spring 2024

of my research group at Mississippi State University. At the undergraduate level, I try to open up opportunities for students to gain exposure to AI and robotics research, whether it is through doing hands-on work on quadruped robots in the lab or writing code to train machine learning models. I do encourage undergraduate students to actively pursue research topics and participate in research publications. For example, I mentored a 2nd-year undergraduate student, Justin Yee, to co-author a paper on digital twins for off-road environments, which was later accepted at the SPIE 2023 conference. Justin also presented the work as a poster at the Undergraduate Research Symposium. I have also served as faculty mentor to undergraduate students in the Honor's college who are interested in doing an Honor's thesis; for example, Austin Cosby, who defended his Honor's thesis in April 2023, and Yuexin Xu, who defended her Honor's thesis in November 2023.

Graduate student mentoring. At the graduate level, I mentor several MS and PhD students across several research areas such as robotics, cybersecurity, computer vision, and machine learning. I meet regularly with graduate students and provide them with guidance on research directions and other skills necessary to have a successful career in academia such as paper writing, paper reviewing, technical documentation, and presenting at conferences. I have mentored 6 graduate students as thesis director for their MS thesis: Erez Meoded, Charles Moore, Joshua Waldbeiser, Sushant Gautam, Malika Dutta, and Jacob Kutch.

My philosophy for mentoring PhD students is to train them to perform individual research, guide them on cross-disciplinary research collaborations, and provide them with

workshop and conference participation opportunities. For example, two of my PhD students traveled to the I3CE 2024 conference in Pittsburgh to present conference papers and two of my PhD students traveled to the ICRA 2025 conference in Atlanta to present workshop papers. My first PhD student to graduate, Kevin Gao, finished defending his PhD dissertation in June 2025. Kevin was the first author on 1 journal paper and 2 conference papers, and has now secured a faculty position at California State University at Sacramento.

Peer Evaluations

As part of continual efforts to enhance and improve my teaching abilities, I have participated in classroom observations with peer instructors and have also invited other professors to attend my classes and evaluate my teaching. The peer evaluation results (one for Fall 2022, one for Fall 2023, and one for Fall 2024) are attached to this dossier as follows:

Do begin with specifics about the number and title of the course, the day and time of the observation, the length of the class period, the space in which the class period took place, the number of students present, etc.

- **CSE - 4643/6643**
- **Nov 1, 3:30-4:15 pm**
- **Room 100, Butler Hall**
- **around 25 students were present, classroom is compact and sufficient**

Describe how the instructor communicated with the students about the course, including how this particular class period follows from, builds on, and prepares for other class periods; ongoing work on assignments and upcoming due dates; etc.

- **full syllabi and roadmap is listed in course website**
- **the instructor had a clear recap at the beginning of the class, including lecture and homework**

Describe how the instructor communicated the content of the day's lesson, including lecture, discussion, group activities, examples, illustrations, simulations, demonstrations, case studies, references to textbooks or PowerPoint slides, use of videos, etc.

- **The lecture content is readable and well-organized, and the balance of figures to text on the slides is appropriate.**
- **clear flow of logic in the slides**
- **intuitive examples were provided to explain mathematics regarding robot movement algorithm, such as avoid obstacle, and reaction to various stimuli**
- **after the theoretical study, a movie with a demo and an explanation of the code were offered**

Describe the students' participation in class, including being on time and attentive, taking notes, asking questions, answering questions, offering responses, interacting in small groups, completing in-class activities, etc.

- **the instructor arrived on time**
- **the majority of students were paying attention while listening to the lecture or taking notes**
- **the instructor actively asked student feedback and questions were fully answered**
- **the instructor stops at several proper timepoints**

Peer Review Form

Instructor of the class:	Dr. Jingdao Chen	Observer:	Maxwell Young
Day and time of observation	3:30pm – 4:45pm on Nov.14, 2023	Length of class:	75 minutes
Location of class and description of the room	McCain 175 (Class is CSE 4643/6643 – AI Robotics). Seating is organized in rows. Approximately 50 students (rough estimate).		
How instructor communicated specifics about the course (review, assignments and homework, summary, etc.):	Dr. Chen speaks at a good pace and is easy to understand. One of the first things he did was review upcoming deadlines for graded items that students should be working on.		
How instructor communicated the content of the day's topic:	Slides were used (presented on a large screen) upon which real-time execution of code was displayed, along with lecture content. Font size was large and easy to read. The Cart Pole game was illustrated, and a student participated in playing the game to convey how it works. Dr. Chen also presented the theoretical/mathematical content behind the solution to the Cart Pole game. The source code for a solution was presented and developed over the course of the lecture, with students participating in offering suggestions for code changes. Periodically throughout the lecture, Dr. Chen queried the students whether they were okay with the material so far presented, and he walked around the classroom to inspect how students were progressing with the interactive exercise.		
Describe the students' participation in class (answering questions, asking questions, working in small groups, etc.):	Dr. Chen posed questions to the students. For example, asking them near the start of class whether they have any questions about upcoming tasks/graded items. Since the class was interactive, the students were following along with the exercise and I can verify that (the several students that I observed) did so. Dr. Chen also invited students to come up to the front of the class and demonstrate how the game is played by a human player.		
Any further comments:	This was a well-explained, interactive lecture on a complex topic. It was apparent that Dr. Chen inspired enthusiasm from the students. I was extremely impressed, and I think we're very lucky to have Dr. Jingdao Chen in our department.		

Peer Review Form

Instructor of the class:	Dr. Jingdao Chen	Observer:	Dr. Yan Sun
Day and time of observation	9/3/2024 3:30pm—4:45pm	Length of class:	75 minutes
Location of class and description of the room	Class location: McCain 185 Seating capacity: 60 Student present: 48		
How instructor communicated specifics about the course (review, assignments and homework, summary, etc.):	- Started with an overview of the class of <i>Turtlebot Practice</i> in terms <u>class activities</u> (connecting a remote PC to a TurtleBot using bringup turtlebot and controlling the Turtlebot with a keyboard from the remote PC), <u>class activity mode and format</u> (students working in nine groups and each group hands-on with a TurtleBot), and <u>the ROS environment</u> for controlling TurtleBot - Talked about homework and its due date: <i>Lab 1</i> published on Canvas requiring students to program a Turtlebot in a simulation environment - Summarized and explained (at the end of the class) with an animated graph how nodes in ROS work and how they work with TurtleBot, and invited students to share their “lessons learned”		
How instructor communicated the content of the day’s topic:	The instructor communicated the content by: - pulling on the projection screen alternately the TurtleBot3 documentation (Bringup and Basic Operation) and roscore from PC for demonstration purposes - Explaining, with a physical TurtleBot, different hardware components and how to connect and tuck away battery etc., and how to connect to a local network through a router and password and how to avoid overloading this local network - Demonstrating how to connect to TurtleBot with its IP address - Checking up on each group on their progress, answering their questions, helping troubleshoot the issues they have, and assisting them to resolve the issues - Answering questions from individual students after class		
Describe the students’ participation in class (answering questions, asking questions, working in small groups, etc.):	Students participation took place at two levels: <u>Intragroup level:</u> - Working in self-formed groups to setup, connect, and control their TurtleBots; and more capable group members helping others in the groups by answering questions and demonstrating the whys and hows. <u>Intergroup level:</u> - Interacting with other groups through controlling their Turtlebots in sitcom types of plots they improvised; more capable groups helping other groups when the instructor was helping other groups		
Any further comments:	The class was built on a student-centered approach giving students more ownership of their learning with instructional scaffolding from the instructor. The hands-on and learning through fun and collaboration experiences were engaging and interactive.		

Sample Syllabus for CSE 8990 Advanced AI Robotics

I have developed a new course in the CSE department titled "Advanced AI Robotics", covering recent topics in intelligent robotics such as autonomous driving, Simultaneous Localization and Mapping, deep reinforcement learning, self-supervised learning, and deep learning for perception. This class was taught in Spring 2022. The syllabus for Advanced AI Robotics is attached to this dossier as follows:

CSE-8990-01-202210 Special Topic in CS (Spring 2022)

[Jump to Today](#)

CSE 8990 - Advanced AI Robotics

Thursday 5:00 – 7:30pm

Swalm 145

Instructor

Jingdao Chen

Email: chenjingdao@cse.msstate.edu

Phone: 662-3257514

Office: Rice Hall 102

Office Hours

Thursday 3:30 pm - 4:45 pm. Other times by email appointment.

Course Description

Advanced Robotics deals with the underlying technology in modern, autonomous, artificially intelligent robots for applications including autonomous vehicles, drones, planetary rovers, as well as their broader impacts in related computer science fields such as artificial intelligence and machine learning. The primary topics covered in the class include real-time localization and mapping, deep learning for perception, deep reinforcement learning, and advanced planning. This class is designed to help graduate students learn and understand the latest trends in software frameworks, sensor technologies and data-driven algorithms in the field of robotics. Students will also learn about how these robotic and automation technologies can be applied to a broad range of industries include agriculture and forestry, construction and mining, field exploration, and search and rescue.

Prerequisites

CSE 4643/6643 (AI Robotics) or CSE 4633 (Artificial Intelligence) or CSE 8673 (Machine Learning) or equivalent courses

Class Schedule

Date	Topic	Reading	Assignment
01/20	0. Intro & ROS Tutorial	Linux Tutorial (Links to an external site.) ROS Tutorial (Links to an external site.) (Links to an external site.)	Paper presentation signup (Links to an external site.)
01/27	1. Localization and Mapping	SLAM (Links to an external site.)	Lab #1 released
02/03	2. Feature Descriptors	SIFT (Links to an external site.) 3D Descriptors (Links to an external site.) ORB-SLAM (Links to an external site.) code (Links to an external site.)	Paper selections due
02/10	3. Visual vs Lidar SLAM	Hector SLAM (Links to an external site.) code (Links to an external site.) LeGO-LOAM (Links to an external site.) code (Links to an external site.)	
02/17	4. Structure from Motion	Structure from Motion (Links to an external site.)	

		code (Links to an external site.)	
02/24	5. Object Recognition Benchmarks	KITTI Vision Benchmark (Links to an external site.) dataset (Links to an external site.) ResNet (Links to an external site.)	Lab #1 due. Lab #2 released
03/03	6. Deep Learning for Object Recognition	PointNet (Links to an external site.) code (Links to an external site.)	Project groups signup (Links to an external site.)
03/10	7. Semantic & Instance Segmentation	Panoptic Segmentation (Links to an external site.)	
03/17	8. Deep Reinforcement Learning	Deep Q-Network (Links to an external site.) BADGR (Links to an external site.)	Project Proposal due
03/24	9. Self-supervised Learning	code (Links to an external site.)	
03/31	10. Exploration		Lab #2 due
04/07	Open Topic		
04/14	Open Topic		Interim Project Report due
04/21	Open Topic		
04/28	CAVS Lab Tour		

The instructor reserves the right to change the course policies and/or schedule in order to facilitate instruction. Any such changes will be updated on Canvas and distributed to students via Announcements on Canvas or through the course email list. Additionally, all changes will be discussed in class.

Learning Objectives

The learning outcomes for students in this course will be as follows: (1) be familiarized with the state-of-the-art algorithms and software frameworks in robotics (2) understand the metrics for robotics algorithms including computational efficiency, accuracy, and real-world applicability (3) implement a robotics project that incorporates multiple algorithms related to sensing, planning, control, and learning (4) develop critical reading skills to understand the strengths and weaknesses of current approaches to research problems and identify interesting open questions and future research directions (5) develop the background and skills necessary to perform research in robotics and related fields such as artificial intelligence and machine learning.

Textbooks

- Probabilistic Robotics - Sebastian Thrun, Wolfram Burgard, Dieter Fox

<https://mitpress.mit.edu/books/probabilistic-robotics> (Links to an external site.)

- Principles of Robot Motion - Choset et al.

<https://mitpress.mit.edu/books/principles-robot-motion> (Links to an external site.)

Additional online readings will be posted on the class schedule on a regular basis.

Individual Lab Assignments

Two lab assignments will be given throughout the semester and is an essential part of understanding the lecture materials. The assignment should be submitted by the specified due date by the end of the day. Assignments will be graded and returned. Students are responsible for assignments turned in on time; even for days when the student has an excused absence. Unless otherwise specified, assignments will be done individually and each student must hand in their

own assignment. It is acceptable, however, for students to collaborate in figuring out solutions and helping each other understand the underlying concepts. When collaborating, the "whiteboard policy" is in effect: You may discuss assignments on a whiteboard, but, at the end of a discussion the whiteboard must be erased, and you must not transcribe or take with you anything that has been written on the board during your discussion. You must be able to reproduce the results solely on your own after any such discussion. Finally, you must report the names of the students you collaborated with on each assignment.

Research Paper Presentations

This class will feature in-class discussion of landmark robotics research papers and provide an opportunity to critique and evaluate the merits of each paper. The paper discussions will also provide an opportunity to examine recent challenges and potential future directions in robotics. Each student will individually make a presentation and lead the discussion of one research paper during the semester. After the paper presentation, all students are expected to take part in class discussions and should contribute at least one question or discussion point. The presentation should make use of one of the topics / papers listed in the syllabus above. Other topics / papers may be allowable (e.g. papers related to your own research or thesis) but will require prior approval from the instructor.

Final Group Project

The course project is an opportunity to explore some of the techniques covered in class and apply them to solve a robotics problem that is of interest. The project topic may be anything related to AI and robotics including navigation, localization, mapping, planning, control, machine learning, etc. Each final project group should consist of 1-3 students. As part of the project deliverables, each group will need to submit a project proposal, interim project report, final project report and make a final presentation.

Grading

5% Class participation

30% Individual lab assignments

15% Research paper presentation

50% Final group project:

- 10% Project proposal
- 15% Interim project report
- 25% Final project report and presentation

Grading Scale:

A = 90 – 100 %

B = 80 – 89.9 %

C = 70 – 70.9 %

D = 60 – 69.9 %

F = Below 60 %

Late assignments: Late assignments will lose 10% per day late, up to a maximum of 50% off.

Add/drop policy: See the Mississippi State University Add/Drop Policy

<https://www.uaac.msstate.edu/faq/how-and-when-can-i-adddrop-class/Links to an external site.>

Attendance Policy: This section is a face-to-face instructional class. Students are expected to attend class regularly and to arrive on time. Although class attendance is mandatory, up to **two** absences are allowed during the semester. Sleeping in class, being late, or leaving early without prior notice may result in an unexcused absence. Each student is responsible for all material and administrative instructions given during the lecture period. Each unexcused absence will result in one percentage point deducted from your final grade up to a maximum of 10%. Please refer to [Academic Operating Policy 12.09](#), regarding detailed attendance expectations and accommodations.

Continuity of Instruction: In the event that face-to-face classes are suspended due to extenuating circumstances, such as weather, the instructor will continue instruction in a manner that best supports the course content and student engagement. In this event, all instructors will notify students of the change via their university email address (the official vehicle for communication with students). At that time, they will provide details about how instruction and communication will continue, how academic integrity will be ensured, and what students may

expect during the time that face-to-face classes are suspended. If a student becomes unable to continue class participation due to extenuating circumstances, (e.g., health and safety, loss of power, etc.) the student should contact their instructor and advisor for guidance. For additional guidance, please refer to [Academic Operating Policy 12.09](#).

Student Honor Code

Mississippi State has an approved Honor Code that applies to all students. The code is as follows: “As a Mississippi State University student, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do.” Upon accepting admission to Mississippi State University, a student immediately assumes a commitment to uphold the Honor Code, to accept responsibility for learning, and to follow the philosophy and rules of the Honor Code. Student will be required to state their commitment on examinations, research papers, and other academic work. Ignorance of the rules does not exclude any member of the MSU community from the requirements or the processes of the Honor Code. For additional information, please visit: <http://honorcode.msstate.edu/policyLinks> [to an external site.](#) [Links to an external site.](#)

Academic Misconduct

Academic misconduct (cheating, plagiarism, forgery, etc) of any form can result in serious consequences. Penalties may be as severe as receiving a grade of “F” in this course, suspension, and/or possible expulsion. Faculty members and students are responsible for reporting cases of academic misconduct. See: [http://students.msstate.edu/honorcode/Links to an external site.](http://students.msstate.edu/honorcode/Links to an external site)

Title IX

MSU is committed to complying with Title IX, a federal law that prohibits discrimination, including violence and harassment, based on sex. This means that MSU’s educational programs and activities must be free from sex discrimination, sexual harassment, and other forms of sexual misconduct. If you or someone you know has experienced sex discrimination, sexual violence and/or harassment by any member of the university community, you are encouraged to report the conduct to MSU’s Director of Title IX/EEO Programs at 325-8124 or by e-mail to titleix@msstate.edu. Additional resources are available at [Dean of Students Sexual Misconduct and Sexual Assault](#).

Disability Resource Center

Mississippi State University is committed to providing equitable access to learning opportunities for all students. The Disability Resource Center (01 Montgomery Hall) collaborates with

students who have disabilities to arrange reasonable accommodations. If you have, or think you may have, a disability, please contact drc@saffairs.msstate.edu or 662-325-3335 to arrange a confidential discussion regarding equitable access and reasonable accommodations. Disabilities may include, but are not limited to, conditions related to mental health, chronic health, attention, learning, autism, brain injury, vision, hearing, mobility, speech, or intellectual disabilities. In the case of short-term disabilities (e.g., broken arm), students and instructors can often work to minimize barriers. If additional assistance is needed, please contact the Disability Resource Center.

University Safety Statement. *Mississippi State University values the safety of all campus community members. Students are encouraged to register for Maroon Alert texts and to download the Everbridge App. Visit the Personal Information section in Banner on your mystate portal to register. To report suspicious activity or to request a courtesy escort via Safe Walk, call University Police at 662-325-2121, or in case of emergency, call 911. For more information regarding safety and to view available training resources, including helpful videos, visit <https://ready.msstate.edu/Links to an external site.> .*

Research Accomplishments

Contents

- Highlights
- Robust off-road navigation for mobile robots
- Terrain segmentation for planetary rovers
- Artifact correction and mitigation for image data
- Digital twin mapping and modeling of building environments
- Security and ethical considerations in robotics
- Competitive External Funding

Highlights

- While at MSU, published a total of 22 full-length peer-reviewed publications including 6 journal papers and 16 conference papers. My overall research record includes a total of 42 full-length peer-reviewed conference publications and 22 journal publications, with more than 2000 citations and a *h*-index of 23 and *i10*-index of 35 on Google Scholar.
 - Involved in PI or Co-PI on successful competitive external funding totaling over 3.5 million, including awards from IARPA, NSF, NASA, and USDA.
 - Received a 5-year CAREER award from NSF in 2025.
-

My primary research focus is intelligent robotic systems. The convergence of robotics, automation, and Artificial Intelligence is a key breakthrough of the 21st century and has the potential to radically change many aspects of industries of the future ranging from transportation, manufacturing, construction, and agriculture. My research combines 3D computer vision, digital twins, and machine learning to achieve robust, accurate, real-time capabilities for intelligent, autonomous agents to operate in challenging environments.

Below, I summarize the research areas where my students and I are active, along with a *subset* of related papers that I have published during my time at MSU. More details are provided in my CV.

Robust off-road navigation for mobile robots

In structured environments, robots can navigate simply by following lane markings or pre-defined routes. However, in unstructured environments associated with construction, mining, agriculture, forestry, rural delivery and disaster sites, robots need to have a deeper comprehension of the semantics of a scene in order to understand the movability, affordances, and hazard-level of different objects. Unfortunately, most mobile robot systems today make little explicit use of higher-level semantic meaning in navigation and exploration and typically only consider geometric maps or 3D object models in structured contexts. To address this fundamental research gap, this project investigates a self-supervised, semantically-aware navigation and exploration framework for field robots in unstructured environments. Major research results from this project include (i) a pipeline for digital twin reconstruction of off-road environments, (ii) analysis of LiDAR configuration on algorithm performance, (iii) uncertainty-aware path planning algorithms, and (iv) self-supervised traversability estimation algorithms. Field experiments have also been conducted at the off-road proving ground at Mississippi State University in collaboration with the Center for Advanced Vehicular Systems (CAVS) at MSU.

This research is funded by a 2-year NSF CRII award (IIS-2153101). Besides research outcomes, this project has also resulted in broader impacts through annual workshops, release of open-source code, outreach to high-school students, and mentorship of undergraduate students.

Overall, this research has resulted in 2 conference publications at SPIE 2023 [2, 14] and 2 conference publications at I3CE [16, 15].

Robotic digital twin mapping and modeling of construction environments

Digital twin modeling is a critical component of automation and robotics for progress monitoring, clash detection, and defect detection. The process of converting a physical building environment into a virtual digital twin involves capturing laser scan data of the site and annotating it with relevant semantic information. Conventional methods for creating these models still rely significant manual annotation of data, which is labor-heavy and time-consuming. My research introduces the idea of leveraging machine learning techniques to automate and expedite the process of creating 3D models for modeling building environments. The 3D models can be used for digital twin simulations, floorplan generation, and detection of damaged elements.

Overall, this research has resulted in 1 journal publication [1] and 2 conference publications [9, 11]. Over the next few years, this research will be supported by the NSF CAREER award secured in 2025.

Terrain segmentation for planetary rovers

Autonomous characterization of terrain is a key technology to enable missions such as the Mars Perseverance and the Lunar Endurance rovers to perform extended autonomous navigation and data collection in hazardous planetary terrain with minimal human intervention. Current state-of-the-art AI/ML models employed to estimate terrain traversability directly from image data are subject to significant power and compute constraints. In addition, ML-trained models frequently suffer from inductive bias due to domain shift when using the same model on data obtained from different time periods and different planetary sites. As planetary rovers traverse longer distances, there is a growing need for ML models that can handle locale changes, and generalize across different planetary bodies, such as the Lunar surface and the Martian surface. Thus, the objective of this study is to investigate an incremental self-supervised learning network paired with knowledge distillation to achieve efficient and adaptive learning of terrain characterization models. This study addresses learning capabilities as well as processing power capabilities and limitations of AI/ML models for enhancing the performance of current and projected autonomous systems.

This research is funded by a 1-year NASA award EPSCOR R3 award in collaboration with Edwin Goh and Deegan Atha at NASA's Jet Propulsion Lab (JPL) at Caltech. We have previously collaborated through a JPL Data Science Working Group project titled CLOVER: Contrastive Learning for Onboard Vision-Enabled Robotics. Through this

continued partnership between MSU and JPL, we hope to enable smoother infusion of AI/ML advances into NASA's Mars Sample Return (MSR) mission and future long-range Mars/Lunar missions.

Overall, this research has resulted in 1 journal publication at the Journal of Spacecraft and Rockets [12], 2 conference publications at IEEE AeroConf [7, 8], and 2 conference publications at ECCV [12, 13].

Artifact correction and mitigation for image data

In robotics, the presence of imaging artifacts from weather, shadows, motion blur, and occlusion constitutes a key challenge that needs to be addressed for effective visualization of a site. This is complicated by the fact that artifacts can originate from many sources including natural artifacts such as rain, snow, fog, shadows, sun glint, motion blur, and occlusion as well as sensing artifacts such as bad pixels, image formation artifacts, blur, compression artifacts, vignetting, burn-in, distortion, backscatter, and noise. This research investigates an artifact correction pipeline that performs segmentation, inpainting, and enhancement to detect visual image artifacts and replace affected regions with a nominal background. Through this research, we created (i) an artifact segmentation algorithm to perform pixel-level foreground and background segmentation to isolate foreground objects, artifact-impacted regions, and background imagery, (ii) an inpainting network that uses masks defined over occluded or undesired image regions ("irrecoverable" artifacts) to replace those regions with a suitable background and (iii) an image enhancement network that takes the inpainted image and corrects for "recoverable" imaging artifacts at the global level such as shadows, blurriness, noise.

This research is funded by a 3.5-year award from IARPA in collaboration with BlueHalo, Princeton University, and Stanford University. The research has resulted in one preprint paper [6].

Security and ethical considerations in robotics

Robotics and Artificial Intelligence (AI) have been inextricably intertwined since their inception. Today, AI-Robotics systems have become an integral part of our daily lives, from robotic vacuum cleaners to semi-autonomous cars. However, while the integration of AI in Robotics systems has enhanced the quality of our lives, it has also presented a serious problem - these systems are vulnerable to security attacks. The physical components, algorithms, and data that make up AI-Robotics systems can be exploited by malicious actors, potentially leading to dire consequences. Motivated by the need to address the security concerns in AI-Robotics systems, this research aims to study security and ethical issues in artificial intelligent robot systems such as trustworthiness, privacy, data poisoning, model inversion, and potential legal issues.

This research is funded by a 3-year NSF SaTC award (DGE-2246920). Besides research tasks, this project will also focus on broader impacts through a new curriculum for cybersecurity in robotics, workshops, and other educational activities.

Overall, this research has resulted in 2 journal publications [10, 5] and 3 conference publications [4, 5, 3].

Competitive External Funding

At MSU, I am grateful to have received grants from various institutions including NSF, IARPA, and NASA to support my research. The following shows a list of external funding received (total funding \$3,538,960):

1. NSF CAREER: Constructing Schematic Maps for Construction Robots in Highly Dynamic Environments. PI. \$510,532. June 2025 – May 2030.
2. NSF Innovative Technology Experiences for Students and Teachers (ITEST). Co-PI. \$1,192,951. September 2024 – August 2027.
3. IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA). Scene Understanding, Reconstruction and Rendering with Varying Environment Altitude and Limited Imagery. PI. \$160,499. June 2023 – June 2024.
4. NSF Computer and Information Science and Engineering Research Initiation Initiative (CRII). Semantic Vector Fields for Robot Navigation and Exploration in Unstructured Environments. PI. \$175,000. April 2022 – April 2024.
5. NSF Secure and Trustworthy Cyberspace (SaTC). Inculcate a culture of preparedness against AI security threats to pervasive robotic systems. Co-PI. \$399,978. June 2023 – May 2026.
6. NASA EPSCoR Rapid Response Research (R3). Incremental learning with knowledge distillation for autonomous rover terrain characterization. PI. \$100,000. September 2023 – August 2024.
7. USDA Agricultural Marketing Service (AMS). Advancing Optical Technologies for Enhanced Quality Evaluation, Grading and Sorting of Sweet Potatoes. Co-PI. \$1,000,000. March 2022 – March 2025.

References

- [1] Jingdao Chen and Yong Kwon Cho. Crackembed: Point feature embedding for crack segmentation from disaster site point clouds with anomaly detection. *Advanced Engineering Informatics*, 52:101550, 2022.
- [2] Jingdao Chen, Mikias Gugssa, Justin Yee, Jun Wang, Christopher Goodin, and Athish Ram Das. Framework for digital twin creation in off-road environments from LiDAR scans. In Christopher L. Howell, Kimberly E. Manser, and Raghuvier M. Rao, editors, *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications*, volume 12529, page 125290F. International Society for Optics and Photonics, SPIE, 2023.

- [3] Ivan A Fernandez, Subash Neupane, Trisha Chakraborty, Shaswata Mitra, Sudip Mittal, Nisha Pillai, Jingdao Chen, and Shahram Rahimi. A survey on privacy attacks against digital twin systems in ai-robotics. In *2024 IEEE 10th International Conference on Collaboration and Internet Computing (CIC)*, pages 70–79. IEEE, 2024.
- [4] Kevin Di Gao, Andrew Haverly, Sudip Mittal, and Jingdao Chen. A bibliometric view of ai ethics development. In *2022 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*, 2023.
- [5] Kevin Di Gao, Andrew Haverly, Sudip Mittal, Jiming Wu, and Jingdao Chen. Ai ethics: A bibliometric analysis, critical issues, and key gaps. *International Journal of Business Analytics*, 11(1):19, 2024.
- [6] Sushant Gautam and Jingdao Chen. X-decode: Extreme deblurring with curriculum optimization and domain equalization, 2025.
- [7] Edwin Goh, Jingdao Chen, and Brian Wilson. Mars terrain segmentation with less labels. In *2022 IEEE Aerospace Conference (AERO)*, pages 1–10, 2022.
- [8] Edwin Goh, Isaac R. Ward, Grace Vincent, Kai Pak, Jingdao Chen, and Brian Wilson. Self-supervised distillation for computer vision onboard planetary robots. In *2023 IEEE Aerospace Conference*, pages 1–11, 2023.
- [9] Seongyong Kim, Yosuke Yajima, Jisoo Park, Jingdao Chen, and Yong Cho. A hybrid semantic-geometric approach for clutter-resistant floorplan generation from building point clouds. In *2022 International Conference on Construction Engineering and Project Management (ICCEPM)*, pages 792–799, 2022.
- [10] Subash Neupane, Shaswata Mitra, Ivan A. Fernandez, Swayamjit Saha, Sudip Mittal, Jingdao Chen, Nisha Pillai, and Shahram Rahimi. Security considerations in ai-robotics: A survey of current methods, challenges, and opportunities. *IEEE Access*, 12:22072–22097, 2024.
- [11] Jeffrey Rugg, Jingdao Chen, Mikias Gugssa, and Jun Wang. Object-level temporal change detection on construction sites with 3d deep learning models. In *International Conference on Computing in Civil Engineering 2024*, 2024.
- [12] Grace M. Vincent, Isaac R. Ward, Charles Moore, Jingdao Chen, Kai Pak, Alice Yepremyan, Brian Wilson, and Edwin Y. Goh. Clover: Contrastive learning for on-board vision-enabled robotics. *Journal of Spacecraft and Rockets*, 0(0):1–13, 2023.
- [13] Isaac Ronald Ward, Charles Moore, Kai Pak, Jingdao Chen, and Edwin Goh. Improving contrastive learning on visually homogeneous mars rover images. In Leonid Karlinsky, Tomer Michaeli, and Ko Nishino, editors, *Computer Vision – ECCV 2022 Workshops*, pages 170–185, Cham, 2023. Springer Nature Switzerland.

- [14] Jinhee Yu, Jingdao Chen, Lalitha Dabbiru, and Christopher T. Goodin. Analysis of LiDAR configurations on off-road semantic segmentation performance. In Michael C. Dudzik, Stephen M. Jameson, and Theresa J. Axenson, editors, *Autonomous Systems: Sensors, Processing, and Security for Ground, Air, Sea, and Space Vehicles and Infrastructure 2023*, volume 12540, page 1254003. International Society for Optics and Photonics, SPIE, 2023.
- [15] Jinhee Yu, Jayakumar Monika, Yilong Chen, Yong Kwon Cho, and Jingdao Chen. Self-supervised learning with lidar-camera fusion for construction site traversability estimation. In *Proceedings of the ASCE 2025 International Conference on Computing in Civil Engineering (i3CE)*, 2025.
- [16] Jinhee Yu, Swayamjit Saha, Monika Jayakumar, Mikias Gugssa, Jingdao Chen, and Jun Wang. Lidar-based traversability estimation for ground robots on construction sites using self-supervised learning. In *Proceedings of the ASCE 2024 International Conference on Computing in Civil Engineering (i3CE)*, 2024.

Service Accomplishments

Contents

- Highlights
- NSF Panel Service
- Workshop Organization
- Broader External Service
- Student Advising
- Departmental Service

Highlights

- Served as reviewer for multiple well-known conferences and journals in robotics including T-RO, RAL, ICRA, IROS, and CASE.
- Served as the lead organizer for annual construction robotics workshop for four years at ICRA'22, ICRA'23, ICRA'24, ICRA'25, the top international conference in robotics.
- Served as Associate Editor for the Journal of Computing in Civil Engineering based on my research in construction robotics.

The following describes my service accomplishments during my time at MSU, both in the broader academic community and within my own department. In addition to the highlights above, I summarize some of this service; more details are provided in my CV.

NSF Panel Service. I served on four NSF review panels in the years 2021, 2023, 2024, and 2025.

Workshop Organization: I co-organized three workshops in total at ICRA as follows:

- Workshop on Future of Construction: Build Faster, Better, Safer - Together with Robots at ICRA'22, Philadelphia, USA, May 23, 2022, <https://construction-robots.github.io/index2022.html>.
- 2nd Workshop on Future of Construction: Robot Perception, Mapping, Navigation, Control in Unstructured and Cluttered Environments at ICRA'23, London, UK, June 2, 2023, <https://construction-robots.github.io/index2023.html>.
- 3rd Workshop on Future of Construction: Lifelong Learning Robots in Changing Construction Sites at ICRA'24, Yokohama, Japan, May 13, 2024, <https://construction-robots.github.io/index2024.html>.
- 4th Workshop on Future of Construction: Safe, Reliable, and Precise Robots in Construction Environments at ICRA'25, Atlanta, USA, May 19, 2025, <https://construction-robots.github.io/index.html>.

Each of these workshops was a one-day event that consisted of invited speakers, poster sessions, sponsored competitions, and panel discussions. The workshops are collaboratively organized together with Georgia Tech, New York University, North Dakota State University, Oxford University, Baidu Research, and HILTI, and is key towards establishing research partnerships with those institutions. Over 4 years, the workshops have hosted 28 invited speakers from both academia and industry. These workshops are usually attended

by 50-100 in person participants and 20-50 virtual participants. Each year, we receive 20-30 paper and poster submissions from researchers worldwide. The workshops have also been successful in receiving competition sponsorship from HILTI and publication sponsorship from IAARC. For example, in 2022, we awarded \$13,000 in prizes for the SLAM challenge and \$1,000 for best paper awards.

Broader External Service: My primary research focus is in intelligent robotic systems, though I have also contributed to related research communities such as construction robotics, disaster relief robotics, and space robotics. I have served as technical reviewer for journals including T-RO, RAL, Automation in Construction, Journal of Computing in Civil Engineering, Advanced Engineering Informatics, as well as conferences such as ICRA, IROS, CASE, and ISARC.

I serve as an Associate Editor (February 2024 - present) for the Journal of Computing in Civil Engineering (ASCE, ISSN: 0887-3801 print version, 1943-5487 electronic version, Scopus Impact Factor: 6.315). I am typically assigned several papers per month, and it is my responsibility to assign reviewers to these papers, synthesize their reports, and correspond with the Editor-in-Chief prior to a final decision.

I have also served as Session Chair for the International Conference on Construction Engineering and Project Management (ICCEPM) in 2022 and the International Conference on Computing in Civil Engineering (i3CE) in 2021 and 2024.

Student Advising: In terms of student advising, I have served in the following capacity:

- Major professor for 16 PhD students.
- Major professor for 15 MS students.
- Committee Member for 18 PhD students and 20 MS students.
- Undergraduate research advisor for 20 students, including 3 Honor's theses.

Departmental Service: Within the Department of Computer Science and Engineering, I have served on the Courses and Curricula Committee as well as the Graduate Studies Committee. I have also served as faculty advisor for the AI Club and the State-Space Robotics Team. Our robotics lab has consistently participated in several annual outreach events including Scholars Recognition Day, Engineering Discovery Day, Academic Insight, and MSU Research Showcase.

Letters of Recommendation: External Reviewers

Example Letter to Reviewers

Response/Rebuttal to Recommendation

Selected Publications

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- Overview
- Selected Publication 1
CrackEmbed: Point Feature Embedding for Crack Segmentation from Disaster Site Point Clouds with Anomaly Detection
- Selected Publication 2
Improving Contrastive Learning on Visually Homogeneous Mars Rover Images
- Selected Publication 3
Security Considerations in AI-Robotics: A Survey of Current Methods, Challenges, and Opportunities

Overview

The following 3 publications are examples of my work in disaster relief robotics, cybersecurity for robotics, and space robotics, and these are appended to my dossier. Both **conference papers and journals** are peer-reviewed and prestigious in the computer science and robotics communities.

1. Chen, J., and Cho, Y. (2022). CrackEmbed: Point Feature Embedding for Crack Segmentation from Disaster Site Point Clouds with Anomaly Detection. Advanced Engineering Informatics, Volume 52, April 2022

Note: This work is an important culmination of my research on point cloud representations in disaster relief robotics and is an extension of my PhD dissertation work.

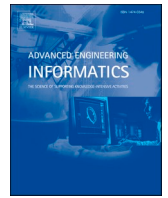
2. Ward, I. R., Moore, C., Pak, K., Chen, J., and Goh, E. (2022). Improving Contrastive Learning on Visually Homogeneous Mars Rover Images. European Conference on Computer Vision (ECCV) Workshop on AI4Space, Tel Aviv, Israel, Oct 23

Note: This work is one of two papers on space robotics we submitted in collaboration with NASA JPL to the AI4Space workshop at ECCV, the second most-highly cited conference in the area of computer vision and recognition. The paper is co-authored by Charles, Moore, my first graduate student advisee.

3. Neupane, S., Mitra, S., Fernandez, I., Saha, S., Mittal, S., Chen, J., Pillai, N., Rahimi, S. (2024). Security Considerations in AI-Robotics: A Survey of Current Methods, Challenges, and Opportunities. IEEE Access, vol. 12, pp. 22072-22097, 2024

Note: This work represents the start of an interdisciplinary collaboration with Dr. Sudip Mittal's lab at MSU to explore AI security considerations in robotic systems.

Selected Publication 1



CrackEmbed: Point feature embedding for crack segmentation from disaster site point clouds with anomaly detection

Jingdao Chen^a, Yong Kwon Cho^{b,*}

^a Department of Computer Science and Engineering, Mississippi State University, 665 George Perry St, Mississippi State, MS 39762, USA

^b School of Civil and Environmental Engineering, Georgia Institute of Technology, 790 Atlanta Dr. N.W., Atlanta, GA 30332-0355, USA

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ABSTRACT

Laser-scanned point clouds can be used to represent the 3D as-damaged condition of building structures in a post-disaster scenario. Performing crack detection from the acquired point clouds is a critical component of disaster relief tasks such as structural damage assessment and risk assessment. Crack detection methods based on intensity or normals commonly result in noisy detections. On the other hand, deep learning methods can achieve higher accuracy but require a large dataset of annotated cracks. This research proposes an unsupervised learning framework based on anomaly detection to segment out cracked regions from disaster site point clouds. First, building components of interest are extracted from the point cloud scene using region growing segmentation. Next, a point-based deep neural network is used to extract discriminative point features using the geometry of the local point neighborhood. The neural network embedding, CrackEmbed, is trained using the triplet loss function on the S3DIS dataset. Then, an anomaly detection algorithm is used to separate out the points belonging to cracked regions based on the distribution of these point features. The proposed method was evaluated on laser-scanned point clouds from the 2015 Nepal earthquake as well as a disaster response training facility in the U.S. Evaluation results based on the point-level precision and recall metrics showed that CrackEmbed in conjunction with the isolation forest algorithm resulted in the best performance overall.

1. Introduction

Natural disasters such as earthquakes and windstorm events result in the loss of thousands of lives, millions of lives affected, and billions of dollars of economic loss annually [1]. In the aftermath of these disasters, a large-scale assessment of damaged building structures is necessary to identify safe locations for emergency shelters, determine risk levels for search and rescue, and detect building structures that need to be repaired. Rapid and timely damage assessment is critical to accelerating the rehabilitation process for urban areas affected by natural disasters [2]. However, a disaster site poses an extremely challenging environment for accurate damage assessment due to the presence of clutter and debris and the irregular position and orientation of building structures.

Conventional methods for building damage assessment rely on manual inspection of damage indicators such as liquefaction, tilting, or cracks [3,4]. The problem with manual inspection is that it is time-consuming, labor-intensive, subjective, and risky. Sensor-based damage assessment methods such as vibration testing [5], acoustic emission testing [6], and guided wave testing [7] can usually provide more

objective measurements of the structural damage. However, the limitation of these sensor-based methods is that they require pre-installation of sensors and physical contact with the damaged structure, which is impractical in high-risk or inaccessible regions of a disaster site. On the other hand, damage assessment can also be performed by observing the surface of damaged structures using a remote sensing approach by acquiring image [8] and point cloud data [9]. Image and point cloud data can efficiently capture the surface condition without physical contact, but the raw data is difficult to parse due to the high dimensionality. Although various learning-based architectures and datasets have been studied for this purpose, classifying damaged and deformed structures is still challenging due to the low availability and possible non-existence of ground truth datasets.

This research focuses on the problem of damage assessment in the point cloud domain since point clouds can directly represent 3D geometry and can also represent additional quantities such as color and reflectivity. Point clouds are easy to visualize and can be used to estimate multiple damage indicators such as tilt, drift, cracks in physical units (i.e., meters). To tackle the problem of automated damage

* Corresponding author.

E-mail addresses: chenjingdao@cse.msstate.edu (J. Chen), yong.cho@ce.gatech.edu (Y.K. Cho).

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assessment from disaster site point clouds, this research proposes a method for crack segmentation using unsupervised machine learning techniques. In particular, the proposed deep neural network, Crack-Embed is trained on the S3DIS dataset [10] and used to compute discriminative point feature embeddings. Moreover, an unsupervised learning technique based on anomaly detection is used to overcome the data deficiency problem for crack detection due to a lack of annotated data for cracks in point clouds. The proposed method is evaluated based on laser-scanned point clouds from (i) the 2015 Nepal earthquake and (ii) a disaster preparedness and tactical training center in Georgia, U.S. A. The Nepal dataset is taken from a publicly-available repository [11] whereas the disaster center dataset is collected by the authors. In this study, training is undertaken on the S3DIS dataset only, and the Nepal dataset and the disaster center dataset are used for inference only, with no fine-tuning carried out in between. The performance of the proposed point feature embedding is compared against that of other methods such as color, intensity, normal vector, curvature, Fast Point Feature Histogram (FPFH), and PointNet++. In addition, the performance of the anomaly detection algorithm is compared against that of other algorithms for segmenting the damaged regions such as K-means clustering and mean-shift clustering.

2. Literature review

2.1. Damage detection using mechanical sensors

A common method for structural health monitoring of civil structures is the use of vibration sensors [5,12]. Vibration sensors can be used to monitor for frequency shifts and mode shape changes that are linked to changes in structural properties. However, this method depends on the pre-installation of a large number of sensors, which is expensive and only practical for critical structures such as bridges. Acoustic emission is another commonly used technique for damage evaluation of concrete structures and steel structures [6,13]. Acoustic emission testing relies on sensors that can listen for acoustic events and measure the component strength and risk of failure. However, acoustic emission testing usually requires physical contact with the damaged structure, which is impractical in high-risk or inaccessible regions of a disaster site. Other methods such as ultrasonic testing [14] and guided wave testing [7] make use of actively emitted pulses instead of passively measuring acoustic waves. Ultrasonic testing is commonly used to detect internal defects but only works on an area in close proximity to the transducer. On the other hand, guided wave testing can be used to inspect large areas of a structure at once but cannot localize the damaged region accurately.

2.2. Damage detection using 2D images

The main benefit of using images for damage detection instead of mechanical sensors is that they can be acquired from longer ranges and are lightweight such that they can be equipped on Unmanned Aerial Vehicles (UAV). There are several works that use deep learning techniques for crack detection from images. Zou et al. [15] used a deep convolutional neural network (CNN) for automatic crack detection by learning high-level features for crack representation. The idea of using CNNs can be extended to use guided filtering and Conditional Random Fields (CRFs) to refine the final prediction results [8]. Another possibility is to pre-process the images using homomorphic filtering and the Otsu thresholding method before passing them to the network [16]. Benz et al. [17] studied the problem of crack segmentation from images acquired by unmanned aircraft systems (UAS). The study uses transfer-learning from an auxiliary dataset to initialize the weights of a neural network for crack segmentation. However, the network still requires fine-tuning in a supervised manner from an actual labeled dataset of cracks. In the industrial domain, Liu et al. [18] used an encoder-decoder network in conjunction with background suppression modules to detect

cracks from images of plastic materials. Whereas, Tabernik et al. [19] studied surface crack detection from images of industrial products. Since the work is targeted for quality control applications, it focused more on the decision problem of identifying images with cracks rather than the detection problem of accurately segmenting the cracks. One common limitation is that these methods require strong supervision in the form of a database of images with labeled cracks. In addition, these methods are only applicable in the 2D domain and do not predict the 3D geometric information that is necessary to determine the crack dimensions.

A subset of works in the literature focus on the highly related problem of crack detection from asphalt pavements. Early studies use the intensity as an indicator feature for cracks [20,21]. The problem with these methods is that the performance is negatively by shadows, lighting effects, non-uniform crack widths, and poor intensity contrast [21]. CrackNet [22] was one of the first large-scale studies into pavement crack detection using deep learning techniques. The study uses a convolutional neural network (CNN) to detect cracks in a dataset of 2000 pavement depth images generated by a laser scanning system. When compared to non-deep learning methods such as 3D shadow modeling and Support Vector Machines (SVM), CrackNet achieved much more accurate crack detection. CrackNet II [23] was introduced as an improvement over the original CrackNet with a deeper network architecture that has fully learnable parameters. This resulted in higher precision and recall as well as processing speed. CrackNet-V [24] further improved the network design with deeper convolutional layers and a more specific activation function, and the performance was also validated using a larger dataset. To overcome the data deficiency problem for crack images, Li et al. [25] proposed semi-supervised learning using a combination of labeled and unlabeled pavement images. In addition, an adversarial learning method was used to improve the segmentation predictions. Several studies [26,27] explore crack detection using fully unsupervised algorithms such as Otsu's thresholding and Random Sample Consensus (RANSAC). However, these algorithms require manual tuning of the threshold hyperparameters to achieve good performance.

2.3. Damage detection using 3D point clouds

A simple method to perform damage detection is to measure the deviation between the post-damage point cloud and the pre-damage point cloud that can be generated from previous scans or from design models [2,28]. Compared to the image domain, this method is more practical in the point cloud domain since point clouds can capture 3D geometry on an absolute scale. The problem with this method is that it requires the pre-damage point cloud, which may not be available for all buildings on a disaster site. A geometry-based method to detect cracks and spalling in point cloud data is to estimate the surface normal and measure the deviation by comparing it to a nominal value [29] or finding outliers after combining the surface normal and color information [9]. Other alternative methods for damage detection include measuring the perpendicular distance from a pre-defined reference plane [30]. However, the limitation of these geometry-based methods is that they require significant parameter tuning and carefully constructed scoring algorithms.

Similar to the image domain, some researchers turned to machine learning techniques for damage detection in the point cloud domain. The problem is that large databases with annotated damages exist in the image domain (e.g., CrackTree260 [15], DeepCrack [8], and UAV 75 [17]) but are difficult to find in the point cloud domain. One possible solution is to synthetically generate deformations in the point cloud data derived from a database of building element Computer-Aided Design (CAD) models [31]. However, this method relies on having the building elements cleanly pre-segmented from a point cloud scene which is difficult to achieve on real disaster sites due to a high amount of occlusion and clutter. Another related method to synthesize deformations such as spalling directly on the point cloud is to apply cubic surface

equations together with added noise [32]. Then, geometry-based classifiers can be trained to segment out damaged regions without reliance on color or intensity information. However, this method still results in a large number of false positives due to surface nonuniformities and sharp features such as edges. In the case of larger-scale damage such as that caused by hurricanes, an airborne LiDAR scanning system can be used to gather point clouds of the disaster site [33]. Building damage indicators can be derived from geometric features such as including roof area and volume, roof orientation, and roof shape. This method works well for macro-level building damage such as destroyed roofs or collapsed structures but does not work for micro-level damage indicators such as cracks.

In summary, there is a research need for further study of 3D segmentation of cracks on building structures as a result of natural disasters such as earthquakes. This is exacerbated by the lack of annotated point cloud datasets for supervised learning. Thus, this research proposes an unsupervised learning approach based on anomaly detection for crack segmentation from point clouds.

3. Research objectives

The objectives of this research are defined as follows: (i) investigate an unsupervised learning method for 3D crack segmentation in the context of disaster relief (ii) examine the ability of deep feature embeddings to represent discriminative information about point cloud data (iii) analyze the effect of different experimental parameters such as crack width, outlier ratio, and point cloud resolution on the algorithm performance.

4. Methodology

The proposed framework for crack segmentation from point cloud data consists of three steps: (i) building component extraction, (ii) point feature embedding computation, and (iii) anomaly detection. Each step is presented in detail in the following subsections:

4.1. Building component extraction

The first step in detecting damaged building elements is to isolate specific building components of interest. This study uses a semi-automated framework for building component extraction from a raw point scene (shown in Fig. 1). The point cloud scene in Fig. 1 is taken from a damaged building in the 2015 Nepal earthquake. First, a region

growing segmentation algorithm is applied to group neighboring points together into clusters such that each cluster contains points from a unique building component. The region growing algorithm works by considering the normal vector and color data from each point and joining them if they have similar values and are within a neighborhood of 0.1 m. Fig. 2 shows a visualization of the region growing segmentation results for two different scenes, where each point cloud cluster is shown in a different color. Fig. 2a is of a damaged building from the Nepal dataset whereas Fig. 2b is of a collapsed concrete structure from a disaster response training facility. Next, the building components of interests are extracted by manually indicating the dimensions for a specific type of building component. For example, the concrete columns from the scene in Fig. 2a can be extracted by filtering for point cloud segments that match dimensions of $0.4 \text{ m} \times 0.4 \text{ m} \times 2.6 \text{ m}$.

4.2. Point feature embedding computation

Once the building components of interest are extracted, point features are computed for each point in that building component. In any machine learning framework, features are important to describe the characteristics of a data sample and are used as a basis for differentiation between samples from different classes. In this study, point features are used to differentiate between cracked regions and undamaged regions in point cloud data. Simple examples for point features are measurements that can be obtained directly from the data acquisition device such as color, intensity and XYZ coordinates. Other examples for point features are features derived from the local geometry such as the normal vector, which describes the orientation of the local surface, and the curvature, which describes whether the local surface is flat or curved.

On the other hand, this study uses the concept of a feature embedding to represent discriminative information about a point. A feature embedding is a transformation of the original space of raw data into a new feature space that is more semantically meaningful. The benefit of having a point feature embedding compared to the raw data is that the embedding is able to capture high-dimensional information in the local geometry and summarize that information succinctly in a feature vector. There exist feature descriptors such as Viewpoint Feature Histogram (VFH) [34], Fast Point Feature Histogram (FPFH) [35], and Spin Images [36] that can compute geometrical features for each point. However, these feature descriptors are manually designed and not data-driven. There are also deep neural nets for point clouds such as PointNet [37], PointNet++ [38], and Recurrent Consolidation Units [39] that can predict feature embeddings, but these embeddings are tuned more for

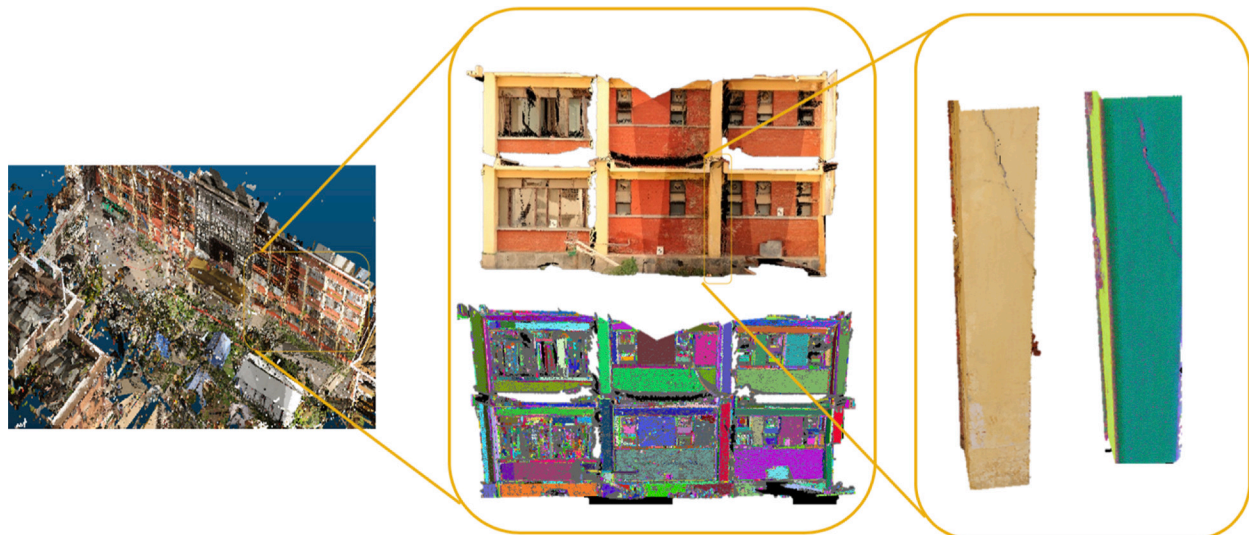


Fig. 1. Visualization of overall framework for building component extraction (e.g., columns).

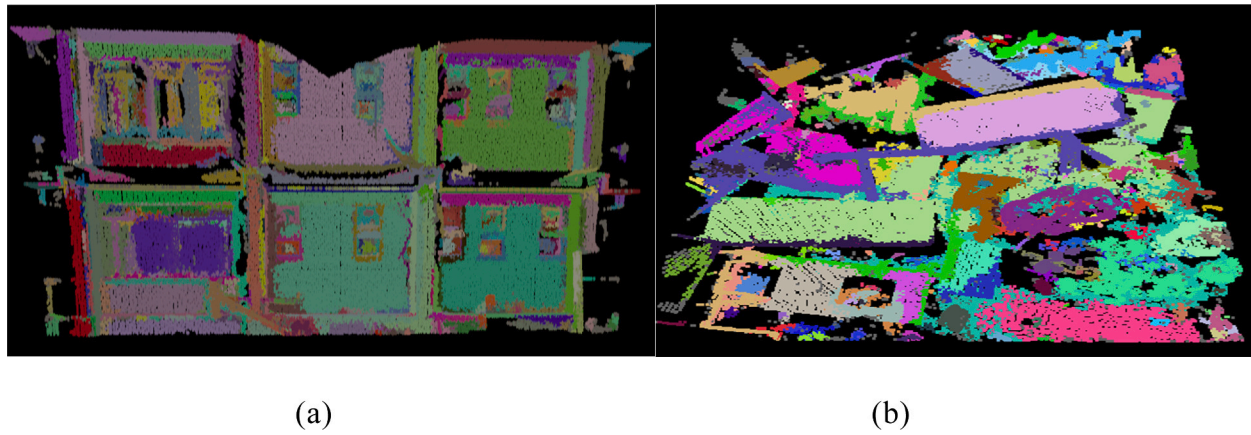


Fig. 2. Region growing segmentation results for (a) concrete column scene and (b) collapsed concrete debris scene.

segmentation of furniture or individual building elements instead of segmentation of surface features such as cracks. Later research introduced architectures such as SqueezeSeg [40], SqueezeSegV2 [41], KPConv [42], Cylinder 3D [43], and AF2S3Net [44] that have improved performance compared to PointNet on public benchmarks such as SemanticKITTI [45]. It should be noted that, however, these methods are mostly applied for sparse point clouds used in autonomous driving and not dense point clouds used for building crack detection.

This research proposes a point-based deep neural network, CrackEmbed, that is used to extract discriminative point features using the geometry of the local point neighborhood. Since annotated data from disaster sites are difficult to obtain, this research uses a network that is trained on regular building scenes and applied to disaster site scenes without the need for further fine-tuning. In this study, the network is trained using the Stanford 3D Indoor Spaces (S3DIS) dataset [46], which contains common building elements such as walls, floors, columns, and beams. The S3DIS dataset contains point clouds from 6 building areas and around 6000 objects acquired using a Matterport scanner. The

S3DIS dataset is annotated with *class labels*, which determine whether a point comes from walls, floors, tables, or chairs, and *instance labels*, which determine which points belong to the same object instances. The problem with *class labels* is that they only involve objects that are found in the environment of the original dataset (i.e. building elements and furniture for the S3DIS dataset) but are not relevant for the target task of crack segmentation. The research hypothesis of this study is that training on *instance labels* is more transferable compared to training on *class labels* for the task of crack segmentation. Thus, CrackEmbed training on the S3DIS dataset is carried out only using the *instance labels* and not the *class labels*. The network is trained to distinguish different types of points without being conditioned to specific classes of objects so that it can generalize to new scenes.

Fig. 3 shows the neural network architecture used to compute the proposed CrackEmbed point feature embeddings in this study. The input to the network is a point cloud with N points represented by a $N \times 6$ matrix, where each row is a point with 6 features, which are the XYZ coordinates and RGB color. The network uses a context pooling module

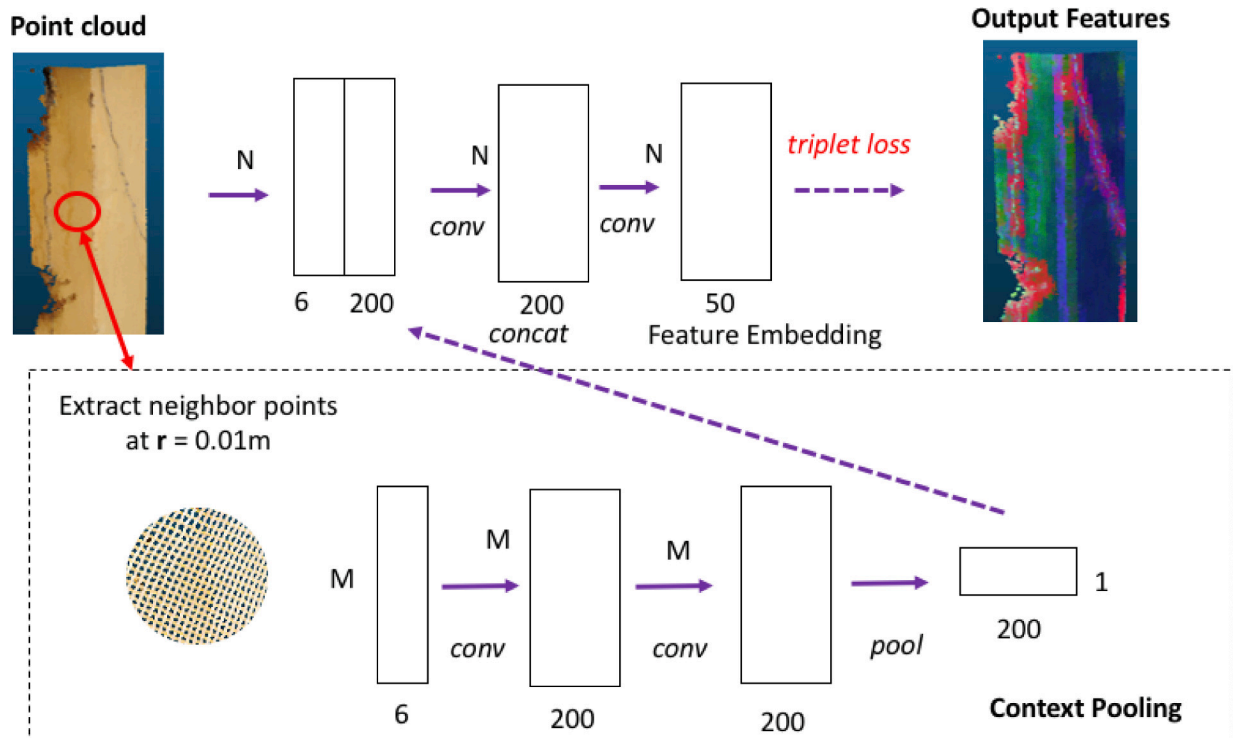


Fig. 3. Network architecture to compute CrackEmbed point feature embeddings based on the triplet loss function.

to extract geometrical information from neighboring points. In particular, for each of the N input points, a set of M neighboring points are sampled at a radius of 0.01 m to be used as context information. In this study, N and M are set to 256 and 50, respectively, after hyperparameter tuning. The input data from the neighboring points are processed using a series of convolutional layers and finally pooled into a 200-dimensional feature vector. This contextual feature vector is concatenated back to the 6-dimensional vector of raw data for each input point. Next, the concatenated features are processed with two more convolutional layers to form a 50-dimensional feature embedding, which means that each input point is represented with a set of 50 features. This feature embedding is trained using the triplet loss function [47], which aims to optimize the weights of the network such that points that are semantically similar are close together in feature space, whereas points that are semantically dissimilar are further apart in feature space. During the training process, triplets of points are sampled from the training dataset with each triplet having two points with the same instance label and one point with a different instance label. The optimizer then updates the network weights so that the feature embeddings of the similarly labeled point pair are closer together compared to the feature embeddings of the differently labeled point pair. This process works under the assumption that points that originate from the same object instance have similar geometry and should have similar feature embeddings, whereas points that originate from different object instances should have dissimilar feature embeddings. Since this triplet loss embedding is trained to separate out points with different types of geometry, it can be transferred to the crack segmentation task because cracked points are likely to have different geometry compared to the surrounding points.

4.3. Anomaly detection

After the point features are computed, an unsupervised learning algorithm is applied to separate out damaged points from non-damaged

points based on their distribution in feature space. This process works based on the intuition that non-damaged points have features that appear at a high frequency, whereas damaged points that features that deviate from normal and appear at a lower frequency. Thus, an unsupervised learning algorithm that can analyze the feature space and identify anomalies will be able to function as a damage detection algorithm. Using unsupervised learning algorithms is advantageous compared to supervised learning algorithms in the context of disaster site data because they do not require training data, which is difficult to acquire and annotate. Unsupervised learning is also a more robust solution compared to applying a fixed threshold on the values of the features since it can adapt to the feature distribution.

This study uses several clustering methods as baseline algorithms for crack segmentation since clustering is a commonly used unsupervised learning algorithm. One such algorithm is K-means clustering [48], which divides the data into K clusters by iteratively computing the cluster centers and assigning data points to the closest cluster. Fig. 4 shows a visualization of K-means clustering used to segment out the cracked region. To show the distribution of points in feature space, Principal Component Analysis (PCA) is used to reduce the feature dimensionality to two dimensions so that it can be plotted. As shown in Fig. 4, the points are color-coded into 5 different colors with corresponds to the $K = 5$ clusters determined by K-means clustering. From these clustering results, the cluster with the smallest number of points is assumed to be the outlier, and the points in that cluster are returned as the cracked region. Another clustering method is Gaussian Mixture Models [49], which considers both the mean and covariance of each cluster and also uses soft assignment of data points to clusters. With Gaussian Mixture Models, the cluster with the smallest number of points is also assumed to be the outlier. A third clustering method considered in this study is mean-shift clustering [50], which aims to discover a set of centroids and updates them so that they represent the mean of their respective regions. With mean-shift clustering, outliers are defined as points that do not fall within the estimated bandwidth of any cluster

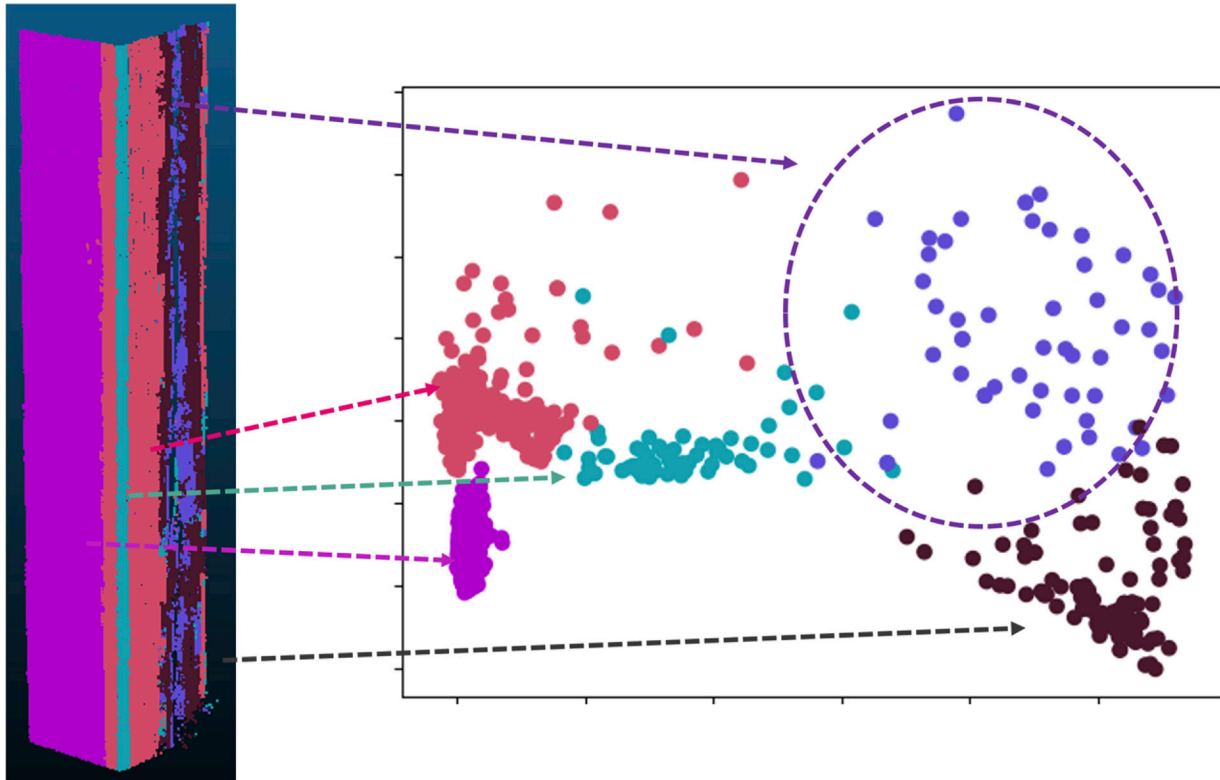


Fig. 4. Visualization of K-means clustering used to segment the points belonging to the cracked region in the feature space.

center.

In order to specifically segment out anomaly points, this study uses anomaly detection methods, which are also known as novelty detection or outlier detection methods. The anomaly detection process assumes that uncommon damaged points (e.g. from cracks) can be separated out as outliers from common non-damaged points based on their distribution in feature space. Situations where the outlier ratio is high such as partially collapsed structures or structures with a deformed surface are more suitable to be processed by deformation detection or debris detection algorithms and not crack detection, which is the focus of this study. This study considers 4 different anomaly detection algorithms which are (i) isolation forest [51], (ii) one-class SVM [52], (iii) robust covariance [53], and (iv) local outlier factor [54]. Isolation forest constructs a decision tree structure over the data points and uses the number of splits required to isolate a point as a measure of normality. On the other hand, one-class SVM works by constructing a frontier over the regular data points and defines anomalies as points that lie outside the frontier. Robust covariance assumes that the regular data points are Gaussian distributed and solves for the covariance of these data points so that the normality can be measured based on the distance to the central mode. Finally, the local outlier factor measures the normality of each data point by comparing the local density to the average density of its nearest neighbors. Fig. 5 shows a visualization of the isolation forest algorithm used to segment out the crack region. Each point is color-coded by its normality score, where red indicates points that are likely to be common points whereas blue indicates points that are likely to be anomaly points. As shown in the figure, the points that have low normality score (i.e., anomaly points) are correlated with points in the cracked region of the original point cloud.

5. Results

To thoroughly evaluate the effectiveness of the proposed crack segmentation method, this research makes use of two point cloud datasets

acquired using terrestrial laser scanners (TLS): (i) a maternity hospital that was damaged during the 2015 Nepal earthquake, and (ii) a damaged concrete structure at a disaster training facility in Georgia, U.S. A. The Nepal dataset (Fig. 6a) contains 7 cracked concrete columns each measuring about $0.4 \text{ m} \times 0.4 \text{ m} \times 2.6 \text{ m}$ whereas the disaster centers dataset (Fig. 6b) contains 2 cracked concrete slabs, and each is about $5 \text{ m} \times 3 \text{ m} \times 0.5 \text{ m}$.

In this study, ground truth labels for the Nepal dataset and the disaster center dataset were manually annotated by the authors so that the accuracy of crack segmentation can be quantitatively evaluated. Note that these ground truth labels are only used for evaluation and not for training the crack segmentation algorithms. To create ground truth annotations of the cracked regions for each building structure, the authors used the CloudCompare software [55] to manually label the point cloud data with point-level segmentation masks to indicate the cracked regions. Due to the time-consuming nature of accurately drawing the segmentation outline in point clouds, only a limited number of building elements could be annotated. The lack of publicly available annotated point clouds of disaster sites and the time-consuming nature of creating those annotations further highlights the need for unsupervised learning methods such as that proposed in this research.

The crack segmentation performance is measured using the point-wise precision and recall metrics, which are determined by how well the predicted segmentation mask matches the ground truth segmentation mask. In addition, the F1-score is used to represent the combined information from both precision and recall metrics (since the F1-score is the harmonic mean of precision and recall). The mean Intersection over Union (mIOU) score is also used to evaluate the crack segmentation performance by taking the ratio of the area of overlap to the area of union. Finally, the length and width of the crack are estimated by taking the extent of all the points within the predicted segmentation mask. Then, the length error and width error are used to measure the absolute error in the predicted dimensions of the overlapped segmentation mask. Note that the length and width errors are measured at the longest and

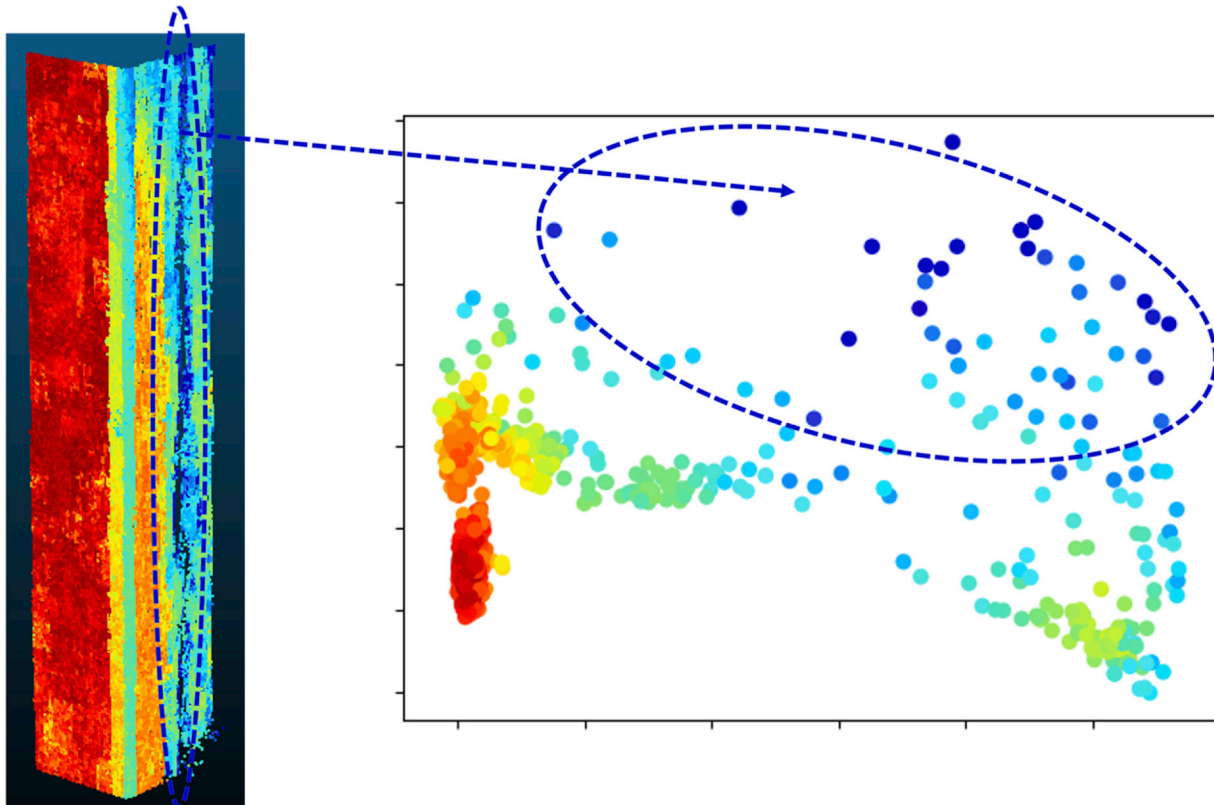
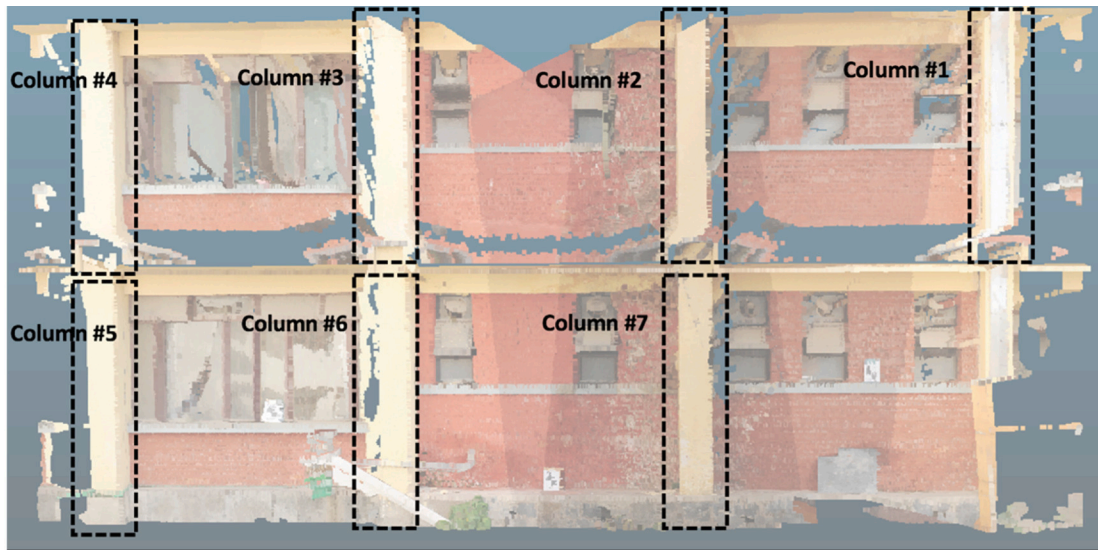
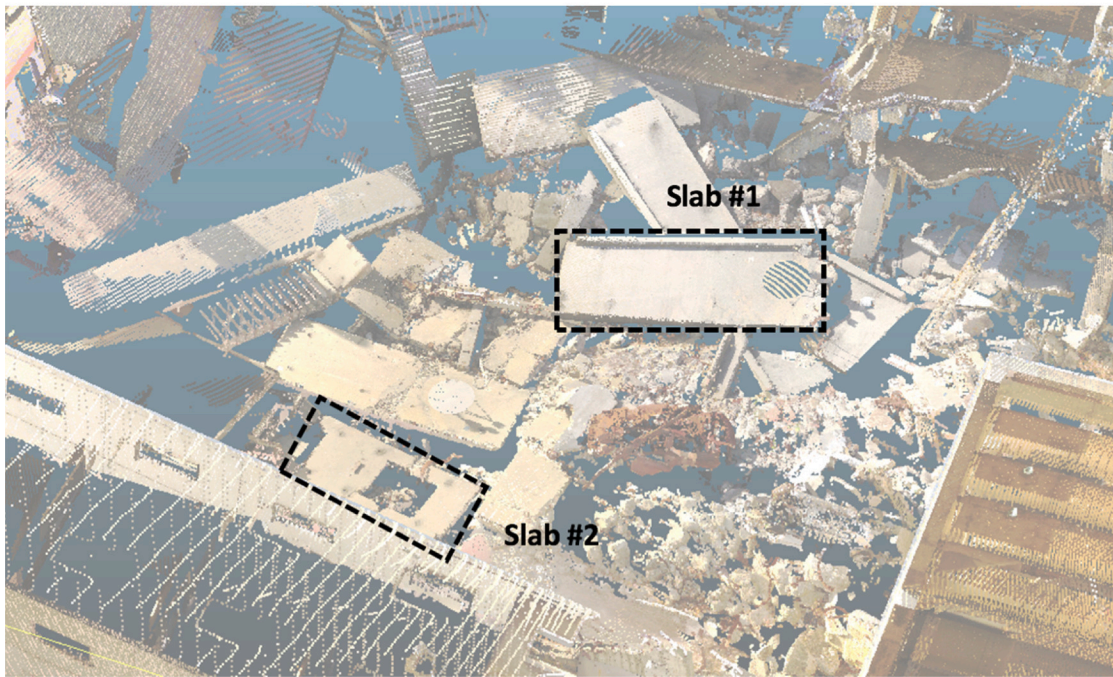


Fig. 5. Visualization of Isolation Forest used to segment the points belonging to the cracked region in the feature space.



(a)



(b)

Fig. 6. Different datasets for damaged structural components: (a) Nepal dataset containing damaged columns and (b) disaster center dataset containing damaged slabs.

widest part of the crack respectively, so the length and width errors could be zero even though the predicted segmentation mask is not perfect. Each evaluation metric is computed using the average over all test samples in a dataset.

Tables 1 and 2 show the performance comparison of different segmentation methods on the Nepal dataset and the disaster center dataset respectively. Crack detection was performed on damaged columns in the Nepal dataset and on slabs in the disaster center dataset. The proposed CrackEmbed is compared with several baselines including segmentation from raw data such as (i) RGB color and (ii) intensity, geometric properties such as (iii) normal and (iv) curvature, feature descriptors from (v) Fast Point Feature Histogram (FPFH) [35] and features from (vi) PointNet++ [38] and (vii) Cylinder3D [43]. Note that CrackEmbed is trained using the instance labels of the S3DIS dataset whereas

Table 1

Performance comparison of different segmentation methods on the Nepal dataset.

Method	Precision	Recall	F1-score	mIOU	Length error (m)	Width error (m)
RGB color	0.32	0.44	0.36	0.23	0.001	0.011
Intensity	0.17	0.33	0.21	0.13	0.003	0.004
Normal	0.26	0.37	0.30	0.18	0.001	0.012
Curvature	0.29	0.42	0.34	0.21	0.001	0.015
FPFH	0.21	0.10	0.13	0.07	0.020	0.018
PointNet++	0.15	0.10	0.12	0.06	0.002	0.045
Cylinder3D	0.07	0.25	0.10	0.06	0.426	0.071
CrackEmbed	0.41	0.70	0.50	0.35	0.000	0.004

Table 2

Performance comparison of different segmentation methods on the disaster center dataset.

Method	Precision	Recall	F1-score	mIOU	Length error (m)	Width error (m)
RGB color	0.72	0.57	0.63	0.47	0.027	0.021
Intensity	0.60	0.61	0.60	0.43	0.006	0.008
Normal	0.52	0.40	0.45	0.29	0.146	0.148
Curvature	0.47	0.44	0.44	0.29	0.049	0.040
FPFH	0.61	0.08	0.14	0.08	0.083	0.086
PointNet++	0.60	0.34	0.41	0.28	0.052	0.068
Cylinder3D	0.18	0.23	0.19	0.11	0.207	0.844
CrackEmbed	0.69	0.75	0.71	0.56	0.004	0.004

PointNet++ and Cylinder3D follow the original implementation of training using the class labels (using the S3DIS dataset and the SemanticKITTI dataset respectively).

In Tables 1 and 2, the output of each method is a set of vectors from either raw data or computed features, and the isolation forest algorithm is then used to process the vectors and extract the cracked region. Results show that CrackEmbed achieved the highest precision, recall, and F1-score as well as the lowest length error and width error. Figs. 7 and 8 show the visualization of the crack segmentation results using different methods where the predicted crack region is highlighted in yellow. Whereas, Figs. 9 and 10 show the visualization of the crack segmentation results on different columns and slabs. The figures show that there are still many false-positive points that are segmented across all the methods, but CrackEmbed resulted in the overall closest match to the ground truth.

Table 3 shows a performance comparison of different unsupervised machine learning algorithms used to separate the cracked points. These include anomaly detection methods such as (i) isolation forest [51], (ii) one-class SVM [52], (iii) robust covariance [53], and (iv) local outlier factor [54]. Clustering methods such as (i) K-means clustering [48], (ii) Gaussian Mixture Model [49], and (iii) mean-shift clustering [50] are also included as baseline methods for comparison. In this study, the anomaly detection and clustering algorithms are implemented using the scikit-learn [56] library. Results in Table 3 show that the anomaly detection methods, with the exception of the local outlier factor, outperform the clustering methods in general. Among the anomaly detection methods, the isolation forest method achieved the highest

recall and the lowest length error and width error. Whereas, the robust covariance method achieved the highest precision and F1-score.

Fig. 11 shows the end result of damage detection, where the segmented cracks from each column can be mapped back to the original point cloud scene. In this way, the affected building can be described in terms of the precise location of each crack as well as the length and width of each crack. This information can be used for further processing by the disaster relief team for structural analysis and risk analysis.

5.1. Analysis of different feature representations

The effectiveness of different point feature representations, when used for the task of crack segmentation, can be analyzed based on the results in Tables 1 and 2. Overall, the proposed CrackEmbed achieved the best performance across the Nepal dataset and the disaster center dataset. This demonstrates that the proposed embedding is able to successfully capture the relevant information in point cloud data and represent them as feature vectors that are useful for segmentation. The RGB color, normal, and curvature data also achieved reasonable performance, whereas the FPFH, PointNet++, and Cylinder3D features resulted in poor performance. This is because FPFH is designed for point cloud registration, whereas PointNet++ and Cylinder3D are designed for semantic segmentation of large objects, and the computed features may not be suitable for segmentation of cracks. In addition, Cylinder3D is designed for use with sparser point clouds found in the SemanticKITTI dataset and not dense point clouds used for building crack detection.

The segmentation performance can also be compared in terms of the qualitative results in Figs. 7 and 8. One caveat of this comparison is that the ground truth outline of a crack is difficult to define in certain situations where the outline of the crack is not visually prominent. However, the performance can still be compared by examining the frequency of false-positive points in regions that clearly do not have cracks. The RGB color, intensity, and CrackEmbed methods have false positives that are mostly caused by shadow effects in the point cloud. Whereas, the normal, curvature, FPFH, and PointNet++ methods have false positives that occur around the edges of the building component. Other confounding factors include uneven surface and uneven resolution of the point cloud caused by incomplete laser scanning. Especially in the Nepal dataset, the performance of the intensity method is negatively affected by false positives, possibly since the intensity measurement could vary

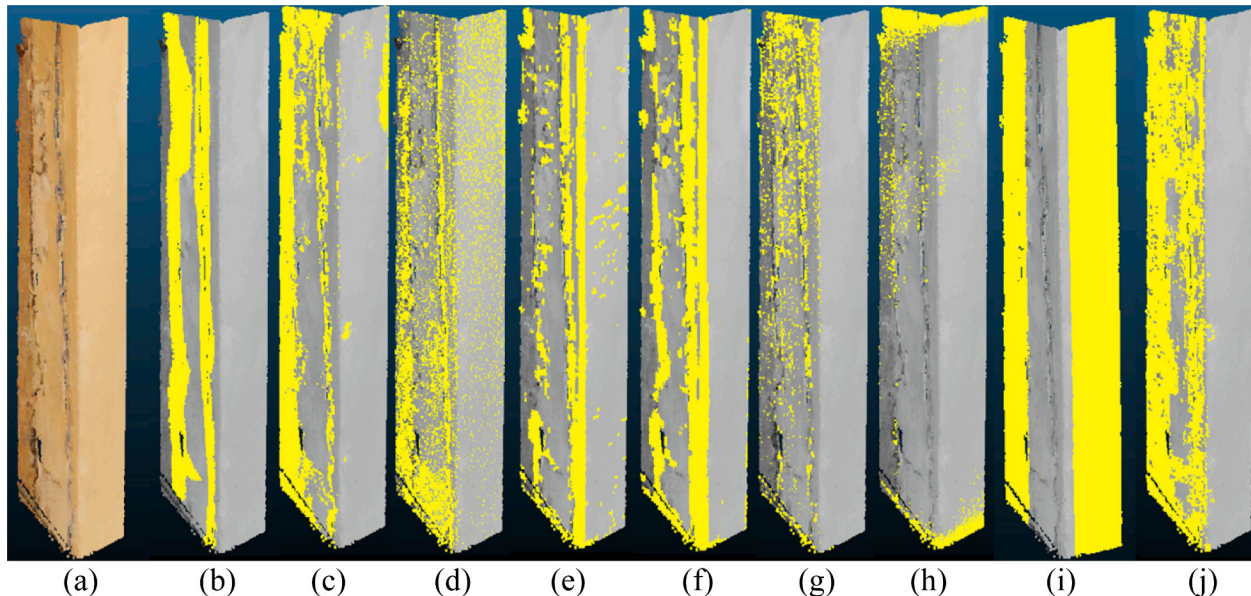


Fig. 7. Visualization of column crack segmentation results with different segmentation methods: (a) input point cloud (b) ground truth (c) RGB color (d) intensity (e) normal (f) curvature (g) FPFH (h) PointNet++ (i) Cylinder3D (j) CrackEmbed.

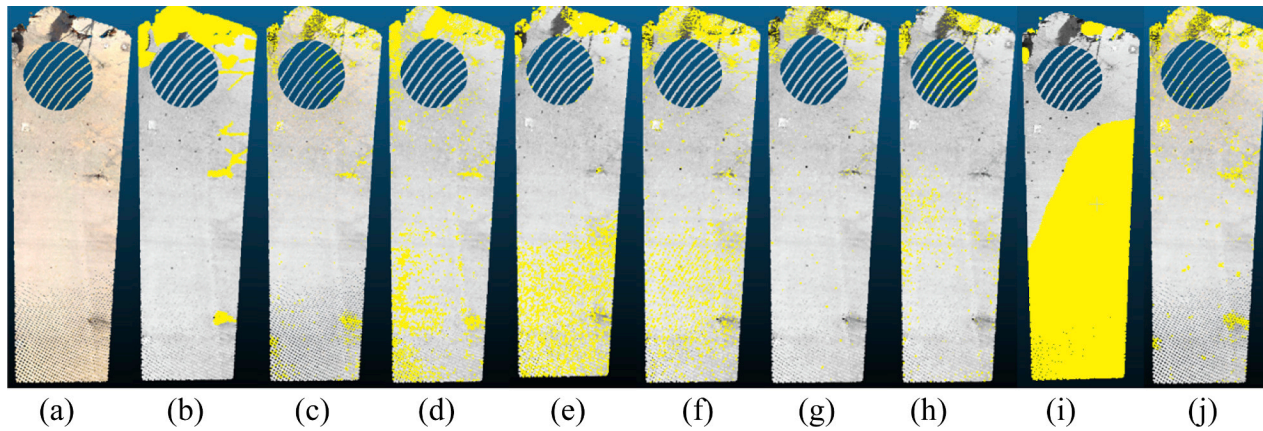


Fig. 8. Visualization of slab crack segmentation results with different segmentation methods: (a) input point cloud (b) ground truth (c) RGB color (d) intensity (e) normal (f) curvature (g) FPFH (h) PointNet++ (i) Cylinder3D (j) CrackEmbed.

depending on the laser incident angle as well as the distance between the scanned structure and the laser scanner.

5.2. Analysis of different unsupervised algorithms

Table 3 analyzes the segmentation performance comparison of different unsupervised algorithms on the Nepal dataset, including anomaly detection algorithms and clustering algorithms. Among the anomaly detection algorithms, the isolation forest and robust covariance algorithms performed the best overall when considering all the metrics. One-class SVM achieved a reasonable performance, whereas the local outlier factor algorithm performed poorly, especially in terms of the F1 score. This could be due to the way the cracked points are distributed in feature space, where they are isolated from other regions but not sparse enough such that the local outlier factor is significant. On the other hand, among the clustering algorithms, K-means clustering achieved the best performance, whereas the other two algorithms did not perform well. One downside of using a clustering algorithm is that the hyper-parameters such as the number of clusters (for the case of K-means clustering and Gaussian Mixture Model) and the bandwidth (for the case of mean-shift clustering) have to be carefully selected to achieve good performance. A clustering algorithm usually creates multiple clusters, one of which is assigned to the anomaly region, whereas an anomaly detection algorithm directly separates the anomaly region and the non-anomaly region. Based on this as well as the results in Table 3, it is better to use anomaly detection algorithms such as isolation forest and robust covariance for the task of crack segmentation.

5.3. Analysis of the effect of crack width

Crack width is an important factor to consider when analyzing the accuracy of crack segmentation since it is related to the visibility of the crack and the difficulty in extracting the crack points from surrounding points. Fig. 12 shows a graph analyzing the trend of the F1 score with varying crack widths when performing crack segmentation on the Nepal dataset. Each data point in the graph represents an individual crack with an associated crack width and the corresponding segmentation F1 score. Note that some data points may be overlapping due to the cracks having similar widths. Another point to note is that the graph shows the maximum crack width for each crack, so even though for a particular crack the width of 20 cm is used for plotting, the actual crack width could vary between 2 cm and 20 cm across the length of the crack. The graph shows that there is a general upward trend in F1 score as the crack width increases, and this trend is observed across different segmentation methods. Larger cracks usually occupy a larger region in the point cloud and have more prominently different color and intensity values, so it is

reasonable that larger cracks can be more accurately detected by the crack segmentation algorithm. There are a few cases where the F1 score decreases with crack width. This is because there are other factors that affect the difficulty of segmenting a particular crack, such as the noise level in the point cloud and the prominence of the crack.

5.4. Analysis of the effect of outlier ratio

Another important factor to consider when analyzing the crack segmentation accuracy is the ratio between the number of points on the crack and the total number of points on a building component. This is because the assumption of the anomaly detection step is that the points on the crack make up a minority of all the points and can be separated as outliers. The effect of this outlier is analyzed in Fig. 13, which shows the relationship between the F1 score and the outlier ratio for different segmentation methods on the Nepal dataset. Based on the results, there is no clear increasing or decreasing trend in the accuracy with different outlier ratios. This is likely because the outlier ratio is mostly low in the Nepal dataset (i.e., points on cracks only make up around 5% – 15% of the point cloud). Thus, it does not significantly alter the feature distribution when performing anomaly detection. There could potentially be situations where the outlier ratio is high such as partially collapsed structures or structures with a deformed surface. However, those situations are more suitable to be processed by deformation detection or debris detection algorithms and not crack detection, which is the focus of this study.

5.5. Analysis of the effect of point cloud resolution

The crack segmentation accuracy can be further analyzed by examining the effect of point cloud resolution. Depending on the laser scanning hardware as well as the distance between the scanned structure and the laser scanner, the point cloud resolution (i.e., the spacing between individual points in the point cloud) could be different. This effect was simulated on the Nepal dataset by applying voxel-grid downsampling to the original laser-scanned point cloud at varying resolutions of 0.001 m, 0.002 m, 0.005 m, 0.007 m, 0.01 m, 0.02 m, 0.05 m, 0.07 m, and 0.1 m. Fig. 14 shows the effect of point cloud resolution on the F1 score, whereas Fig. 15 shows the effect of point cloud resolution on the crack width estimation error, averaged over all the columns in the dataset. Note that in both graphs, the X and Y axes are plotted on a logarithmic scale for easier visualization. Results show that the segmentation performance remains mostly unchanged in the 0.001 m to 0.01 m range. Whereas, the F1 score decreases significantly, and the width error increases significantly when the resolution approaches the 0.1 m range. This trend makes sense because when the point spacing is large, there is



Fig. 9. Visualization of crack segmentation results for all 7 columns in the Nepal dataset: (top row) input point cloud (middle row) ground truth (bottom row) CrackEmbed results.

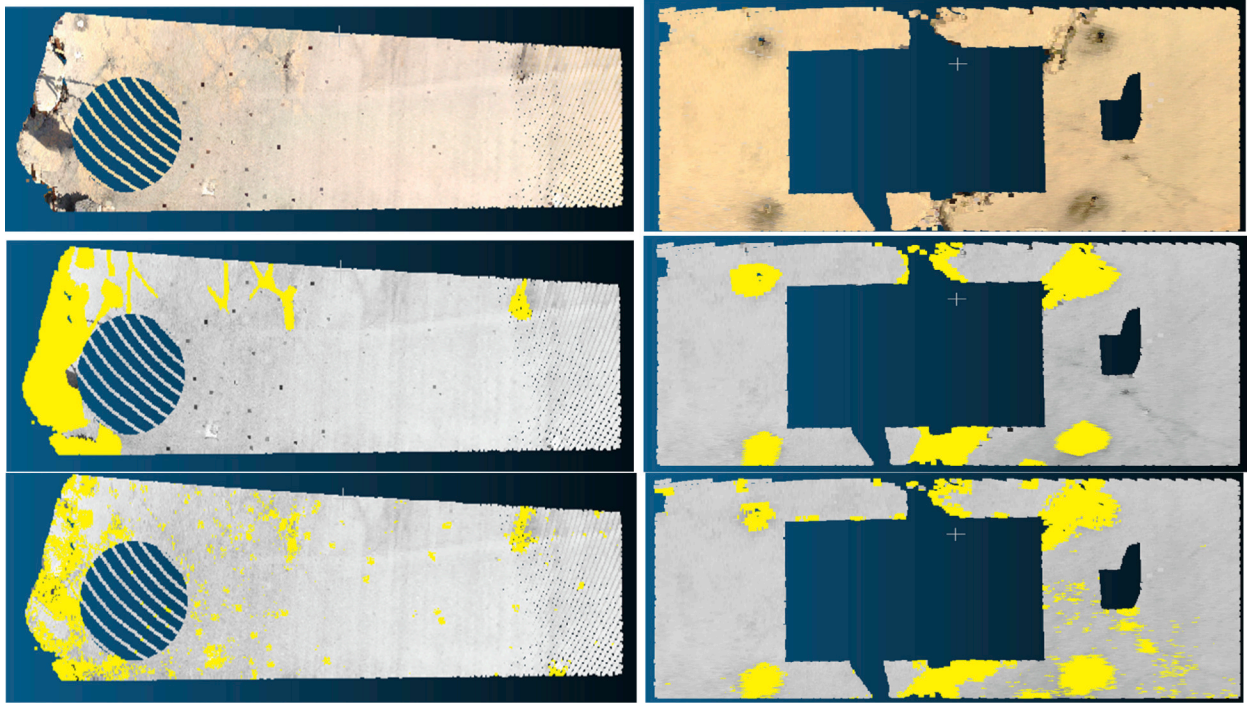


Fig. 10. Visualization of crack segmentation results for 2 slabs in the disaster center dataset: (top row) input point cloud (middle row) ground truth (bottom row) CrackEmbed results.

Table 3

Performance comparison of different unsupervised algorithms on the Nepal dataset.

Method	Precision	Recall	F1-score	Length error (m)	Width error (m)
<i>Anomaly detection</i>					
Isolation Forest	0.39	0.76	0.50	0.000	0.004
One-class SVM	0.38	0.62	0.41	0.027	0.010
Robust covariance	0.59	0.50	0.52	0.040	0.010
Local Outlier Factor	0.13	0.12	0.12	0.018	0.008
<i>Clustering</i>					
K-means	0.58	0.36	0.43	0.287	0.010
Gaussian Mixture Model	0.27	0.15	0.19	0.386	0.020
Mean-shift	0.34	0.24	0.27	0.032	0.012

less information in the point cloud for the segmentation algorithm to clearly identify the outline of the crack. On a practical level, the results also suggest that when acquiring the point cloud data, having a resolution (point spacing) in the 1 mm to 1 cm range is important to maintain accurate crack segmentation and crack width estimation.

5.6. Fine crack segmentation with synthetic data

Since the datasets shown in Fig. 6 do not have fine cracks that are scanned with sufficient resolution, this study makes use of a synthetic point cloud dataset with a manually created crack. This is to demonstrate the applicability of the proposed crack segmentation algorithm to fine cracks in the range of 1 mm – 5 mm in width. Fig. 16 shows the process of generating a synthetic fine crack based on existing point cloud data. First, a small subsection is extracted from a laser-scanned point cloud of a reinforced concrete column. Next, the outline of a fine crack is manually segmented from the point cloud and a mesh model is generated across the surface using Poisson mesh reconstruction. Finally, a new

synthetic point cloud is obtained from the mesh model by sampling points on the surface with a sampling resolution of 0.1 mm. The synthetically-generated crack has a width of 2 mm at its narrowest point and a depth of 2 mm. One caveat with this synthetically generated point cloud is that it may not be representative of fine cracks that are deeper than they are wide, in which case the scanned point cloud would be missing many of the internal surface points. However, the proposed method can still partially segment the outer surface of the crack as long as most of the external crack is visible (as shown in the results in Figs. 9 and 10 with many partially-scanned cracks).

Fig. 17 shows the results of applying the proposed crack segmentation method to the synthetic point cloud data. First, the point cloud is processed by the CrackEmbed network to compute a feature embedding for each point on the surface, as visualized in Fig. 17a. Next, the Isolation Forest algorithm is applied on the feature embeddings to determine the normality score. This is visualized in Fig. 17b, where points with reddish hues have high normality scores whereas points with blueish hues have low normality scores. Finally, the points with low normality scores which correspond to outliers are segmented out from the surface, as can be seen with the yellow points in Fig. 17c.

6. Discussion

6.1. Contributions to knowledge

Overall, this research provides several contributions to knowledge in the areas of disaster relief, crack detection, unsupervised machine learning, and deep learning. The main contribution of this research is CrackEmbed, a deep neural network that can compute crack feature embeddings for 3D crack segmentation in the context of disaster relief tasks. This research also investigates and implements unsupervised machine learning techniques for crack segmentation to overcome the problem of lack of annotated training data. Results in Table 3 show that anomaly detection algorithms can be effectively used for crack segmentation based on the property that points with low normality score (i. e., anomaly points) are correlated with points in the cracked region of

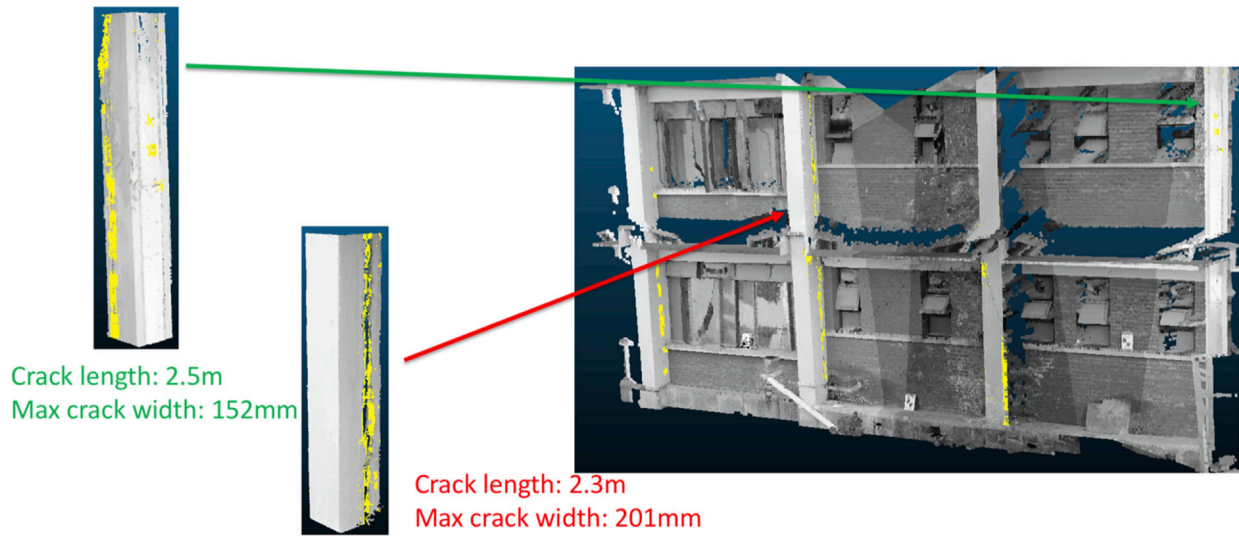


Fig. 11. Visualization of crack segmentation results for concrete columns mapped back to the original scene.

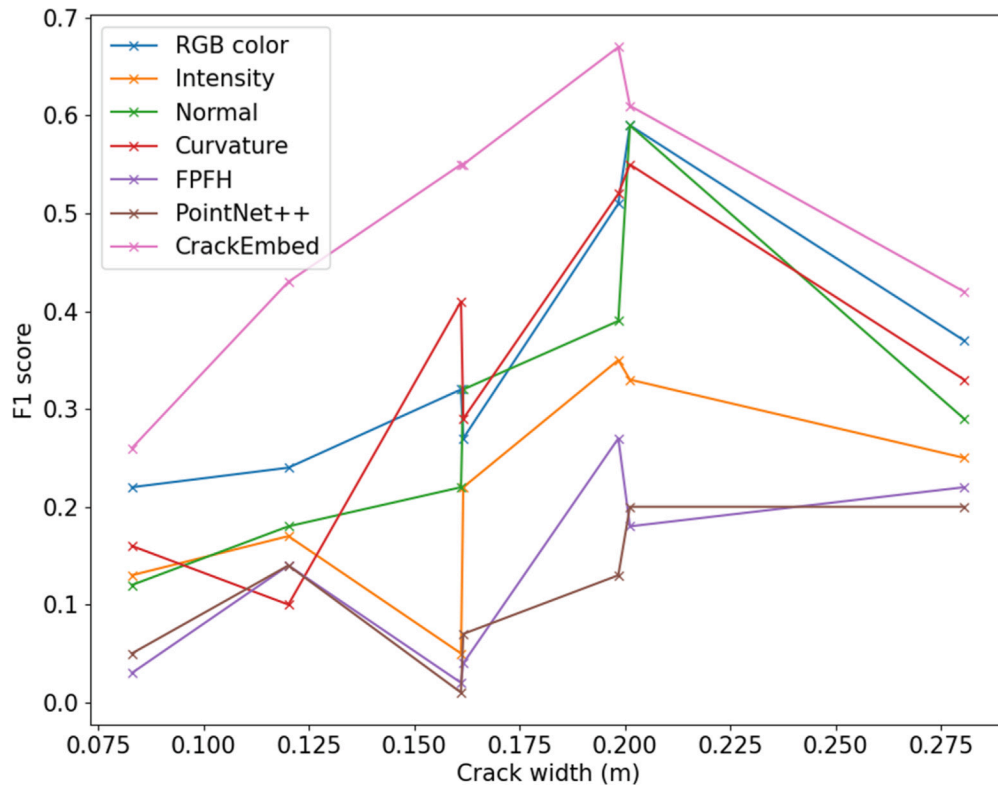


Fig. 12. Graph of F1 score vs. crack width for different segmentation methods.

the point cloud. The viability of using deep feature embeddings to represent discriminative information about point cloud data is in turn supported by experimental results in Tables 1-2, which show that CrackEmbed is able to achieve accurate crack segmentation results compared to other feature representations. The experimental results in Figs. 12-15 also present valuable contributions to knowledge in terms of the effect of different experimental parameters on the algorithm performance. For example, the ablation study shows that the crack segmentation performance increases with crack size, whereas the effect of the outlier ratio is less significant. In addition, having a resolution (point spacing) in the 1 mm to 1 cm range is important to maintain accurate crack segmentation and crack width estimation.

6.2. Practical considerations

One important practical consideration is the applicability of the proposed crack segmentation method across different sizes of cracks. The proposed method is targeted for disaster relief applications where it is important to detect immediate structural failure or collapse in the form of large cracks for rapid disaster relief activities instead of fine cracks which would be more relevant for structural health monitoring. From an algorithmic perspective, the proposed method can still work with point cloud data with fine cracks as long as the scanned point cloud has adequate resolution and detail. Even though the experimental results in Fig. 12 show that segmentation of fine cracks is less accurate, it is

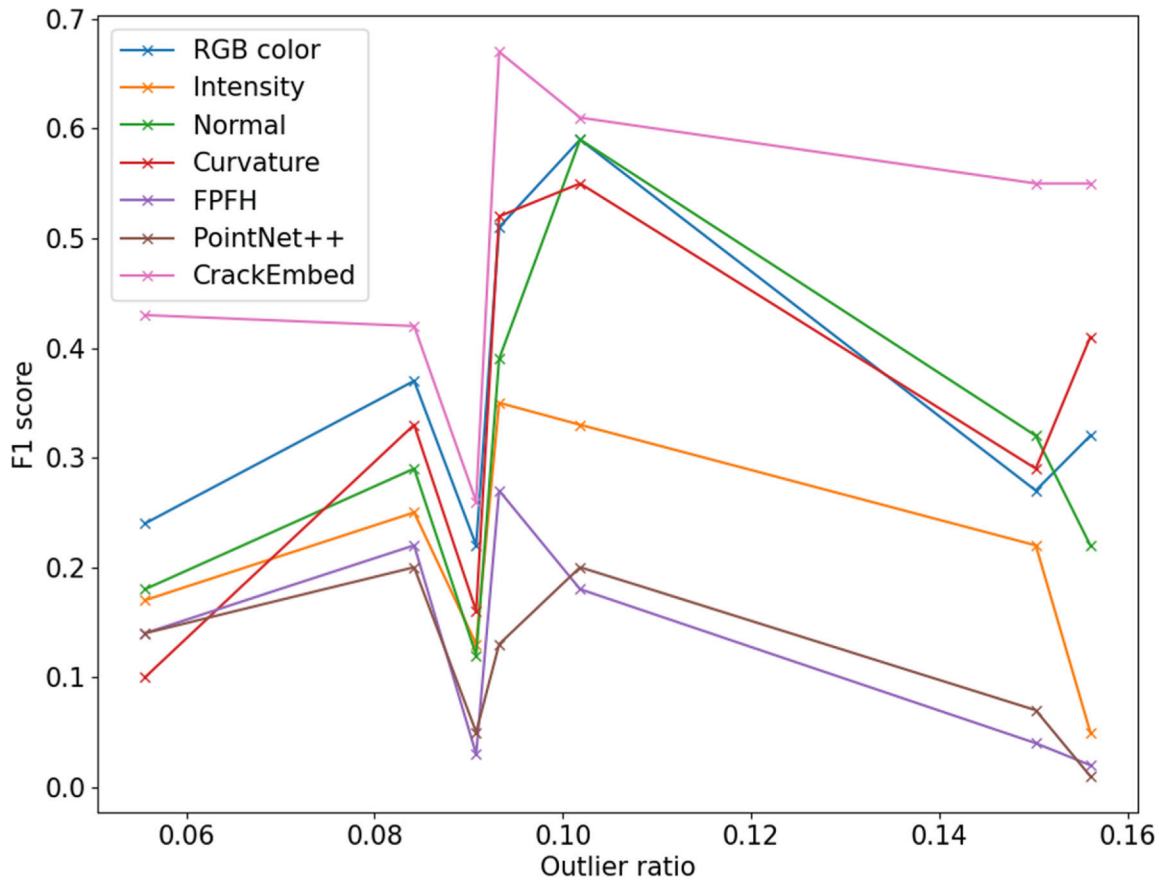


Fig. 13. Graph of F1 score vs outlier ratio for different segmentation methods.

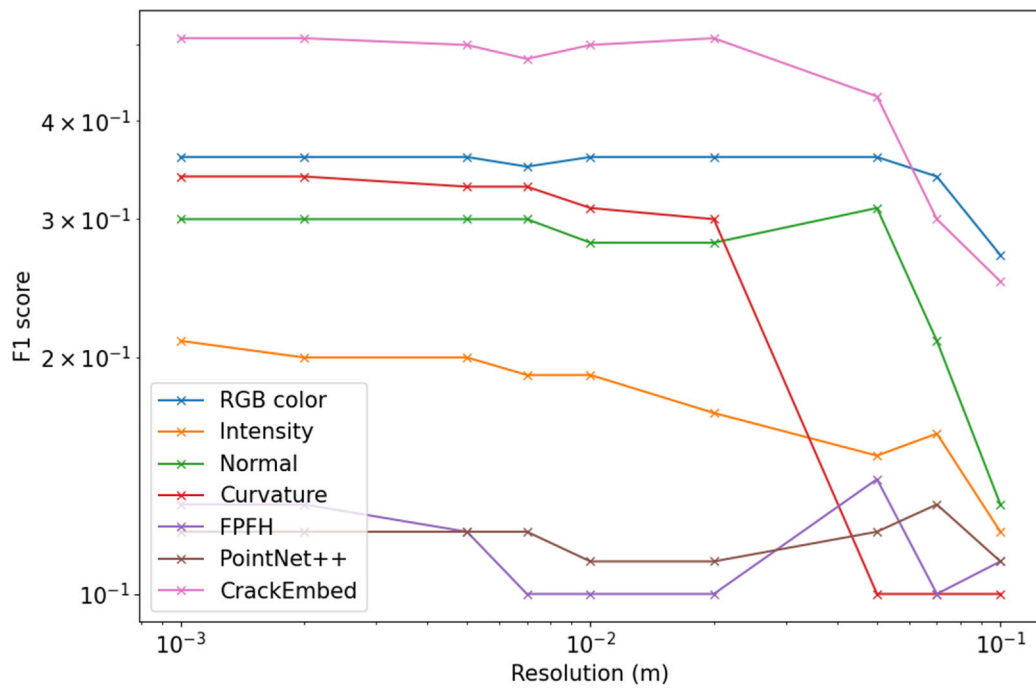


Fig. 14. Graph of F1 score vs. point cloud resolution for different segmentation methods.

expected that this is due to the limitations of the scanning device and the scan data in terms of resolution and scan distance rather than an inherent limitation of the proposed algorithm. Based on building

inspection guidelines [57], cracks that are less than 0.1 mm in width are negligible and the typical range of fine cracks is 1–5 mm. Thus, in the Results section, a synthetic dataset is used to demonstrate the

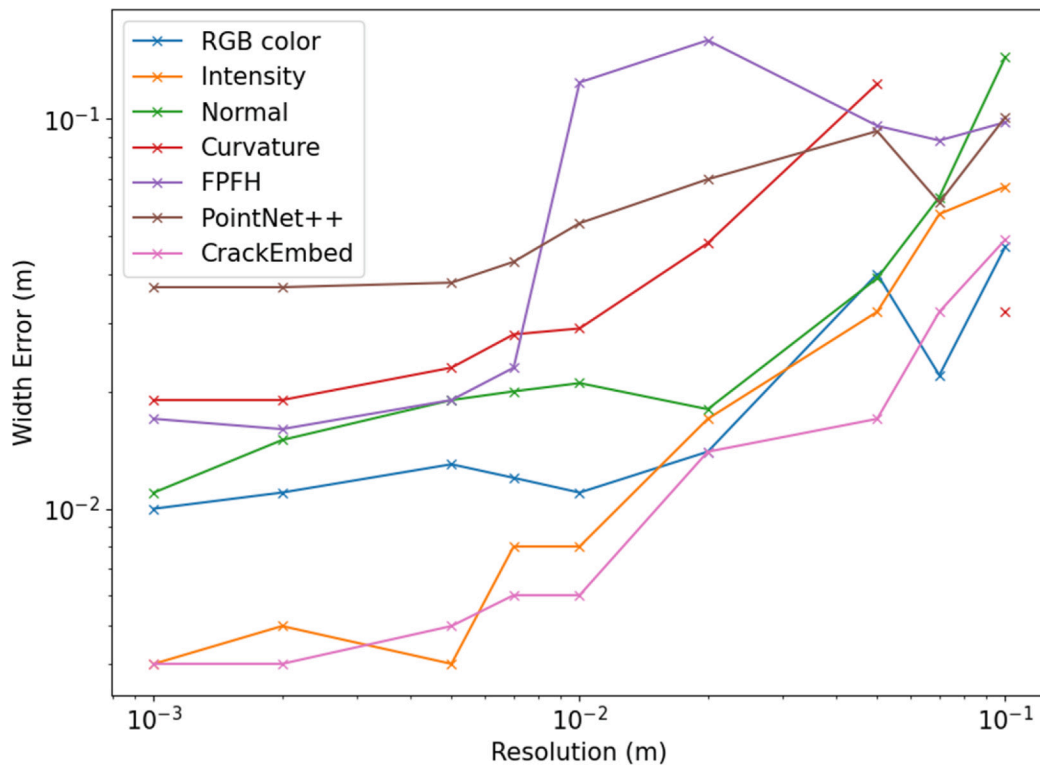


Fig. 15. Graph of width error vs. point cloud resolution for different segmentation methods.

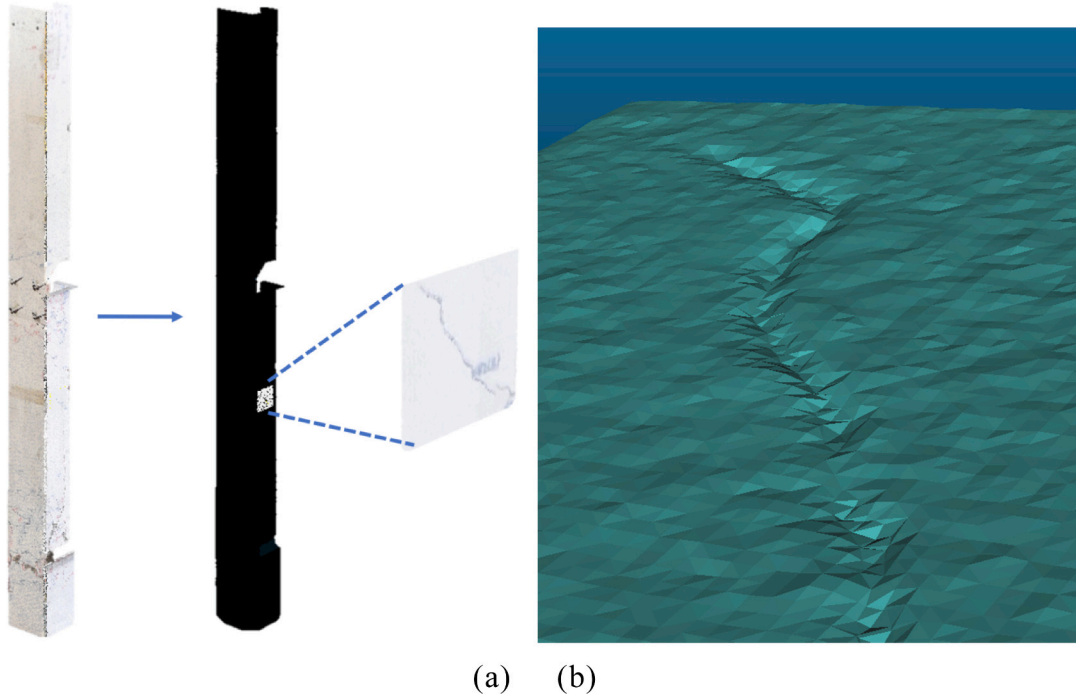


Fig. 16. Generation of synthetic fine crack with 2 mm width and 2 mm depth: (a) extraction of point cloud segment (b) mesh reconstruction from point cloud surface.

applicability to fine crack segmentation at a width and depth of around 2 mm. This result shows that fine cracks can be detected by CrackEmbed algorithm if the point cloud has sufficient resolution if acquired with a high-resolution handheld laser scanner from a close-up position, which typically provides 0.025 mm resolution and 0.035 mm accuracy.

Another practical consideration is the feasibility of structural damage assessment based on external crack segmentation. The Federal

Emergency Management Agency (FEMA) guidelines [58] show that observed damage such as cracks are important indicators for disaster relief in post-earthquake scenarios. In general, the seriousness of structural damage is related to its observability since the crack width is indicative of the maximum reinforcement strain and the level of damage severity. Most established post-earthquake inspection guidelines use the method of inspecting external cracks as a form of safety evaluation

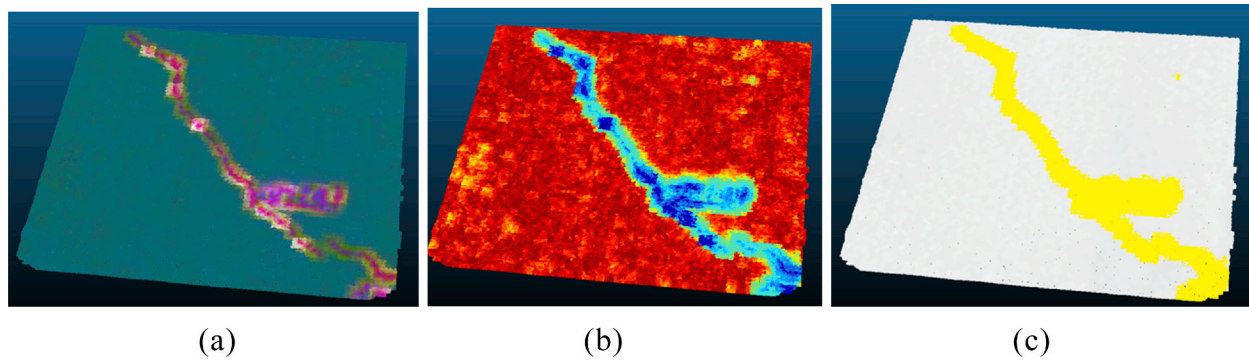


Fig. 17. Visualization of fine crack segmentation results: (a) feature embedding computed by CrackEmbed (b) normality score for anomaly detection (c) final segmented crack.

[58,59]. The limitation of only inspecting external structural damage (e.g. surface cracks) is that it may not reflect internal structural damage that cannot be observed. One form of internal damage in concrete structures is caused by thermal attacks such as freezing or fire events. However, these are less relevant in earthquake scenarios, where stress forces will usually result in external damage [60]. Even in certain situations where earthquakes may cause hidden damage such as a buckled rebar, it is still cost-prohibitive and even dangerous to intrusively test every structure in a building for hidden damage [58]. In view of these considerations, the proposed point cloud-based approach is a reasonable solution for non-intrusive damage assessment since it can be achieved through remote sensing and laser scanning.

6.3. Qualitative comparison

This section aims to qualitatively discuss the proposed crack segmentation method compared to other conventional methods used for crack detection. The proposed method works with 3D point cloud data whereas the problem of crack detection has been more widely researched in the 2D image domain. However, there are certain scenarios such as night-time or low lighting conditions where image-based crack detection methods do not work whereas the proposed method can still be applicable. Moreover, existing methods for image-based crack detection require images that are taken from a close-up view of the structure which is not always possible. 3D point clouds also allow the crack width, length, and especially depth to be directly measured from the scan data compared to 2D images which require additional calibration and processing steps to convert pixel measurements to physical measurements. It should be noted that many researchers used 2D images rather than 3D laser scanners because of the low cost and convenience of using cameras, which does not necessarily mean that 2D image-based crack detection performs better than 3D-LiDAR-based methods. If fine-crack detection is the main purpose, there exists high-resolution handheld LiDAR scanners that perform much better than a camera-based approach for depth measurements. Handheld LiDAR scanners, such as the ScanTech [61] and HandyScan [62] devices, typically have a resolution of 0.02 – 0.03 mm but with a range of only 0.3 m. Thus, these devices require close-up data collection, unlike the terrestrial laser scanners (TLS) which were mainly used in our study.

More generally, the proposed method for crack segmentation from point cloud data has several advantages over conventional methods for crack detection including the ability to be deployed as a remote sensing solution, not requiring training annotations, and the ability to measure the crack size in 3D. These qualitative properties are summarized in Table 4 below:

7. Conclusions

To tackle the problem of automated damage assessment with point

Table 4

Qualitative comparison of crack detection methods.

Category	Example works	Remote sensing	Require training annotations	Measure crack size
Vibration sensing	[5,11]	No	No	No
Acoustic sensing	[6,7,12]	No	No	Yes
Image + unsupervised learning	[15,16,17]	Yes	No	Indirect
Image + supervised learning	[8,14,25]	Yes	Yes	Indirect
Point cloud + geometry-based thresholding	[9,28]	Yes	No	Yes
Point cloud + unsupervised learning	Proposed CrackEmbed	Yes	No	Yes

clouds of disaster sites, this research proposes a method for crack segmentation using unsupervised machine learning techniques. A deep neural network is used to compute a discriminative point feature embedding which is then processed by anomaly detection algorithms to detect cracks in point clouds. The performance of the proposed approach was evaluated based on experimental studies on laser-scanned point clouds from two different sites. Findings show that anomaly detection algorithms such as isolation forest and robust covariance were advantageous compared to clustering algorithms for unsupervised crack segmentation. The proposed CrackEmbed is a viable form of deep feature embedding to represent discriminative information about point cloud data for crack segmentation purposes. Results on a synthetic test set also show the robust performance of CrackEmbed for fine crack detection as long as the point cloud has sufficient resolution. In spite of limitations such as only detecting external cracks and requiring high-resolution point clouds, the proposed point cloud-based approach can still be a reasonable solution for non-intrusive damage assessment since it can be achieved through remote sensing and laser scanning. Findings from this study are expected to contribute to better knowledge and understanding of how damage detection from point cloud data can be performed under low supervision conditions. Future work in this research involves combining the feature computation and anomaly detection components into a single network that can be trained end-to-end. The authors will also explore the segmentation of a wider variety of surface damage, including concrete spalling and water damage.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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References

- [1] CRED, Natural disasters 2019: Now is the time of not give up, Cent. Res. Epidemiol. Disasters - CRED. (2020) 1–8. <https://www.cred.be/publications> (accessed October 5, 2020).
- [2] J. Chen, Y.K. Cho, Detection of Damaged Infrastructure on Disaster Sites using Mobile Robots, in: 2019 16th Int. Conf. Ubiquitous Robot., 2019: pp. 648–653. Doi: 10.1109/URAI.2019.8768770.
- [3] M. Kazama, T. Noda, off the Pacific Coast of Tohoku Earthquake damage), Soils Found. 52 (5) (2012) 780–792. <https://doi.org/10.1016/j.sandf.2012.11.003>.
- [4] H.Y. Noh, D. Lallemand, A.S. Kiremidjian, Development of empirical and analytical fragility functions using kernel smoothing methods, Earthq. Eng. Struct. Dyn. (2015) 1163–1180. <https://doi.org/10.1002/eqe>.
- [5] H. Noh, R. Rajagopal, A.S. Kiremidjian, Sequential structural damage diagnosis algorithm using a change point detection method, J. Sound Vib. 332 (24) (2013) 6419–6433. <https://doi.org/10.1016/j.jsv.2013.07.005>.
- [6] T.M. Roberts, M. Talebzadeh, Acoustic emission monitoring of fatigue crack propagation, J. Constr. Steel Res. 59 (6) (2003) 695–712. [https://doi.org/10.1016/S0143-974X\(02\)00064-0](https://doi.org/10.1016/S0143-974X(02)00064-0).
- [7] J.B. Harley, Predictive Guided Wave Models Through Sparse Modal Representations, Proc. IEEE. 104 (8) (2016) 1604–1619.
- [8] Y. Liu, J. Yao, X. Lu, R. Xie, L. Li, DeepCrack: A deep hierarchical feature learning architecture for crack segmentation, Neurocomputing. 338 (2019) 139–153. <https://doi.org/10.1016/j.neucom.2019.01.036>.
- [9] B.G. Erkal, J.F. Hajjar, Laser-based surface damage detection and quantification using predicted surface properties, Autom. Constr. 83 (2017) 285–302. <https://doi.org/10.1016/j.autcon.2017.08.004>.
- [10] I. Armeni, O. Sener, A.R. Zamir, H. Jiang, I. Brilakis, M. Fischer, S. Savarese, 3D Semantic Parsing of Large-Scale Indoor Spaces, 2016 IEEE Conf. Comput. Vis. Pattern Recognit. (2016) 1534–1543. Doi: 10.1109/CVPR.2016.170.
- [11] B. Brando, D. Rapone, E. Spacone, M.S. O'Banion, M.J. Olsen, A.R. Barbosa, M. Faggella, R. Gigliotti, D. Liberatore, S. Russo, L. Sorrentino, S. Bose, A. Stravidis, Damage Reconnaissance of Unreinforced Masonry Bearing Wall Buildings After the 2015 Gorkha, Nepal, Earthquake, Earthq. Spectra. 33 (2017) S243–S273. Doi: 10.1193/010817EQS009M.
- [12] S.W. Doehling, C.R. Farrar, M.B. Prime, D.W. Shevitz, Damage identification and health monitoring of structural and mechanical systems from changes in their vibration characteristics: A literature review, U.S. Off. Sci. Tech. Inf. (1996). <https://doi.org/10.2172/249299>.
- [13] T. Suzuki, M. Ohtsu, Damage estimation of concrete canal due to earthquake effects by acoustic emission method, Constr. Build. Mater. 67 (2014) 186–191. <https://doi.org/10.1016/j.conbuildmat.2014.05.016>.
- [14] S. Popovics, J.L. Rose, J.S. Popovics, The behaviour of ultrasonic pulses in concrete, Cem. Concr. Res. 20 (2) (1990) 259–270. [https://doi.org/10.1016/0008-8846\(90\)90079-D](https://doi.org/10.1016/0008-8846(90)90079-D).
- [15] Q. Zou, Z. Zhang, Q. Li, X. Qi, Q. Wang, S. Wang, DeepCrack: Learning Hierarchical Convolutional Features for Crack Detection, IEEE Trans. Image Process. 28 (3) (2019) 1498–1512.
- [16] R. Fu, H. Xu, Z. Wang, L. Shen, M. Cao, T. Liu, D. Novák, Enhanced Intelligent Identification of Concrete Cracks Using Multi-Layered Image Preprocessing-Aided Convolutional Neural Networks, Sensors. 20 (2020) 2021. <https://doi.org/10.3390/s20072021>.
- [17] C. Benz, P. Debus, H.K. Ha, V. Rodehorst, Crack Segmentation on UAS-based Imagery using Transfer Learning, in: 2019 Int. Conf. Image Vis. Comput. New Zeal., 2019: pp. 1–6. Doi: 10.1109/IVCNZ48456.2019.8960998.
- [18] G. Liu, N. Yang, L. Guo, S. Guo, Z. Chen, A One-Stage Approach for Surface Anomaly Detection with Background Suppression Strategies, Sensors. 20 (2020) 1829. <https://doi.org/10.3390/s20071829>.
- [19] D. Tabernik, S. Sela, J. Skvarč, D. Skočaj, Segmentation-based deep-learning approach for surface-defect detection, J. Intell. Manuf. 31 (3) (2020) 759–776. <https://doi.org/10.1007/s10845-019-01476-x>.
- [20] Y.-C. Tsai, F. Li, Critical Assessment of Detecting Asphalt Pavement Cracks under Different Lighting and Low Intensity Contrast Conditions Using Emerging 3D Laser Technology, J. Transp. Eng. 138 (5) (2012) 649–656. [https://doi.org/10.1061/\(ASCE\)TE.1943-5436.0000353](https://doi.org/10.1061/(ASCE)TE.1943-5436.0000353).
- [21] V. Kaul, Y. (James) Tsai, R.M. Mersereau, Quantitative Performance Evaluation Algorithms for Pavement Distress Segmentation, Transp. Res. Rec. 2153 (1) (2010) 106–113. <https://doi.org/10.3141/2153-12>.
- [22] A. Zhang, K.C.P. Wang, B. Li, E. Yang, X. Dai, Y.i. Peng, Y. Fei, Y. Liu, J.Q. Li, C. Chen, Automated Pixel-Level Pavement Crack Detection on 3D Asphalt Surfaces Using a Deep-Learning Network, Comput. Civ. Infrastruct. Eng. 32 (10) (2017) 805–819.
- [23] A. Zhang, K.C.P. Wang, Y. Fei, Y. Liu, S. Tao, C. Chen, J.Q. Li, B. Li, Deep Learning Based Fully Automated Pavement Crack Detection on 3D Asphalt Surfaces with an Improved CrackNet, J. Comput. Civ. Eng. 32 (5) (2018) 04018041. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000775](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000775).
- [24] Y. Fei, K.C.P. Wang, A. Zhang, C. Chen, J.Q. Li, Y. Liu, G. Yang, B. Li, Pixel-Level Cracking Detection on 3D Asphalt Pavement Images Through Deep-Learning-Based CrackNet-V, IEEE Trans. Intell. Transp. Syst. 21 (1) (2020) 273–284.
- [25] G. Li, J. Wan, S. He, Q. Liu, B. Ma, Semi-Supervised Semantic Segmentation Using Adversarial Learning for Pavement Crack Detection, IEEE Access. 8 (2020) 51446–51459.
- [26] M.R. Jahanshahi, F. Jazizadeh, S.F. Masri, B. Becerik-Gerber, Unsupervised Approach for Autonomous Pavement-Defect Detection and Quantification Using an Inexpensive Depth Sensor, J. Comput. Civ. Eng. 27 (6) (2013) 743–754. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000245](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000245).
- [27] A. Akagic, E. Buza, S. Omanovic, A. Karabegovic, Pavement crack detection using Otsu thresholding for image segmentation, in: 2018 41st Int. Conv. Inf. Commun. Technol. Electron. Microelectron., 2018: pp. 1092–1097.
- [28] J. Chen, Y.K. Cho, Point-to-point Comparison Method for Automated Scan-vs-BIM Deviation Detection, in: Proc. - 17th Int. Conf. Comput. Civ. Build. Eng., 2018. <http://programme.exordo.com/iccbe2018/delegates/presentation/303/>.
- [29] S. Bose, A. Nozari, M.E. Mohammadi, A. Stavridis, M. Babak, R. Wood, D. Gillins, A. Barbosa, Structural Assessment of a School Building in Sankhu, Nepal Damaged Due to Torsional Response During the 2015 Gorkha Earthquake, in: S. Pakzad, C. Juan (Eds.), Dyn. Civ. Struct. Vol. 2, Springer International Publishing, Cham, 2016: pp. 31–41.
- [30] W. Liu, S. Chen, E. Hauser, LiDAR-based Bridge Structure Defect Detection, Exp. Tech. 35 (6) (2011) 27–34. <https://doi.org/10.1111/j.1747-1567.2010.00644.x>.
- [31] J. Chen, Y.K. Cho, J. Ueda, Sampled-Point Network for Classification of Deformed Building Element Point Clouds, in: Proc. 2018 IEEE Conf. Robot. Autom., 2018.
- [32] M.E. Mohammadi, R.L. Wood, C.E. Wittich, Non-Temporal Point Cloud Analysis for Surface Damage in Civil Structures, ISPRS Int. J. Geo-Information. 8 (12) (2019) 527. <https://doi.org/10.3390/ijgi8120527>.
- [33] Z. Zhou, J. Gong, X. Hu, Community-scale multi-level post-hurricane damage assessment of residential buildings using multi-temporal airborne LiDAR data, Autom. Constr. 98 (2019) 30–45. <https://doi.org/10.1016/j.autcon.2018.10.018>.
- [34] R.B. Rusu, G. Bradski, R. Thibaux, J. Hsu, Fast 3D Recognition and Pose Using the Viewpoint Feature Histogram, in: Intell. Robot. Syst., 2010: pp. 2155–2162.
- [35] R.B. Rusu, N. Blodow, M. Beetz, Fast Point Feature Histograms (FPFH) for 3D registration, 2009 IEEE Int. Conf. Robot. Autom. (2009) 3212–3217. Doi: 10.1109/ROBOT.2009.5152473.
- [36] A.E. Johnson, M. Hebert, Using spin images for efficient object recognition in cluttered 3D scenes, IEEE Trans. Pattern Anal. Mach. Intell. 21 (1999) 433–449. <https://doi.org/10.1109/34.765655>.
- [37] C. Qi, H. Su, K. Mo, L.J. Guibas, PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation, Proc. Comput. Vis. Pattern Recognit. (CVPR), IEEE, 2017. Doi: 10.1109/3DV.2016.68.
- [38] C.R. Qi, L. Yi, H. Su, L.J. Guibas, PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space, in: NIPS, 2017. <http://arxiv.org/abs/1706.02413>.
- [39] F. Engelmann, T. Kontogianni, A. Hermans, B. Leibe, Exploring Spatial Context for 3D Semantic Segmentation of Point Clouds, (2017). Doi: 10.1109/ICCVW.2017.90.
- [40] B. Wu, A. Wan, X. Yue, K. Keutzer, SqueezeSeg: Convolutional neural nets with recurrent crf for real-time road-object segmentation from 3d lidar point cloud, ICRA, 2018.
- [41] B. Wu, X. Zhou, S. Zhao, X. Yue, K. Keutzer, SqueezeSegV2: Improved Model Structure and Unsupervised Domain Adaptation for Road-Object Segmentation from a LiDAR Point Cloud, in: ICRA, 2019.
- [42] H. Thomas, C.R. Qi, J.-E. Deschaud, B. Marcotegui, F. Goulette, L.J. Guibas, KPConv: Flexible and Deformable Convolution for Point Clouds, Proc. IEEE Int. Conf. Comput. Vis. (2019).
- [43] X. Zhu, H. Zhou, T. Wang, F. Hong, Y. Ma, W. Li, H. Li, D. Lin, Cylindrical and Asymmetrical 3D Convolution Networks for LiDAR Segmentation, ArXiv Prepr. ArXiv2011.10033, 2020.
- [44] R. Cheng, R. Razani, E. Taghavi, E. Li, B. Liu, (AF)2-S3Net: Attentive Feature Fusion With Adaptive Feature Selection for Sparse Semantic Segmentation Network, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2021: pp. 12547–12556.
- [45] J. Behley, M. Garbade, A. Milioto, J. Quenzel, S. Behnke, C. Stachniss, J. Gall, SemanticKITTI: A Dataset for Semantic Scene Understanding of LiDAR Sequences, in: Proc. IEEE/CVF Int. Conf. ~on Comput. Vis., 2019.
- [46] I. Armeni, O. Sener, A.R. Zamir, H. Jiang, I. Brilakis, M. Fischer, S. Savarese, 3D Semantic Parsing of Large-Scale Indoor Spaces, in: Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit., 2016.
- [47] E. Hoffer, N. Ailon, Deep Metric Learning Using Triplet Network, in: A. Feragen, M. Pelillo, M. Loog (Eds.), Similarity-Based Pattern Recognit, Springer International Publishing, Cham, 2015, pp. 84–92.
- [48] D. Arthur, S. Vassilvitskii, K-Means++: The Advantages of Careful Seeding, in: Proc. Eighteenth Annu. ACM-SIAM Symp. Discret. Algorithms, Society for Industrial and Applied Mathematics, USA, 2007: pp. 1027–1035.
- [49] D.A. Reynolds, Gaussian Mixture Models, Encycl. Biometrics. 741 (2009).
- [50] D. Comaniciu, P. Meer, Mean shift: a robust approach toward feature space analysis, IEEE Trans. Pattern Anal. Mach. Intell. 24 (2002) 603–619.
- [51] F.T. Liu, K.M. Ting, Z. Zhou, Isolation Forest, in: 2008 Eighth IEEE Int. Conf. Data Min., 2008: pp. 413–422.

- [52] B. Schölkopf, J.C. Platt, J. Shawe-Taylor, A.J. Smola, R.C. Williamson, Estimating the Support of a High-Dimensional Distribution, *Neural Comput.* 13 (7) (2001) 1443–1471, <https://doi.org/10.1162/089976601750264965>.
- [53] P.J. Rousseeuw, K.V. Driessen, A Fast Algorithm for the Minimum Covariance Determinant Estimator, *Technometrics*. 41 (3) (1999) 212–223, <https://doi.org/10.1080/00401706.1999.10485670>.
- [54] M.M. Breunig, H.-P. Kriegel, R.T. Ng, J. Sander, LOF: Identifying Density-Based Local Outliers, in: *Proc. 2000 ACM SIGMOD Int. Conf. Manag. Data*, Association for Computing Machinery, New York, NY, USA, 2000: pp. 93–104. Doi: 10.1145/342009.335388.
- [55] CloudCompare (version 2.10), GPL Softw., 2020. <http://www.cloudcompare.org/>.
- [56] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, E. Duchesnay, Scikit-learn: Machine Learning in {P}ython, *J. Mach. Learn. Res.* 12 (2011) 2825–2830.
- [57] M. Son, E.J. Cording, Estimation of Building Damage Due to Excavation-Induced Ground Movements, *J. Geotech. Geoenviron. Eng.* 131 (2) (2005) 162–177, [https://doi.org/10.1061/\(ASCE\)1090-0241\(2005\)131:2\(162\)](https://doi.org/10.1061/(ASCE)1090-0241(2005)131:2(162)).
- [58] FEMA, FEMA 306: Evaluation of Earthquake Damaged Concrete and Masonry Wall Buildings, FEMA, 1998.
- [59] Z. Zhu, S. German, I. Brilakis, Visual retrieval of concrete crack properties for automated post-earthquake structural safety evaluation, *Autom. Constr.* 20 (7) (2011) 874–883, <https://doi.org/10.1016/j.autcon.2011.03.004>.
- [60] K. Kovler, V. Chernov, Types of damage in concrete structures, in: N. Delatte (Ed.), *Fail. Distress Repair Concr. Struct.*, Woodhead Publishing, 2009: pp. 32–56. Doi: 10.1533/9781845697037.1.32.
- [61] ScanTech, (n.d.). <https://www.aniwaa.com/brands/scantech/> (accessed January 12, 2022).
- [62] HandyScan, (n.d.). <https://www.creaform3d.com/en/portable-3d-scanner-handy-scan-3d/technical-specifications> (accessed January 12, 2022).

Selected Publication 2



Improving Contrastive Learning on Visually Homogeneous Mars Rover Images

Isaac Ronald Ward^{1,3} , Charles Moore², Kai Pak¹ , Jingdao Chen²,
and Edwin Goh¹

¹ Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA
{isaac.r.ward,kai.pak,edwin.y.goh}@jpl.nasa.gov

² Computer Science and Engineering, Mississippi State University,
Starkville, MS, USA
cam1271@msstate.edu, chenjingdao@cse.msstate.edu

³ Department of Computer Science, University of Southern California,
Los Angeles, CA, USA

Abstract. Contrastive learning has recently demonstrated superior performance to supervised learning, despite requiring no training labels. We explore how contrastive learning can be applied to hundreds of thousands of unlabeled Mars terrain images, collected from the Mars rovers Curiosity and Perseverance, and from the Mars Reconnaissance Orbiter. Such methods are appealing since the vast majority of Mars images are unlabeled as manual annotation is labor intensive and requires extensive domain knowledge. Contrastive learning, however, assumes that any given pair of distinct images contain distinct semantic content. This is an issue for Mars image datasets, as any two pairs of Mars images are far more likely to be semantically similar due to the lack of visual diversity on the planet's surface. Making the assumption that pairs of images will be in visual contrast — when they are in fact not — results in pairs that are falsely considered as negatives, impacting training performance. In this study, we propose two approaches to resolve this: 1) an unsupervised deep clustering step on the Mars datasets, which identifies clusters of images containing similar semantic content and corrects false negative errors during training, and 2) a simple approach which mixes data from different domains to increase visual diversity of the total training dataset. Both cases reduce the rate of false negative pairs, thus minimizing the rate in which the model is incorrectly penalized during contrastive training. These modified approaches remain fully unsupervised end-to-end. To evaluate their performance, we add a single linear layer trained to generate class predictions based on these contrastively-learned features and demonstrate increased performance compared to supervised models; observing an improvement in classification accuracy of 3.06% using only 10% of the labeled data.

Keywords: Self-supervised learning · Contrastive learning · Unsupervised learning · Unlabeled images · Multi-task learning · Planetary science · Astrogeology · Space exploration · Representation learning · Robotic perception · Mars rovers

1 Introduction

The primary goal for exploring Mars is to collect data pertaining to the planet’s geology and climate, identify potential biological markers to find evidence of past or present life, and study the planet in preparation for eventual human exploration [15]. As it is currently infeasible for humans to do this work, autonomous rovers have emerged as the primary means to explore the Martian surface and collect images and other data. The deluge of images produced from these rovers — on the order of hundreds of thousands of images — has provided the opportunity to apply deep learning (DL) based computer vision methods to tackle a variety of science and engineering challenges.

Existing DL-based approaches have tackled the aforementioned challenge using supervised (transfer) learning and typically require thousands of annotated images to achieve reasonable performance [20, 21]. Although a manual labeling approach may seem viable, Mars images contain subtle class differences and fine-grained geological features that require highly specialized scientific knowledge and expertise, meaning that scaling manual efforts to the volume of required data is difficult, if not impractical. Several efforts at training citizen scientists to annotate such datasets have been met with some success [18]. However, these logistically complex efforts can introduce inconsistencies that must be resolved by experts [19] to prevent ambiguities/bias.

Self-supervised learning (SSL) can help to circumvent the need for large-scale labeling efforts, since these techniques do not require labeled data to train. Contrastive learning (CL) — a type of SSL — has demonstrated much success in this domain in the past few years, and continues to gain research momentum thanks to the promise of leveraging unlabeled images to achieve state-of-the-art performance on large-scale vision benchmarks [3]. These techniques generally create pseudo-labels by leveraging the intrinsic properties of images, such as prior knowledge that two views (augmented crops) originating from the same source image must belong to the same semantic class. These pseudo-labels are then used in conjunction with a contrastive loss function to help the model learn representations that attract similar (positive) samples and repel different (negative) samples, and in doing so, identify an image’s defining visual features.

The contrastive loss function implicitly assumes that an image and its views define a unique semantic classes. While this implicit assumption (herein referred to as the *contrastive assumption*) may be practical in diverse vision datasets with a large number of balanced classes, it is problematic for planetary science applications due to the homogeneity of planetary images. This, in combination with small batch sizes, increases the probability that multiple source images within the same batch are semantically similar, which further invalidates this assumption. Mars rover images, for example, are regularly taken immediately after one another, and 90% of their content may be overlapping. This results in the incorrect assignment of pseudo-labels, diminishing performance overall. **In short, any two images from a Mars rover dataset may not necessarily provide the contrast required for effective contrastive learning.**

In this work, we demonstrate how contrastive learning can be used to extract useful features from Mars rover images without the need for labels. We also

explore how simple modifications to the contrastive learning process can improve performance when the contrastive assumption is violated, without introducing the need for any human supervision. Our key contributions are thus:

1. An improved method of contrastive training with a cluster-aware approach that improves the contrastive loss formulation.
2. Evidence that mixed-domain datasets can markedly improve the performance of downstream vision tasks by increasing semantic diversity.
3. Demonstrations of how SSL followed by supervised fine-tuning/linear evaluation with limited labels can exceed the performance of published supervised baselines on Mars-related vision tasks.

2 Related Work

Supervised Learning for Mars Images. Wagstaff et al. proposed a fully supervised approach to training AlexNet-based [13] Mars classification models [20, 21]. In these works, benchmark datasets were created to validate the classification performance of the trained models: the Mars Science Laboratory (MSL) dataset, and the High Resolution Imaging Science Experiment (HiRISE) dataset. In this work, we use these datasets to benchmark and compare the downstream task performance of our trained feature extractors (more details in Sect. 3). Other supervised approaches have since improved on the initial results reported by Wagstaff et al. by using attention-based models [2].

Semi-supervised Learning for Mars Images. Wang et al. engineered a semi-supervised learning approach tailored for the semantic content of Mars rover images [22]. Their approach ignores problematic (redundant) training samples encountered during contrastive learning by making use of labels. Their proposed multiterm loss function contains both supervised and unsupervised terms, thus creating a semi-supervised approach.

Self-supervised Learning for Mars Images. Panambur et al. extract granular geological terrain information without the use of labels for the purpose of clustering sedimentary textures encountered in 30,000 of Curiosity’s Mast camera (Mastcam) images [16]. They modify a neural network architecture that was originally designed for texture classification so that it can support self-supervised training, and use a metric learning objective that leverages triplet loss to generate a learning signal. The K-nearest neighbors (KNN) algorithm is then used on the embeddings to cluster, and thus support the querying of the data. The results of this deep clustering are validated by planetary scientists, and a new taxonomy for geological classification is presented.

Self-supervised Learning for Earth Observations. Learning based approaches for Earth observation predictions have long been of interest to the space community [11, 14]. Wang et al. introduced the ‘Self-Supervised Learning for Earth Observation, Sentinel-1/2’ (SSL4EO-S12) dataset and illustrate how

SSL can be used to achieve comparable or superior performance to fully supervised counterparts [23]. This work uses techniques such as MoCo [6,9] and DINO [1], the former of which considers contrastive learning as a dictionary lookup problem, and builds a dynamic dictionary solution that leverages a queue with a moving-averaged encoder, and the latter of which leverages the properties of vision transformers [7] trained with large batch sizes.

Contrastive Learning for Mars Images. A Simple Framework for Contrastive Learning (SimCLR, proposed in [3] and improved upon in [4]) has been used to train discriminant and performant feature extractors in a self-supervised manner across many domains. In [8], a deep segmentation network is pretrained on unlabeled images using SimCLR and trained further in a supervised manner on a limited set of labeled segmentation data (only 161 images). This approach outperforms fully supervised learning approaches by 2–10%.

Relevant Improvements to Contrastive Learning. A number of approaches to detect and counteract issues relating to violating the contrastive assumption are outlined in [5,12]. Such techniques generate a support set of views for any given image, and use cosine similarities between the support set images and an incoming view to detect if the potential pair will be a false negative. Techniques that use this manner of false negative detection have demonstrably outperformed standard contrastive learning techniques on the ImageNet dataset [17].

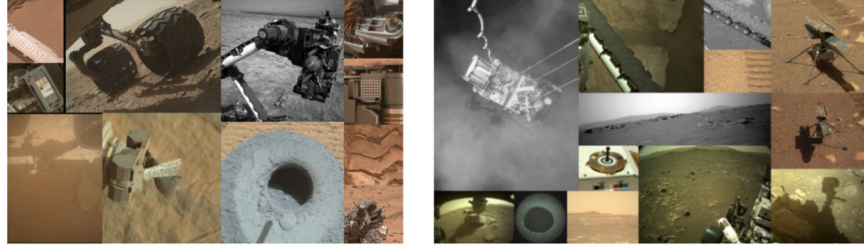
3 Datasets

Table 1. Statistics and descriptions of the datasets used in this work.

Dataset name	Description	Images (or train/test/val.)	Labels
Perseverance	Mars terrain images captured from the Perseverance rover across 19 cameras	112,535	No
Curiosity	Mars terrain images captured from the Curiosity rover across 17 cameras	100,000	No
MSL v2.1 [20,21]	Mars terrain images captured from the Curiosity rover, partitioned into 19 classes	5920/300/600	Yes
HiRISE v3.2 [20,21]	Orbital images of Mars captured from the Mars Reconnaissance Orbiter, partitioned into 8 classes	6997/2025/1793	Yes

3.1 Perseverance Rover Images

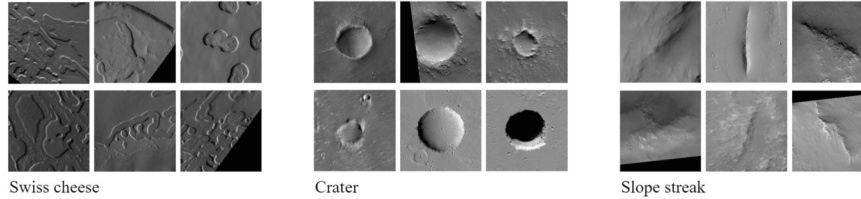
The primary source of training data in this work are unlabeled images from the Perseverance rover. We downloaded 112, 535 Perseverance rover images, captured



(a) Unlabeled Curiosity images (left), and (b) unlabeled Perseverance images (right)



(c) Labeled MSL v2.1 images [21,20], with 3 of the 19 classes shown.



(d) Labeled HiRISE v3.2 images [21,20], with 3 of the 8 classes shown.

Fig. 1. The various datasets used in this work. The two unlabeled datasets (a and b) are entirely unstructured, but can be leveraged by contrastive learning techniques for training feature extractors. The two labeled datasets (c and d) are used for linear evaluation and benchmarking. We note that three of the four datasets are in the domain of Mars rover images (a, b, and c), whereas the HiRISE v3.2 dataset (d) is in the domain of orbital images. Further outlined in Table 1.

during its traversal of the planet Mars from Sols 10–400. These images can be accessed through NASA’s Planetary Data System (PDS)¹. Each image is accompanied by metadata that outlines which of the rover’s 19 different cameras was used to take the image, the time of capture, and more. This results in a set of images that vary in resolution, colour profile, and semantic content. Image contents include the Mars surface, landscapes, terrain features, geologic features, the night sky, astronomical targets, calibration targets, the Perseverance rover itself, and more (as seen in Fig. 1).

¹ <https://pds.nasa.gov/>.

3.2 Curiosity Rover Images

We consider two sets of Curiosity rover images in this work; an unlabeled set, and a labeled set. The unlabeled set consists of 100,000 raw rover images, and is similar in semantic content to the Perseverance rover images. This dataset is used for contrastive learning in the same domain as the Perseverance rover images. The labeled dataset (herein referred to as MSL v2.1) was compiled and annotated by Wagstaff et al.’s [20,21] and consists of 6820 images divided into 19 classes of interest.

The MSL v2.1 dataset is further divided by Sol into train, validation, and test subsets (see Table 1). We use MSL v2.1’s testing subset to benchmark the performance of the feature extraction networks that we train with contrastive learning, thus permitting the comparison of our results with the fully supervised approaches outlined in Wagstaff et al.’s work. We note that the MSL v2.1 dataset is imbalanced: one of the 19 classes (entitled ‘nearby surface’) accounts for 34.76% of the images, whereas the ‘arm cover’ class accounts for only 0.34% of the images.

3.3 Mars Reconnaissance Orbiter Images

We additionally use a labeled dataset of Mars images taken from the Mars Reconnaissance Orbiter’s HiRISE camera, again compiled and annotated by Wagstaff et al. and split into train, validation, and test subsets (see Table 1). This dataset (herein referred to as HiRISE v3.2) is again used entirely for benchmarking, but instead focuses on a separate visual domain (orbital images of Mars rather than rover-based images of Mars). Similarly to MSL v2.1, the HiRISE v3.2 dataset is imbalanced: one of the 8 classes (entitled ‘other’) accounts for 81.39% of the images, whereas the ‘impact ejecta’ class accounts for only 0.68% of the images.

4 Methods

4.1 Contrastive Learning

Our baseline method in this work is based on SimCLR (v2) [3,4], which is a contrastive learning framework that enables deep learning models to learn efficient representations without the need for labeled images. We introduce two approaches to address violations of the contrastive assumption (illustrated in Fig. 2). The first approach modifies the SimCLR framework to use the results from an unsupervised clustering process to identify violations, and the second modification addresses the same issue by altering the distribution of the data that the model ingests.

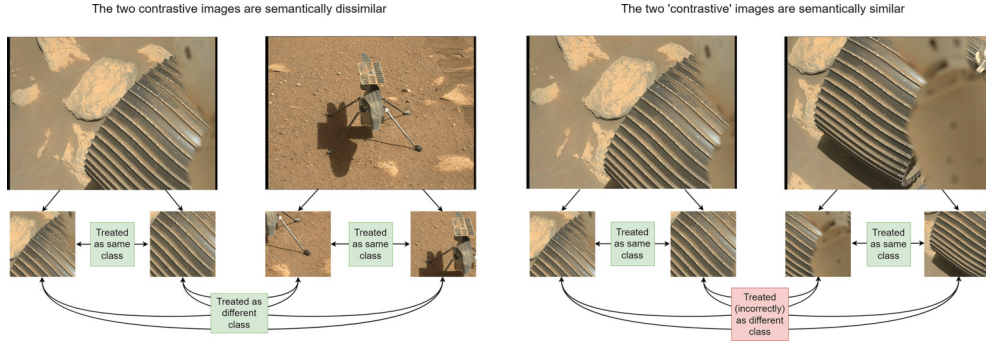


Fig. 2. The contrastive assumption is being held (left), as the two sampled images *are* in visual contrast. The contrastive assumption is being violated (right) as the two sampled images are not in contrast, *but they are assumed to be by the baseline contrastive learning framework*. The violating pairs are thus given an incorrect pseudo-label during training, causing reduced performance. This issue is exacerbated as more semantically similar images are sampled into a training batch.

Whereas traditional supervised learning techniques attach a manually generated annotation or label to an input image, SimCLR operates by automatically annotating pairs of images. Two views from each source image in a batch are taken and augmented. All pairs of views are analysed and labeled as a *positive* pair if they came from the same source image, otherwise they are labeled as a *negative* pair. This work follows SimCLR in generating augmented views of each training image by taking a random crop of the source image and applying the following transformations at random: horizontal flipping, color jittering, and grayscaling [4].

A neural network then completes a forward pass over each of the views in the batch, creating a batch of embedding vectors. For any given positive pair of views, we know that their embeddings should be similar, and for any given negative pair of views, their embeddings should be dissimilar. In practice, cosine similarity is used as the similarity measure for the embedding vectors, and this defines an objective which can be optimised using gradient descent (Eq. 1), thus allowing the network to be trained.

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\text{sim}(z_i, z_j) / \tau)}{\sum_{k=1}^M \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k) / \tau)} \quad (1)$$

where z_x refers to the x^{th} view's representation, $\text{sim}(u, v)$ is the cosine similarity of some embedding vectors u and v , B is the batch size, M is the number of views in the batch ($M = 2 \cdot B$), τ is a temperature parameter, and i, j , and k are view indices. Note how identical views ($k = i$) are ignored when calculating the loss.

4.2 Cluster-Aware Contrastive Learning

The assumptions of standard SimCLR fail when two views from different images actually contain the same or highly similar visual content. In this case, they will be labeled incorrectly as a negative pair; they are a *false negative* (see Fig. 2, right). Our first modification to SimCLR addresses this using prior information relating to if two images actually contain similar visual content.

We gather this information using the unsupervised clustering technique outlined in [16], based on a ResNet-18 backbone. We input the unlabeled training dataset, and the output is a partition of this dataset into some manually defined number of clusters K . Clusters are defined by the similarity of their semantic content, with a specific focus on geological texture as a result of the deep texture encoding module that is incorporated into the architecture [25]. Since this clustering approach is unsupervised, the end-to-end contrastive training process still requires no supervision.

During training, we define positive view pairs as views which came from the same source image *or cluster*. This has the effect of turning false negatives into true positives; taking view pairs that would have otherwise been incorrectly labeled during training, and converting them into correctly labeled instances.

$$\mathcal{L}_{\text{cluster-aware}} = -\log \frac{\exp(\text{sim}(z_i, z_j) / \tau)}{\sum_{k=1}^M \mathbb{1}_{[c_k \neq c_i]} \exp(\text{sim}(z_i, z_k) / \tau)} \quad (2)$$

The modified loss function for the cluster-aware contrastive learning method is outlined in Eq. 2, where c_x refers to the cluster index of the x^{th} view in the batch. The key difference being that view pairs are considered negative based on their cluster index, rather than their view index.

4.3 Mixed-Domain Contrastive Learning

We observe that the rate of false negative view pairs is related to the semantic homogeneity of a training dataset, and as such, we hypothesize that increasing the visual variance of the dataset should decrease the rate of false negatives encountered during contrastive learning. We achieve this by simply concatenating and shuffling two visually different datasets and then performing baseline SimCLR contrastive training on the resultant dataset.

However, mixed-domain training may also have negative side-effects in that injecting too many out-of-domain images may ultimately reduce performance on Mars-related vision tasks. In this study, to achieve a reduced rate of false negative view pairs while preserving a sufficient amount of in-domain images, we perform mixed-domain contrastive learning by combining unlabeled Curiosity images and ImageNet images during pretraining.

4.4 Benchmarking Learned Feature Extraction with Linear Evaluation

Both SimCLR and our modified variants take a set of unlabeled training images and train a neural network to extract discriminant features from such images. Naturally, it is our desire to quantify the success of this training paradigm, and the resultant feature extractor. To do so, we apply the contrastively-trained backbone to supervised Mars-related vision tasks, namely MSL v2.1 and HiRISE v3.2, which represent in-domain and out-of-domain challenges respectively (relative to the unlabeled training dataset which is comprised of Mars rover images).

When benchmarking, we precompute the set of features for the benchmark images using the contrastively-trained model, and then use a single dense/linear layer of 128 neurons to learn a mapping from the features to the images' class. We train this evaluation layer using the features we extracted from the MSL v2.1 and HiRISE v3.2 training subsets and their corresponding labels, before benchmarking the performance on the datasets' corresponding testing subsets. Importantly, the parameters of the contrastively-trained model remain untouched in this process; we are only testing how useful the extracted features are when completing downstream Mars-related vision tasks. This form of linear evaluation protocol is consistent with other contrastive-learning works [3, 4, 6, 9, 23].

5 Results and Discussion

5.1 Adhering to the Contrastive Assumption Strengthens Performance

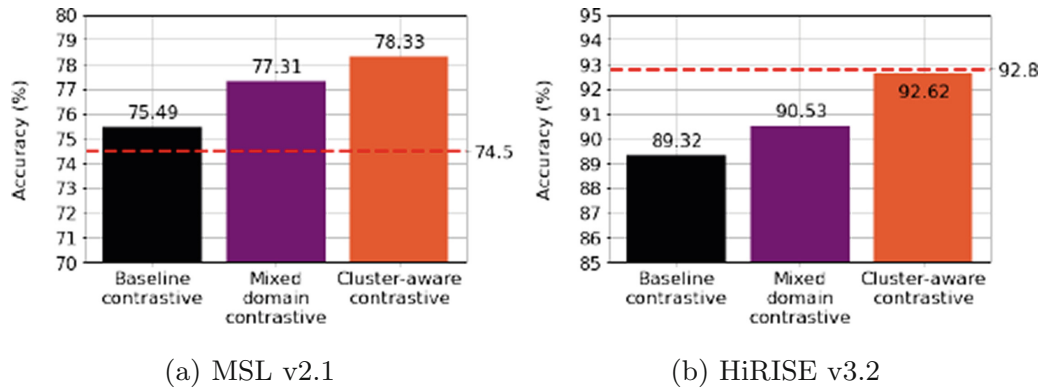


Fig. 3. Performance improvements with mixed-domain contrastive learning and cluster-aware contrastive learning. With a pretraining process that requires no labels, contrastive pretraining generally outperforms the supervised transfer learning baseline (marked with a dotted line) with a model one third the size.

We began by investigating how addressing violations of the contrastive assumption improve the quality of learned features. We trained a feature extractor using the baseline SimCLR and the two modified SimCLR approaches on the unlabeled Perseverance and Curiosity datasets outlined in Sect. 3, and then compared their performance on the MSL v2.1 classification benchmark, with the results outlined in Fig. 3.

Figure 3 shows that each of the contrastive learning approaches (baseline, mixed-domain, cluster-aware) exceeds Wagstaff et al.’s published baseline results (obtained using supervised transfer learning) on the MSL v2.1 benchmark [20].

We note that other published techniques do outperform our model on the MSL v2.1 classification benchmark, but these approaches incorporate labels into a semi-supervised learning approach (95.86% test accuracy on MSL v2.1) [22], or take a fully supervised approach with attention-based models (81.53% test accuracy on MSL v2.1) [2].

Importantly, no labels were required to train our contrastive feature extractors, whereas Wagstaff et al.’s model required a supervised dataset in addition to 1M+ labeled ImageNet images (for pretraining). Moreover, we note that Wagstaff et al.’s fully supervised approach used a neural network backbone with approximately 60M parameters (AlexNet [13]), whereas our approaches use a backbone with approximately 21M parameters (ResNet-50 [10, 24]).

We see that the baseline contrastive approach is outperformed by the mixed-domain approach, which is then outperformed by the cluster-aware approach. A potential explanation for this is that the mixed-domain approach *minimises* contrastive assumption violations, and the cluster-aware approach *corrects* said violations, hence the performance difference.

We note that the performance gains associated with our proposed modified contrastive learning approaches are comparable to those reported in literature using similar approaches designed to mitigate incorrectly labeled view pairs during contrastive training. These works report accuracy increases ranging from approximately 3–4% [5, 12].

5.2 Learned Features Generalise to Out-of-domain Tasks

We extended our testing to the out-of-domain HiRISE v3.2 benchmark — a classification task comprised entirely of orbital images of Mars, reporting the results in Fig. 3b. Wagstaff et al.’s fully supervised model was trained on HiRISE data (orbital images), whereas our contrastive models were trained on Perseverance data (rover images) — the HiRISE v3.2 task is thus out-of-domain for our contrastive approaches, but in-domain for the fully supervised technique.

We observe that the fully supervised technique outperforms our contrastive approaches, likely due to the in-domain/out-of-domain training difference. Regardless, the contrastive methods still demonstrate comparable performance to a fully supervised approach with only a linear classification head, thus illustrating the general nature of the learned feature extraction; performance is maintained even under a domain shift and without any fine tuning of the feature extractor’s parameters. Put simply: our model has never encountered an orbital

image of Mars during training, yet still performs competitively with a supervised model that was trained specifically on Martian orbital images.

As in Sect. 5.1, we observe a performance increase from the baseline contrastive method, to the mixed-domain method, to the cluster-aware method.

5.3 Mixed-Domain Approaches Increase Dataset Variability and Performance

The ‘mixed domain contrastive’ results in Fig. 3 indicate that increasing the semantic heterogeneity of the training data by mixing datasets from different visual domains increases the performance of the trained feature extractors with respect to downstream classification tasks.

Table 2. Analysis of mixed-domain contrastive learning

Domain(s)/training datasets	FN per batch (%)	MSL Acc. (%)	HiRISE Acc. (%)
100K Curiosity	15.0	75.60	89.70
100K ImageNet	0.08	76.54	91.89
50K Curiosity + 50K ImageNet	0.30	77.31	90.53

Table 2 quantifies this effect; each row represents a different training set (or combination of training sets) that were used during training. The first two rows represent baseline contrastive learning, and the last row represents *mixed-domain* contrastive learning — as the training dataset is a mix of two domains. The *number* of training images remains constant in all cases. More details on this simulation are provided in this work’s supplementary file. We expect that performance will be worst when the model is trained on pure datasets that are highly semantically homogeneous (e.g. 100,000 Curiosity images). We also expect the inverse to be true — diverse datasets (e.g. ImageNet) will result in a more performant model, as the contrastive assumption will be violated less often during training.

Moreover, we expect to find a trade off in performance; training on data that is similar to the benchmark data will result in increased performance with respect to that benchmark, but performance will still be hampered due to semantic homogeneity. A mix of two domains, one which shares the domain of the benchmark dataset, and one which provides semantic heterogeneity, should then result in increased performance.

Column two of Table 2 shows the average percentage of false negatives view pairs out of all the view pairs encountered in a given training batch. These results are derived from a simulation which samples multiple batches and checks for FNs using labels. Note that the more diverse dataset (ImageNet) has a far lower false negative view pairs rate than the less diverse Curiosity dataset, and mixing these datasets 50–50 results in a FN rate equal to a ‘diversity-weighted’

average of the two constituent datasets (ImageNet is far more diverse than the Curiosity images, with 1000 classes and 19 classes respectively).

Column three shows our results on the MSL v2.1 benchmark. We note that the model trained on pure Curiosity data (i.e., the same domain as the benchmark dataset) is the *least* performant model. Training on pure ImageNet increases the performance further, and training on an equal mix of both domains (i.e., mixed-domain contrastive learning) provides the best performance.

In column four, we see equivalent results, but on a benchmark domain which is *not* represented in any training dataset (HiRISE v3.2’s orbital images). In this case, the benefit of having in-domain training images (i.e. Curiosity images) is removed; the most semantically heterogenous training dataset (ImageNet) provides the best performance and out-of-domain generalisation. In fact, using any Curiosity images in this case *reduces performance*. This raises a useful conclusion: if no data from the target testing domain is available, use the most semantically heterogeneous dataset available (when using contrastive learning). If data from the target testing domain is available, then using a mix of training data may provide increased performance.

6 Ablation Studies

6.1 10% of the Labels Is Sufficient to Transfer to Other Tasks

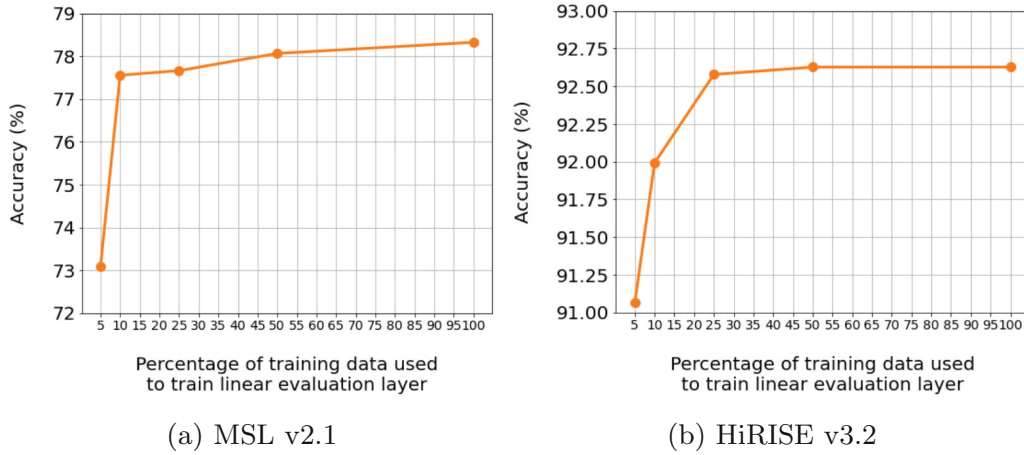


Fig. 4. The task performance as a function of different percentages of training data used during linear evaluation. In all cases, we pretrain on the full unlabeled Perseverance dataset using our proposed cluster-aware contrastive learning, then perform linear evaluation with limited task-specific training data, then benchmark and report the results on the task’s testing set. We note that the accuracy return for increasing amounts of training data quickly diminishes.

Expanding on the results outlined in Sect. 6.2, we observe in Fig. 4 that performance remains relatively unchanged even when a fractional amount of the

benchmark training data is used. On the MSL v2.1 benchmark, accuracy is within 1% of the best score even when only 10% of the task-specific training data is being used, though a distinct performance drop is noted when this proportion is reduced to 5%. On the HiRISE v3.2 benchmark, accuracy is within 0.5% of the best score even at 25% of the task-specific training data. The relatively higher proportion of training data required on the out-of-domain HiRISE v3.2 classification task may suggest that more training data is required to specialise to an out-of-domain task.

Overall, these results suggest that the majority of the learning has occurred during contrastive training and without labels, necessitating only a small number of task-specific labels to leverage the features extracted by the contrastively-trained neural network. This demonstrates key benefits of contrastive learning; the resulting models are relatively general and can be optimised for specific tasks with only a limited number of training samples (especially when compared to a fully supervised training pipeline).

6.2 Cluster-Aware Contrastive Learning Is Sensitive to the Number of Clusters Chosen

We altered the number of clusters (K) during the unsupervised clustering stage of our cluster-aware contrastive approach and benchmarked the results.

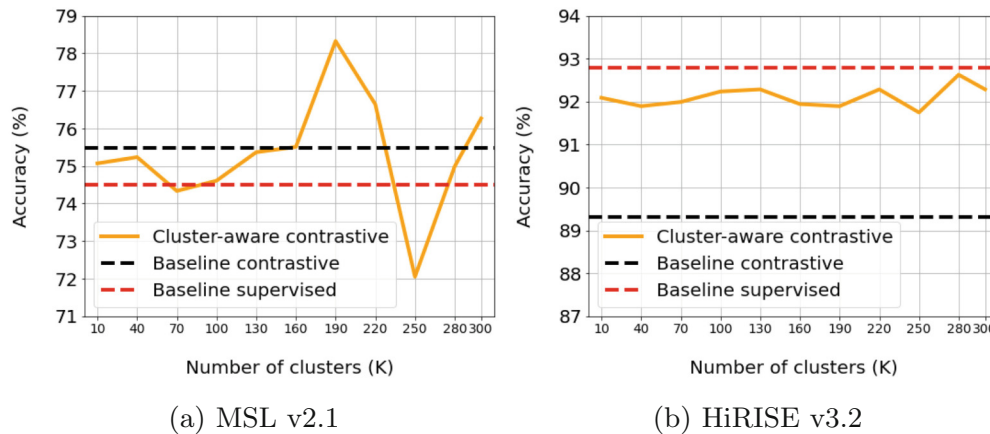


Fig. 5. The performance of the contrastively-trained feature extractors, as measured with respect to an in-domain (left) and out-of-domain (right) downstream task, is dependent on the number of clusters computed prior to training. Too few clusters and images that are dissimilar in content may be assigned to the same cluster, too many clusters and images that are similar in content may be assigned to different clusters — both will cause incorrect pseudo-labels during contrastive training.

We note that the hyperparameter K requires careful optimisation, otherwise the cluster-aware method may not improve (and may decrease) performance. We posit that too few clusters may fail to adequately address violations of the

contrastive assumption; images with different semantic content may be placed into the same clusters due to the coarseness of the clustering, thus failing to solve the issue at hand. With too many clusters, the effectiveness of the method may diminish, with only weakly contrasting images being labeled as negatives.

We observe that for the out-of-domain task HiRISE v3.2, the tuning of the hyperparameter K is less important, with any choice of K producing meaningful performance gains when compared to the baseline contrastive case. We hypothesize that our modified contrastive methods only attend to features inherent to the types of images that were encountered during contrastive training, thus resulting in nuances of orbital images being ignored.

7 Conclusion

In this work, we have explored and quantified a common issue native to contrastive learning on semantically homogeneous datasets; violations of the contrastive assumption leading to false negative view pairs. We analysed this issue within the purview of Mars images taken from the rovers Curiosity and Perseverance, and from the Mars Reconnaissance Orbiter, and we proposed two modifications to contrastive learning tailored for this domain.

Our experiments demonstrate how these modifications — cluster-aware and mixed-domain contrastive learning — improve upon baseline contrastive learning by training more discriminant and powerful feature extractors. In each case, we compare our results to a fully supervised baseline, and note that for in-domain classification tasks our methods result in feature extractors that exceed the performance of their fully supervised counterparts, even when using only 10% of the available labels. It is our hope that this work illustrates how the benefits of contrastive learning can be applied to the large scale space datasets — even when the underlying visual data is semantically homogeneous or not sufficiently contrastive — thus increasing the scope for contrastive learning in the domain of space images.

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References

1. Caron, M., et al.: Emerging properties in self-supervised vision transformers. In: Proceedings of the International Conference on Computer Vision (ICCV) (2021)
2. Chakravarthy, A.S., Roy, R., Ravirathinam, P.: MRSCAtt: a spatio-channel attention-guided network for mars rover image classification. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1961–1970 (2021)

3. Chen, T., Kornblith, S., Norouzi, M., Hinton, G.: A simple framework for contrastive learning of visual representations. In: International Conference on Machine Learning, pp. 1597–1607. PMLR (2020)
4. Chen, T., Kornblith, S., Swersky, K., Norouzi, M., Hinton, G.: Big self-supervised models are strong semi-supervised learners. arXiv preprint [arXiv:2006.10029](https://arxiv.org/abs/2006.10029) (2020)
5. Chen, T.S., Hung, W.C., Tseng, H.Y., Chien, S.Y., Yang, M.H.: Incremental false negative detection for contrastive learning. arXiv preprint [arXiv:2106.03719](https://arxiv.org/abs/2106.03719) (2021)
6. Chen, X., Fan, H., Girshick, R., He, K.: Improved baselines with momentum contrastive learning. arXiv preprint [arXiv:2003.04297](https://arxiv.org/abs/2003.04297) (2020)
7. Dosovitskiy, A., et al.: An image is worth 16x16 words: transformers for image recognition at scale. arXiv preprint [arXiv:2010.11929](https://arxiv.org/abs/2010.11929) (2020)
8. Goh, E., Chen, J., Wilson, B.: Mars terrain segmentation with less labels. arXiv preprint [arXiv:2202.00791](https://arxiv.org/abs/2202.00791) (2022)
9. He, K., Fan, H., Wu, Y., Xie, S., Girshick, R.: Momentum contrast for unsupervised visual representation learning. arXiv preprint [arXiv:1911.05722](https://arxiv.org/abs/1911.05722) (2019)
10. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 770–778 (2016)
11. Helber, P., Bischke, B., Dengel, A., Borth, D.: Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE J. Select. Top. Appl. Earth Obser Remote Sens.* **12**(7), 2217–2226 (2019)
12. Huynh, T., Kornblith, S., Walter, M.R., Maire, M., Khademi, M.: Boosting contrastive self-supervised learning with false negative cancellation. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 2785–2795 (2022)
13. Krizhevsky, A., Sutskever, I., Hinton, G.E.: ImageNet classification with deep convolutional neural networks. In: 25th Proceedings of Advances in Neural Information Processing Systems (2012)
14. Kucik, A.S., Meoni, G.: Investigating spiking neural networks for energy-efficient on-board AI applications. a case study in land cover and land use classification. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 2020–2030 (2021)
15. NASA: Mars exploration rover mission goals, <http://mars.nasa.gov/mer/mission/science/goals>
16. Panambur, T., Chakraborty, D., Meyer, M., Milliken, R., Learned-Miller, E., Parente, M.: Self-supervised learning to guide scientifically relevant categorization of Martian terrain images. arXiv preprint [arXiv:2204.09854](https://arxiv.org/abs/2204.09854) (2022)
17. Russakovsky, O., et al.: ImageNet large scale visual recognition challenge. *Int. J. Comput. Vision* **115**(3), 211–252 (2015)
18. Sprinks, J.C., Wardlaw, J., Houghton, R., Bamford, S., Marsh, S.: Mars in motion: an online citizen science platform looking for changes on the surface of mars. In: AAS/Division for Planetary Sciences Meeting Abstracts# 48. vol. 48, pp. 426–401 (2016)
19. Swan, R.M., et al.: Ai4mars: a dataset for terrain-aware autonomous driving on mars. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1982–1991 (2021)
20. Wagstaff, K., et al.: Mars image content classification: three years of NASA deployment and recent advances. arXiv preprint [arXiv:2102.05011](https://arxiv.org/abs/2102.05011) (2021)
21. Wagstaff, K.L., Lu, Y., Stanboli, A., Grimes, K., Gowda, T., Padams, J.: Deep mars: CNN classification of mars imagery for the PDS imaging atlas. In: Thirty-Second AAAI Conference on Artificial Intelligence (2018)

22. Wang, W., Lin, L., Fan, Z., Liu, J.: Semi-supervised learning for mars imagery classification. In: 2021 IEEE International Conference on Image Processing (ICIP), pp. 499–503. IEEE (2021)
23. Wang, Y., Braham, N.A.A.A., Albrecht, C.M., Xiong, Z., Liu, C., Zhu, X.X.: Ssl4eo-s12: a large-scale multimodal multitemporal dataset for self-supervised learning in earth observation (2022)
24. Wightman, R.: Pytorch image models (2019). <http://github.com/rwightman/pytorch-image-models>. <https://doi.org/10.5281/zenodo.4414861>
25. Xue, J., Zhang, H., Dana, K.: Deep texture manifold for ground terrain recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 558–567 (2018)

Selected Publication 3

SURVEY

Security Considerations in AI-Robotics: A Survey of Current Methods, Challenges, and Opportunities

**SUBASH NEUPANE^{ID}, SHASWATA MITRA, IVAN A. FERNANDEZ, SWAYAMJIT SAHA^{ID},
SUDIP MITTAL^{ID}, (Member, IEEE), JINGDAO CHEN^{ID}, (Member, IEEE),
NISHA PILLAI^{ID}, (Member, IEEE), AND SHAHRAM RAHIMI, (Member, IEEE)**

Department of Computer Science and Engineering, Mississippi State University, Starkville, MS 39762, USA

Corresponding author: Subash Neupane (sn922@msstate.edu)

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ABSTRACT Robotics and Artificial Intelligence (AI) have been inextricably intertwined since their inception. Today, AI-Robotics systems have become an integral part of our daily lives, from robotic vacuum cleaners to semi-autonomous cars. These systems are built upon three fundamental architectural elements: *perception, navigation and planning*, and *control*. However, while the integration of AI in Robotics systems has enhanced the quality of our lives, it has also presented a serious problem - these systems are vulnerable to security attacks. The physical components, algorithms, and data that makeup AI-Robotics systems can be exploited by malicious actors, potentially leading to dire consequences. Motivated by the need to address the security concerns in AI-Robotics systems, this paper presents a comprehensive survey and taxonomy across three dimensions: *attack surfaces, ethical and legal concerns*, and *Human-Robot Interaction (HRI) security*. Our goal is to provide readers, developers and other stakeholders with a holistic understanding of these areas to enhance the overall AI-Robotics system security. We begin by identifying potential attack surfaces and provide mitigating defensive strategies. We then delve into ethical issues, such as dependency and psychological impact, as well as the legal concerns regarding accountability for these systems. Besides, emerging trends such as HRI are discussed, considering privacy, integrity, safety, trustworthiness, and explainability concerns. Finally, we present our vision for future research directions in this dynamic and promising field.

INDEX TERMS AI-robotics, cybersecurity, attack surfaces, ethical and legal concerns, human-robot interaction (HRI) security.

I. INTRODUCTION AND MOTIVATION

Robotics is a multidisciplinary field of engineering that deals with the design, construction, and operation of intelligent machines called robots, capable of sensing the environment and executing tasks autonomously or through human guidance. Robotic systems are comprised of sensors and actuators that can interact with the environment to accomplish a set of tasks [1]. The field has made remarkable strides in reshaping industries such as transportation, manufacturing, healthcare,

agriculture, entertainment, space exploration, military, and beyond. On the other hand, Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems. In this context, AI-Robotics systems may be referred to as integration and collaboration between AI and robotic technologies. These systems combine AI algorithms, which enable machines to perform tasks that typically require human intelligence, with robotic platforms designed to interact with and manipulate the physical world. The recent advances in Machine Learning (ML) and Deep Learning (DL) techniques have paved the way for the development of robots with cognitive abilities to execute functions

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such as *perception, navigation and planning*, and *control*. This empowerment enables robots to perceive, learn, deduce, and make real-time decisions, closely mimicking human behaviors. With ML/DL algorithms continuing to advance, the synergy with robotics propels us toward a future where robots play pivotal roles in societies, offering unprecedented levels of automation, efficiency, and adaptability.

As AI-Robotics systems become increasingly sophisticated, versatile, and integrated into our daily lives, it also revolutionize how we work, live, and explore the world. For example, in the domain of *autonomous vehicles and transportation*, self-driving cars [2], drones, and Unmanned Aerial Vehicles (UAVs) are revolutionizing transportation systems. Similarly, in *industrial automation and manufacturing processes*, to improve efficiency, productivity, and safety- collaborative robots [3] perform tasks such as assembly, welding, material handling, and quality inspection. In *healthcare*, AI-Robotics systems have also demonstrated their potential in revolutionizing surgery as evidenced by Smart Tissue Autonomous Robot (STAR) [4], and rehabilitation processes as discussed in [5]. In *agriculture*, where they assist in diverse tasks like planting, harvesting, and crop health monitoring, enabling the realization of precision agriculture [6], [7], [8]. Alongside day-to-day civilian activities, defense and military operations are equally utilizing AI-Robotics systems to improve strategic capabilities while minimizing human exposure to danger [9]. From autonomous surveillance drones that enhance situational awareness and warfare missions to robotic vehicles for logistics transportation and reconnaissance in hazardous environments, AI-Robotics is revolutionizing how military forces plan, execute, and adapt.

The security of robotic systems has been an active research paradigm for a long time [10], [11], [12], [13], [14], [15], [16], [17], [18]. Due to their pervasive nature, AI-Robotics systems are susceptible to diverse security attacks by threat actors aiming to exploit vulnerabilities in the underlying AI technologies. These attacks are capable of compromising the functionality, safety, and reliability of a system. Adversarial attacks, data breaches, and system manipulations can lead AI-Robotics systems to cause accidents, property damage, and even endanger human lives.

Recently, there have been several examples of adversarial attacks on semi-autonomous vehicles that demonstrate the security risk in AI-Robotics. In 2020, researchers manipulated Tesla cars into speeding up by 50 miles per hour by subtly altering a speed limit sign on the side of a road, fooling the car's Mobileye EyeQ3 camera system [19]. While, in 2015, a Jeep Cherokee got remotely hacked while being driven, causing Fiat Chrysler to recall 1.4 million vehicles and pay \$105 million in fines to the National Highway Traffic and Safety Administration [20]. The vulnerability occurred due to a lack of code signing in the head unit of the car or Electronic Control Unit (ECU), allowing the gateway chip to get reprogrammed and send instructions to vehicle components, such as the steering wheel and brakes [21]. Furthermore,

attacks on *confidentiality, privacy, integrity*, and *availability* can affect trust and acceptance among users [10], [22], [23], [24]. Therefore, understanding the security dimension of AI-Robotics systems and familiarizing oneself with the existing methods to defend against in the event of an attack holds the utmost importance.

Furthermore, the integration of robots into various aspects of our lives has given rise to pressing *ethical and legal* concerns. These concerns extend beyond security attacks and are intertwined with the growing cognitive capabilities of AI-Robotics systems. Ethically, the influence of robots on human behavior and emotions is a significant issue. In scenarios involving medical, social, and assistive robots, the potential for robots to form emotional bonds with users raises questions about the risk of dependence on machines and their psychological impact on individuals [25]. Also, the overt reliance on autonomous robotic agents for decision-making processes can have negative impacts on human cognitive thinking and reasoning capacities, ultimately leading to a loss of human autonomy [26]. On the legal front, navigating liability and responsibility within robotic systems presents a complex challenge. For instance, determining accountability when a "remote-assisted telesurgery" procedure goes awry requires careful consideration. Moreover, the exchange of personal data and information between robots and humans raises data privacy concerns, necessitating adherence to legal frameworks like the 2020 California Consumer Privacy Act (CCPA) [27] and the 2016 General Data Protection Regulation (GDPR) [28].

Besides, growing cognitive capabilities of AI-Robotics systems have allowed robots to have direct cognitive interactions with humans' raising security concerns over Human-Robot Interaction (HRI). The ethical and legal dimensions of these interactions revolve around integrity, privacy, and safety. Addressing these ethical and legal intricacies is paramount to ensuring responsible, respectful, and secure interactions between robots and humans.

In contrast to prior surveys that focus on specific security aspects in AI-Robotic systems, ranging from attacks on operating systems, networks, and the physical layer [29], information faults [30], to ensuring secure communication in IoTs [30], [31], and securing trust between operators and cobots in automation and construction domains [32], our paper sets itself apart by seamlessly integrating *attack surfaces and mitigation strategies, ethical and legal concerns*, and *HRI security* dimensions, providing a comprehensive review across these critical areas. On a taxonomic front, Jahan et al., [33] employs a classification approach akin to our paper, albeit with a focus solely on autonomous systems.

To the best of our knowledge, this paper is the first to dissect the AI-enabled attack surfaces in a generic hybrid AI-Robotics architecture 1. This unique perspective enhances the overall security understanding of AI-robotics systems. Furthermore, we also investigate the ethical and legal concerns surrounding the accessibility, fairness, trustworthiness,

explainability, and accountability of robotic applications, along with the security studies of HRI.

The major contributions of this paper are as follows:

- We present a taxonomy structured around three dimensions of AI-Robotics systems security: *attack surfaces and mitigation strategies*, *ethical and legal concerns*, and *security of HRI*. This taxonomy is designed to augment the overall security of these systems by offering stakeholders comprehensive insights across these dimensions.
- We provide a comprehensive survey of the attack surfaces within hybrid AI-robotics systems, encompassing architectural layers such as *perception*, *navigation and planning*, and *control*. In addition, we present state-of-the-art defenses aimed at significantly reducing the risk of attacks across these layers.
- We discuss the challenges inherent within AI models, frameworks, software and explain how the existing vulnerabilities within these tools can inadvertently contribute to undermining the integrity of the final robotic systems.
- We present the multifaceted exploration of the ethical and legal dimensions associated with AI-Robotics systems.
- We address the intricate security issues pertaining to HRI, with the focus on privacy concerns. Within this context, we recommend strategic defensive solutions.
- We provide research recommendations organized according to the three dimensions mentioned earlier, with the goal of enhancing the robustness and resilience of future AI-Robotic systems.

The remainder of the sections are as follows: Section II provides background on robotics architecture. Section III summarizes our survey and taxonomy. Following that, we investigate on attack surfaces on various layer of robotic architecture in Section IV. Then, we discuss ethical and legal concerns surrounding the use of AI-Robotic systems in Section V. Section VI discusses security studies pertaining human robot interaction. Section VII identifies research challenges and provides future direction in developing secured robotic systems. Finally, Section VIII concludes our survey.

II. AI-ROBOTICS SYSTEMS ARCHITECTURE

Robotic systems can be designed using a variety of architectural paradigms. In this section, we will explore these paradigms further. Murphy et al. [35] describe a range of current architectures such as *hierarchical*, *reactive*, and *hybrid deliberative-reactive* in terms of the relationships between three primitives (i.e., sense, plan, and act) for organizing intelligence in robots. Table 1 attempts to characterize these three fundamental building blocks based on their inputs and outputs. Hierarchical paradigms in robotics follow a top-down approach, where the robot first senses the world, then plans its next action and finally executes it. A hierarchical

paradigm is also known as a deliberative architecture. Each layer in the hierarchy depends on the layer below it for information and guidance. Reactive paradigms prioritize immediate responses directly connected to sensor input, without extensive planning. Hybrid deliberative-reactive paradigms strike a balance between reactive responses and higher-level decision-making, utilizing both immediate reactions and planned actions based on sensory information.

TABLE 1. Robot primitives defined in terms of input and output. Adapted from [35].

Primitives	Input	Output
Sense	Sensor data from LiDAR, Cameras, GPS etc.	Sensed information such as Obstacle location, Occupancy grids etc.
Plan	Map information	Waypoints, Path, Trajectory
Act	Waypoints	Actuator commands

Modern AI-Robotics systems commonly adopt a hybrid architecture that combines both deliberative and reactive modules. Figure 1 illustrates a generic three-layered hybrid architecture that corresponds to the sense-plan-act paradigm, with *Perception*, *Navigation & Planning*, and *Control* as its constituent layers. In the subsequent subsection, we will provide a detailed explanation of each of these layers.

A. ROBOTIC PERCEPTION

The perception stack of the robot architecture involves sensing and understanding relevant information about the state of the operating environment by leveraging several sensors, for example, GPS, accelerometer, inertial measurement unit (IMU), gyroscope, magnetometer, camera, and tactile sensors among others [36]. The combination of multiple sensors as depicted in Fig 1 make up perception stack. These sensors play a pivotal role in actively sensing the raw data that represent the physical state of robotics systems in the environment. Furthermore, this sensory data aids in planning and calculation of actuator signals in the subsequent stages. [37].

Modern AI-Robotics systems require high-level sensing abilities, such as object detection, localization, positional information, and interaction with humans. An important aspect of achieving successful perception functions in robotics is gaining a deep understanding of the operating environment. One highly influential approach for environment understanding, especially in the context of mobile robots and autonomous vehicles, is the representation of the environment through 2D mappings known as occupancy grid mapping [38]. However, attention has now shifted towards 3D representations. Recent technological advancements, including Octomaps [39], and the widespread use of RGBD sensors with depth capabilities have significantly enhanced the construction of larger and more detailed 3D maps for detailed 3D-like environment representations. Efforts have been focused on semantically labeling these maps by integrating them with Simultaneous Localization and Mapping (SLAM) frameworks. This integration enables

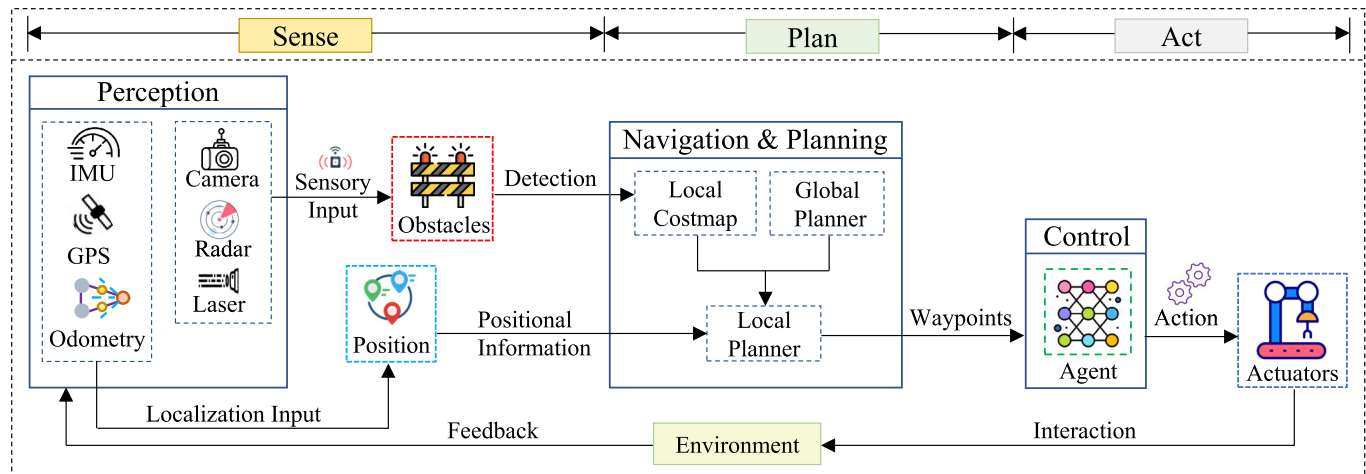


FIGURE 1. A generic robot architecture consisting of perception, navigation, and control stacks. Icons from [34].

autonomous navigation and precise self-localization based on the generated map.

Furthermore, the advances in AI have enabled the gathering, analysis, and prediction of visual, audio, depth, and tactile information that is required for perception functions [40]. Object detection and localization are mostly performed through visual information (image) and depth information using Convolution Neural Network (CNN) architecture [41], [42], [43], and its variants Recurrent CNN [44], [45], and convolutional auto-encoder [46]. More recently, the state-of-the-art Visual Transformer [47] and its variants have also been used to improve perception performance [48], object detection, and tracking [49] among others.

B. ROBOTIC PLANNING AND NAVIGATION

The input to the navigation stack is the sensory information and map obtained by the various sensors from the perception stack [II-A] as depicted in Fig 1. Navigation algorithms are responsible for the generation of a series of waypoints, which are then passed on to the control stack [II-C]. Classic robotic navigation and planning algorithms typically involve explicit rule-based methods and predefined algorithms for path planning and obstacle avoidance. These algorithms often rely on geometric models and predefined maps of the environment. They can be categorized into four groups, including *graph search algorithms* (for example, A* [50]), *sampling-based algorithms* such as Rapidly-exploring Random Tree (RRT [51]), *interpolating curve algorithms* [52], and *reaction-based algorithms* such as Dynamic Window Approach (DWA [53]).

AI/ML algorithms, on the other hand, offer a distinct advantage in object and environment identification as they can autonomously learn patterns and features directly from sensor data, eliminating the need for manual feature engineering or predefined models. This capability has resulted in remarkable enhancements in the performance of robotic navigation and planning, leading to its adoption for generating local costmaps as well as for both global and local

planning tasks, thereby paving the way for more adaptive and efficient robotic systems.

The process of navigating towards a long-term goal is referred to as global planning. Local and global planners work together to achieve navigation goals. Using Deep Reinforcement Learning (DRL) technique, Kastner et al. [54], for example, combine two planners that operate at different scales: a global planner that optimizes long-term navigation by producing high-level waypoints and a local planner that navigates from one-waypoint to the next while avoiding obstacles. On the other hand, local path planning and costmap generation refers to the process of short-term path planning to avoid obstacles and is used in mobile robots, where the navigation is solely based on local landmarks and environmental factors. Costmaps are utilized in objection tasks in small-scale robotic applications. Work that leverages CNN and costmaps can be found in [55], while map-based Deep Reinforcement Learning (DRL) for mobile robot navigation for the obstacle detection task based on a generated local probabilistic costmap is presented in [56]. Deep learning models that employ CNNs for local path planning by circumventing objects for autonomous navigation can be found in the works of [57]. Similarly, Ritcher et al. [58], proposed a combination of a fully connected feed-forward network and an autoencoder to model collision probability, and more recently, Regler et al. [59] presented a DRL-based local planner with a conventional navigation stack, and Chen et al. [60] presented DRL for path planning, dynamic obstacle avoidance, and goal approaching. However, an AI-Robotic systems should also employ navigation algorithms and planning that are responsive to unexpected changes such as grasping novel objects [61] and moving objects [60] in the dynamic environment.

C. ROBOTIC CONTROL

The input to the control stack is the waypoint information from the navigation and planning stack II-B, as illustrated in Fig. 1. The control stack is responsible for generating control

signals by leveraging various algorithms to manipulate the behavior of robots. It involves designing control systems that enable robots to perform specific tasks such as moving actuators based on the planned path given by the planning and navigation stack. The goal of this is to develop algorithms that can effectively and efficiently command the robot's actuators based on sensor feedback.

Some of the classic control algorithms include Proportional-Integral-Derivative (PID) control [62] and Linear-Quadratic-Regulator (LQR) [63]. PID adjusts control signals based on proportional, integral, and derivative terms. It provides stable and robust control by continuously comparing desired and actual states and minimizing the error. LQR on the other hand, is used to control non-linear dynamical systems by computing the optimal control signal needed to move the system to a desired state. These algorithms, while effective for many applications, often rely on explicit mathematical models and predefined control strategies.

In recent years, there has been a significant shift towards leveraging AI to enhance robotic control capabilities. In particular, Deep Reinforcement Learning (DRL), has gained considerable attention in robotic control, including locomotion, manipulation, and autonomous vehicle control [64]. DRL algorithms allow robots to learn control policies through interactions with the environment, receiving rewards or penalties based on their actions. Kalashnikov et al. [65] demonstrated the use of DRL for controlling a robotic arm to grasp and manipulate objects with a high success rates in their study. In another line of work, the authors in [66] leveraged DRL techniques for robot controllers to learn a range of dynamic manipulation behaviors, such as stacking large Lego blocks at a range of locations, threading wooden rings onto a tight-fitting peg, inserting a shoe tree into a shoe, screwing bottle caps onto bottles, and assembling wheels in a toy airplane.

In the next section, we discuss our survey approach and taxonomy.

III. SURVEY APPROACH AND TAXONOMY

This survey builds upon the intricate relationship between physical and digital vulnerabilities within AI-Robotic systems. It adopts a comprehensive approach by providing a taxonomy that explores the existing attack surfaces along with defensive strategies that are relevant to AI-Robotics systems. The primary objective is to understand how various attack vectors impact different components of robotic architecture. To achieve this, we first introduce the hybrid AI-Robotics architecture, comprising three layers: *perception*, *navigation and planning*, and *control* (see Section II). The foundational understanding of different architectural layers of robotic systems allows us to identify the attack surfaces and their impact on these components.

We begin the survey by categorizing the AI-Robotic systems into three dimensions, including *attack surfaces*, *ethical and legal concerns*, and *human-robot interaction security*, as illustrated in our taxonomy (see Fig. 2). The

first category explores attack surfaces and their impact on the architectural components of robotic systems. The second category, on the other hand, discusses the ethical and legal concerns surrounding the use of robotics systems in various domains such as healthcare, the military, autonomous vehicles, manufacturing, and agriculture. Finally, we explore the security landscape of human-robot interaction and recommend defensive strategies to safeguard privacy concerns.

The attack surfaces are divided into three distinct domains: *physical attack*, *digital attack*, and *other relevant attacks*. Physical attack encompasses attacks targeting the perception layer (attack in input sensors, see Section IV-A) within the hybrid robotic architecture. Digital attack, on the other hand, pertains to attacks directed at the navigation and planning, as well as the control layer. The interconnectedness between physical and digital attacks becomes evident, with physical attacks having a direct impact on digital systems. For example, if input sensors such as cameras, LiDAR, or RGBD sensors are tampered with, they could produce poisoned data. When these compromised data (see Section IV-B3a) are used by digital systems like object recognition algorithms, it may lead to misclassifications, thereby affecting navigation and planning (see Section IV-B1). Furthermore, it can also cause mistimed actuation commands within the control systems (see Section IV-B2). On the other hand, the manipulation of physical objects within the environment can introduce a new set of challenges, potentially leading to deception, malfunctioning, or erratic behaviors in robotic systems. A compelling illustration of this is the work by Eykholt et al. [67], who showcased physical attacks using adversarial stickers to deliberately deceive an autonomous vehicle's object recognition system. This illustrates how vulnerabilities in physical components (both environment and sensors) can cascade into the digital realm, causing major disruptions and interruptions in robotic system operation. Furthermore, AI-Robotics systems can also suffer from inherent challenges of AI/ML models itself such as model training attacks (see Section IV-B3a) and model inference attacks (see Section IV-B3b).

The third category within the attack surfaces is other relevant attacks. These include backdoors, malware, ransomware, as well as network and communication-based attacks on robotics systems. We discuss these attacks in greater detail in Section IV-C.

Following this, our focus extends to the ethical and legal concerns (see Section V) surrounding the use of these systems. We explore the ethical dilemmas arising from robotics technology in various domains such as healthcare, autonomous vehicles, assistive robots, military, and manufacturing. The focus is to showcase the importance of ethical considerations and their influence in the development, deployment, and use of AI-Robotics systems in the aforementioned domains. For instance, healthcare robotics systems such as surgical robots must be developed that align with medical ethics, ensuring patient safety and privacy. Another important issue to consider is how to program

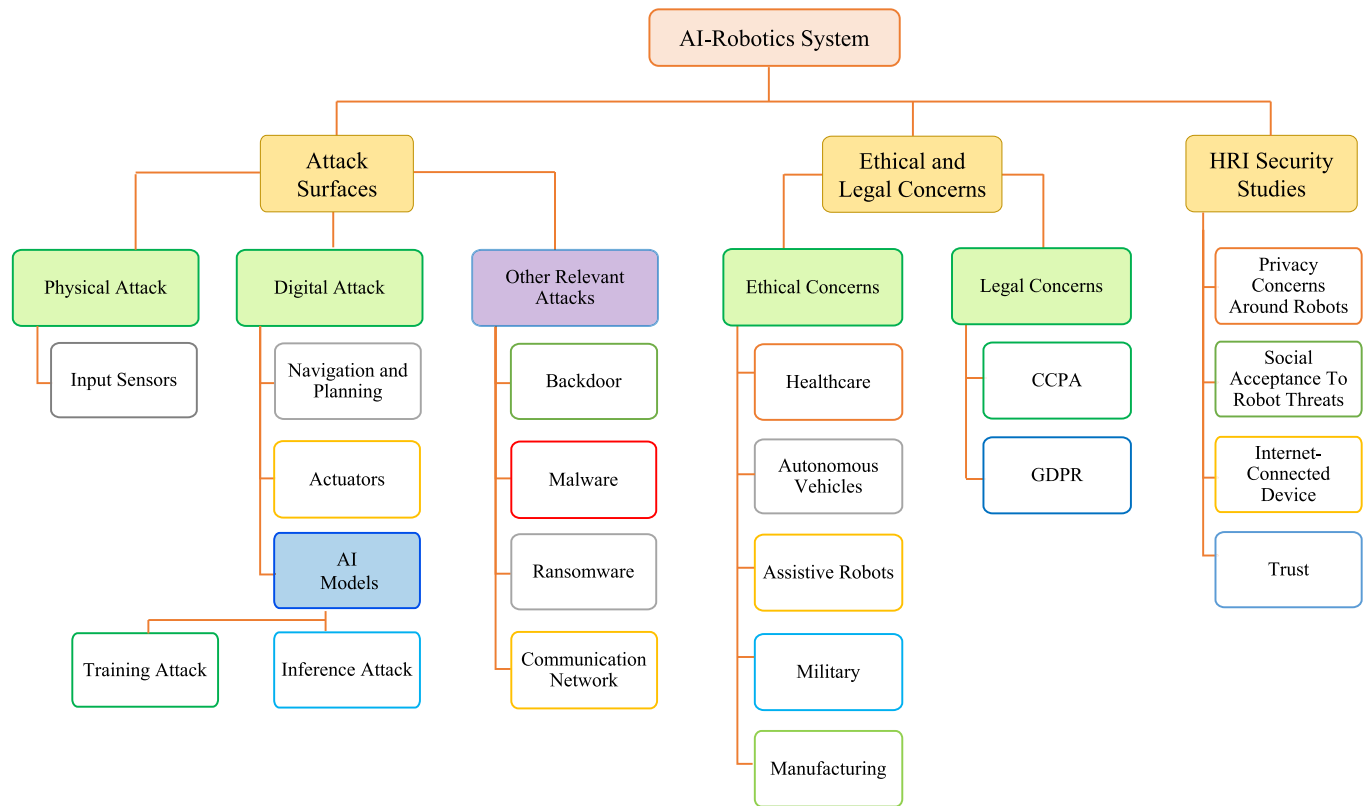


FIGURE 2. An overview of our proposed taxonomy. The central parent node represents the overarching AI Robotic System, from which three primary branches emerge: Attack Surface, Ethical and Legal Concerns, and Human-Robot Interaction. Attack surfaces is further into two families: physical attack and digital attack. Physical attacks encompass attacks on the perception layer (input sensors). Digital attacks include attacks on navigation and planning, attacks on actuators, and attacks on AI models (training and inference attacks).

AI agents in complex ethical scenarios. Examples of this include the trolley problem and the self-sacrifice dilemma as depicted in Fig 7 in relation to Autonomous Vehicles (AVs). Aside from these concerns, the overt reliance on autonomous robotic agents for decision-making processes can have adverse effects on human cognitive thinking and reasoning abilities, ultimately leading to a reduction in human autonomy [26]. From legal point of view, we explore two regulations that ensures privacy protection to personal data. The first one is California Consumer Privacy Act (CCPA [27] in case of the United States (US) and General Data Protection Regulations (GDPR) [28] in case of Europe (EU). Both CCPA and GDPR provide similar privacy protections to consumers.

Furthermore, we expand our survey to encompass the realm of Human-Robot Interaction (HRI) security (see Section VI), delving into the intricacies of physical, cognitive and social interactions between people and robots. We survey current approaches that are aimed at safeguarding interactions between humans and AI-Robotic systems.

The survey of attack surfaces and their defensive solutions based on the taxonomy is available in Section IV while the survey of ethical and legal concerns surrounding AI-Robotic systems is in Section V, and human-robot interaction in Section VI.

IV. ATTACK SURFACES

In the context of robotic systems, an attack surface is a set of potentially vulnerable points on the boundary of a system, a system element, or an environment, through which an adversary attempts to compromise or manipulate the robot's functionalities [68]. Attack surfaces on robotic platforms can be broadly divided into two categories such as *physical attack* and *digital attack* [69] as illustrated in Fig 2.

A. PHYSICAL ATTACK

Physical attacks involve the act of physically manipulating hardware or components with the aim of either disrupting its normal functioning or obtaining unauthorized access. For example, tampering, destruction, removal, theft, or manipulation of embedded sensors disrupt the perception of robotic systems. Humanoid and social robots that interact with human counterparts are susceptible to these attacks, as evidenced by the authors in [70]. On the other hand, hardware-level trojans presented in [71] can manipulate physical components such as integrated circuits and cause robotic systems to malfunction abruptly.

Over the next few paragraphs, we will investigate an array of attack vectors encompassing techniques like jamming and spoofing in relation to input sensors. Our analysis will also extend to how these vectors influence the perception layer within hybrid AI-Robotics systems architecture.

The perception layer is the first component of AI-Robotics architecture. It serves as a vital pillar in AI-Robotics systems, utilizing an array of proprioceptive and exteroceptive sensors. Proprioceptive sensors such as Inertial Measurement Unit (IMU), Global Positioning System (GPS), encoders, etc. are responsible for measuring values internal to the robotic system and exteroceptive sensors such as cameras, Light Detection And Ranging (LiDAR), radar, and others, are responsible for collecting information regarding the robot's environment. These sensors' information is utilized to comprehend and operate through the environments. These sensors constitute physical devices, and any attempts to compromise their integrity fall within the realm of physical attacks, as elaborated in our taxonomy. They play a pivotal role in enabling AI-Robotics systems to create a comprehensive world model, facilitating collision avoidance, precise path and trajectory planning, and accurate location mapping. However, the perception layer becomes a natural target for adversaries, who aim to tamper with or compromise the quality of sensor data [72]. Manipulation of sensor data affects the integrity and reliability of the perception process, thereby degrading the performance of the navigation and planning layers, and ultimately influencing the behaviors in the control layer. A summary of the attack surface, affected sensors, attack summary, and proposed defensive strategies can be found in Table 2.

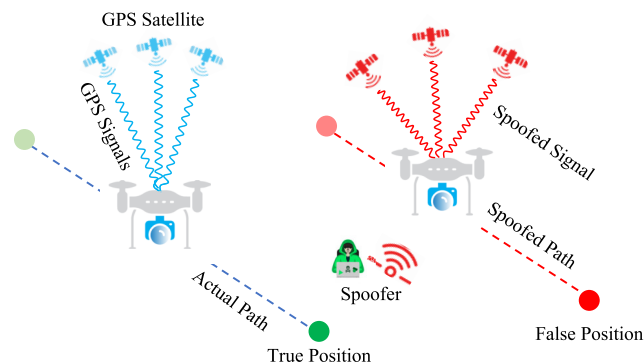


FIGURE 3. An illustration of GPS spoofing attack on UAVs, in which an adversary redirects the UAV to a false location by transmitting deceptive GPS signals.

In the realm of adversarial attacks, adversaries can engage in various tactics such as jamming, spoofing, manipulating, or software vulnerability exploitation for proprioceptive and exteroceptive input sensors, causing disruptions, impairments, and deceptions that hinder the accurate perception of the environment by these sensors.

1) JAMMING ATTACKS

In [73], the authors discuss the use of intense light to jam camera sensors, leading to a significant deterioration in the quality of captured images and subsequently affecting object recognition. Another study in [74] focuses on blinding attacks targeted at LiDAR sensors, where a bright light source with the same wavelength as the LiDAR renders the sensor

ineffective in perceiving objects from the direction of the light source. Similarly, in [75] researchers simulate jamming attacks on ultrasonic sensors, while the vulnerability of GPS signals to jamming devices is investigated in [76]. These attacks could also lead to other types of attacks, such as a Denial-of-Service (DoS) attack wherein the perception sensors become unavailable for authorized users as a result of malicious jamming.

2) SPOOFING ATTACKS

Recent studies suggest adversaries can also spoof LiDAR sensors including three techniques such as laser projection techniques [74], [77], [78], shape manipulation strategies [79], and object manipulation approaches [80]. An example of GPS spoofing attack is presented in [85] wherein a GPS spoofing attack is meticulously designed to manipulate navigation systems. For instance, in Figure 3, a GPS spoofing attack on a UAV is depicted. In this scenario, the attacker manipulates the path trajectory and alters the true position of the UAV by transmitting false or spoofed GPS signals.

3) MANIPULATION ATTACKS

In addition, some studies have brought to light the vulnerability of IMUs employed in robotic systems to targeted attacks. These attacks aim to manipulate the gyroscope, accelerometer, and magnetometer sensors [90]. Such attacks aim to disrupt the accurate measurement of linear and angular velocity, consequently causing the robotic systems to lose control and in some cases deceive object detection tasks [86].

4) SOFTWARE VULNERABILITY ATTACKS

In addition to the aforementioned attacks, robotic systems are susceptible to software vulnerabilities within programs responsible for processing sensor data, such as OpenCV [87], TensorFlow Image [88], and Open3D [89]. Despite their widespread use, these frameworks have certain limitations and vulnerabilities, as they often rely on assumptions regarding the structure and format of sensor data. As a result, if these assumptions are not met, these frameworks may fail [91]. For example, OpenCV has been discovered to be vulnerable to a heap-based buffer overflow (CVE-2018-5268) [92], which can be triggered by parsing a malicious image during the loading process. Similarly, vulnerabilities can arise from the lack of, or incomplete, or invalid validation of input parameters (CVE-2021-37650) [93] in the TensorFlow library, assuming that all records in the examined dataset are strings.

To mitigate blinding attacks on cameras, a defense strategy involves incorporating a near-infrared-cut filter during daytime operations [94]. Also, the utilizing photochromic lenses to selectively filter certain types of light can enhance the overall image quality. In a separate study [76], the authors propose a method for detecting GPS jamming incidents. Similarly, countermeasures against spoofing attacks encompass

TABLE 2. Overview of attacks on input sensors along with defensive strategies.

Attack Surface	Medium	Attack Summary	Defensive Strategy	Research
Jamming	Camera, GPS, LiDAR, Ultrasonic sensors	• Use of intense light to jam camera sensors. • Using bright light source (blinding light) to render LiDAR renders the sensor ineffective in perceiving objects.	• Incorporating a near-infrared-cut filter and photochromic lenses to selectively filter certain types of light.	[73]–[76]
Spoofing	GPS, LiDAR	• GPS spoofing attack designed to manipulate navigation system. • Adversary may spoof LiDAR sensors using techniques such as laser projection [74], [77], [78], shape manipulation [79], and object manipulation [80].	• Filtering [81], Randomization [74], and Fusion [82]. • Anomaly detection systems [83]. • Spoofing detection systems for radars-based spoofing attacks [84].	[85], [74], [77]–[79]
Manipulation	IMUs	• Manipulate the gyroscope, accelerometer, and magnetometer sensors. • Disrupt the accurate measurement of linear and angular velocity and cause errors in control system.	• Inertial sensors safeguarding. • Sensor fusion techniques.	[86]
Software vulnerabilities	OpenCV, TensorFlow Image, Open3D	• Adversary may exploit software vulnerabilities within programs responsible for processing sensor data.	• Data verification and validation. • Data Sanitization. • Patching and updating for known vulnerabilities.	[87]–[89]

techniques such as filtering [81], randomization [74], and fusion [82] to enhance the robustness and reliability of the sensors. Additionally, the advancement of machine learning techniques has led to the development of spoofing detection methods, as demonstrated in [83]. This research paper introduces an anomaly detection approach that employs an auto-encoder architecture specifically designed for LiDAR sensors to identify sensor spoofing attacks. Furthermore, a proposal in [84] outlines a strategy to detect and counter radar-based spoofing attacks by implementing a spatio-temporal challenge-response technique. This method verifies analog domain physical signals, effectively preventing sensor spoofing incidents. Besides, verification and sanity checking of input sensor data should be conducted to ensure input data are free from adversarial inputs.

Next, we delve into digital attacks, where we explore attack patterns on the navigation and planning layers and control layers (actuators) of a hybrid AI-Robotic architecture. Furthermore, we also discuss the inherent challenges of ML models.

B. DIGITAL ATTACK

Digital attack pertains to software-related attacks that aim to exploit the software vulnerability within the robot's operating system, algorithms, firmware, communication channels, or other software component (for example AI-based agents) to manipulate navigation & planning (see Section IV-B1) and control layer (see Section IV-B2) of a robotic systems. The literature also shows AI-specific attacks such as training attacks (see Section IV-B3a) and inference attacks (see Section IV-B3b) that impact robotic systems.

In the subsequent subsections, we present a comprehensive analysis of attack patterns and their associated security strategies within the diverse components of AI-Robotics systems.

1) NAVIGATION AND PLANNING ATTACKS WITH SECURITY STRATEGIES

Attacks on the navigation and planning layer of robotic system architecture present a complex and concerning challenge in the realm of AI, automation, and cybersecurity. Contrary to the perception-oriented attacks through robotic input sensors, where the objective is to tamper with perception sensors. Attacks on navigation and planning relate to manipulating the AI-Robotics system's navigation systems by leveraging the poisoned data generated from compromised sensors. With the rapid integration of autonomous robots into daily human lives, from self-driving cars to industrial automation, from military applications to health care, ensuring the security and integrity of their navigation and planning systems becomes paramount. Table 3 provides an overview of various forms of attacks on the navigation and planning layers of the hybrid AI-Robotics system architecture, along with defense strategies.

Navigational attacks can manifest in various forms, ranging from simple and localized disruptions to controlled, sophisticated, widespread manipulations. One of the most common forms of attack involves sensor spoofing or jamming [95], [96], where adversaries intentionally deceive the robot's sensors, such as GPS, camera, LiDAR, etc, to provide false or distorted information. Doing so can mislead the robot's perception of its environment, leading to potential collisions, misrouting, or compromised decision-making. Furthermore, navigational systems could be vulnerable to data manipulation attacks [97]. These attacks involve altering the mapping or localization data robots use, causing incorrect interpretations of the environment. For example, in an urban setting, this could result in a robot delivering packages to the wrong address or navigating hazardous routes, endangering pedestrians and other vehicles.

TABLE 3. Research summary on adversarial attacks on robotic navigation and planning systems with proposed defense strategies.

Attack Surface	Medium	Attack Summary	Defense Strategies	Research
Navigation	Communication Sensor (GPS, Radio, etc.)	• Transmit spoofed positional data. • Block communication channel.	• Need minimal control input for scope limitation. • Need secured recovery algorithm.	[85], [95], [98]–[104]
	Visual Sensor (Camera, LiDAR, etc.)	• Falsify environment information for incorrect perception. • Manipulate operational environment.	• Incorporate adversarial training. • Need to use advanced robust AI algorithm.	[74], [79], [80], [86], [94], [95], [105]–[108]
	Environmental Force Sensor (IMU, etc.)	• Alters operational environment.	• Reduce single-sensor decision dependency.	[90], [95], [109]–[111]
Planning	Black-box DRL models and planners (Image Segmentation, 3-D point cloud, etc.)	• Exploit and optimize the cost function for controlled degradation.	• Through careful tuning and environment-specific modifications are needed. • Development on non-iterative planning schemes.	[15], [104], [112]–[117]
	Coordination algorithms (Robust [118], [119], Adaptive and reactive [120], [121])	• Break communication for coordinated systems and prevent recovery. • Exploit risk-reward trade-off algorithms.	• Utilize GNN algorithms for decentralized information propagation. • Research on risk/uncertainty-aware search algorithms.	[122], [102]–[104], [123]–[128]

To prevent sensor spoofing by the adversary and enable secure and optimized navigation, Bianchin et al. [98] developed an algorithm to detect spoofed sensor readings and falsified control inputs intended to maximize the traversing trajectory. The authors consider an undetectable attack over Global Navigation Satellite System (GNSS) sensor and a Radio Signal Strength Indicator (RSSI) sensor. The detection algorithm works on the principle of nominal control input. In another work by Yang et al. [99], the authors introduced timing-based adversarial strategies against a DRL-based navigation system by jamming in physical noise patterns on the selected time frames. The attack imposes adversarial timing based on three principle scenarios: pulsed zero-out attack (noise through off-the-shelf hardware), Gaussian average on sensor fusion (noisy sensor fusion system developed by Gaussian filter), and adversarial noise patterns (fast gradient sign method to generate adversarial pattern against trained Deep Q-network (DQN)). Through the attack, both value and policy-based DRL algorithms are easily manipulated by a black-box adversarial attacking agent. As a preventive measure, the authors suggested adapting adversarial training to train DQN and Asynchronous Advantage Actor-Critic (A3C) models by noisy states.

On the other hand, attacks on planning phase are equally disruptive. It targets algorithms responsible for determining a robot's actions and decisions based on sensory information. Adversaries could introduce subtle changes to the robot's decision-making process, causing it to make sub-optimal choices or even hazardous actions [116], [129]. For example, an autonomous drone might be tricked into flying into a restricted area or performing unauthorized surveillance. Such actions can be in the form of a targeted signal or sensor distortion or in the form of data manipulation. In the work of Wang et al. [130], the authors proposed jamming attacks using Reinforcement Learning (RL) to influence the Unmanned Aerial Vehicle (UAV) performance and then developed anti-jamming strategies using

double DQN to mitigate the attack in UAV trajectory planning.

Similarly, Bhandarkar et al. [114] proposed DRL techniques to launch Sybil attacks and transmit spoofed beacon signals to disrupt the path planning logic. Hickling et al. [115], on the other hand, proposed to utilize the explainability of DL methods. In their approach, the planning agent is trained with a Deep Deterministic Policy Gradient (DDPG) with Prioritized Experience Replay (PER) DRL scheme, that utilizes Artificial Potential Field (APF) to improve training time and obstacle avoidance performance. Along the same lines, Peng et al. [116] proposed an Online Path Planning (OPP) framework to continuously update the environmental information for the planner in adversarial settings. For intelligently selecting the best path from the OPP output, the Bayesian network and fuzzy logic are used to quantify the bias of each optimization objective.

Additionally, SLAM is the problem of planning and controlling the motion of a robot to build the most accurate and complete model of the surrounding environment [14]. Active SLAM problems fall under Partially Observable Markov Decision Processes (POMDPs), where both action and observation uncertainties affect navigation and planning. In several cases, adversaries exploit such uncertainty to disrupt the autonomous navigation and planning system. In the work by Wang et al. [117], the authors introduced an infrared light-based attack that can alter environment perception results and introduce SLAM errors to an autonomous vehicle. For defensive measures, the authors developed a software-based detection module. Similarly, Li et al. [128] demonstrated an attack that exploits timing-sensitive multi-modal data from different sensors to cause destabilization of the SLAM perception model in cyber-physical systems. Researchers have emphasized various recommendations to prevent and mitigate beyond-line-of-sight and active spatial perception errors from getting exploited [14] and require active research to mature and achieve real-world impact.

In the next section, we explore attacks on the control layer of hybrid AI-Robotics systems architecture with a focus on attacks on actuators and recommend corresponding security strategies.

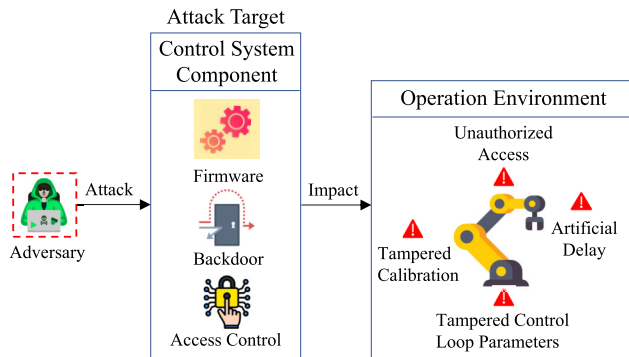


FIGURE 4. An overview of attack surfaces in control systems and their impact on actuators within the operating environment. Icons from [34].

2) ACTUATOR ATTACKS WITH SECURITY STRATEGIES

In the robotic systems architecture, robots act on the environment by leveraging its joint manipulators (in the case of robot arms or humanoids), or wheels (in the case of ground mobile robots) or propellers (in the case of aerial and marine robots). To act, these systems need actuators. An Actuator is a mechanical device that utilizes various sources of energy, such as hydraulics, external signals, pneumatics, and electricity, to convert them into motion that can be manipulated according to specified control parameters [131]. Electric motors, hydraulic actuators, and pneumatic actuators are the most common types of actuators used in robotics. As highlighted in the Section III and depicted in the taxonomy Figure 2, attacks on actuators are classified as digital attacks.

Potential attack surfaces for robot actuators include actuator hardware and firmware, communication protocols, and network components such as those used by tele-operated robots. Fig 4 illustrates the attack scenario wherein the adversary targets the firmware, manipulates the access control mechanism, or employs backdoor techniques to influence the behavior of robotic systems within their operating environment. Table 4 summarizes various types of attacks on actuators along with defensive strategies. To gain access to the robot control systems, attackers can install hardware-level backdoors or kill switches to bypass control protection mechanisms [132]. For example, Sasi [133] developed a backdoor program that allows attackers to remotely control UAV robotic systems. Another area that is vulnerable to attack is actuator firmware, largely due to insecure communication (for example, using the public internet to update a robot's firmware). Several studies [134], [135], [136] indicate that attackers can exploit communication vulnerabilities of public networks to gain full control of robotic systems.

Authentication (for example Sybil attack [137]) and access control are often overlooked in robotic systems. Researchers indicate that some robotic application interfaces are designed

without the need for a login portal, allowing anyone to access them remotely [138], while others that are equipped with access control systems are also known to suffer from vulnerabilities. These include communication packets containing credentials in clear text or weakly encrypted, systems with default or hard-coded credentials that can be found in publicly accessible user manuals, or in some cases, a lack of access rights configuration allowing adversaries with low privileges to control robotic systems [139].

Recent scientific research has shown the potential vulnerability of the control systems and actuators of teleoperated surgical robots to malicious attacks. In their study, Bonaci et al. [140], demonstrated the ability to exert unauthorized control over various surgical functions. Furthermore, various research has revealed targeted attacks that can bypass the robot's input sensors, including the surgeon's input, and directly impact the actuators [141]. These attacks result in behaviors such as the involuntary halting of the system or the abrupt movement of robotic arms. Notably, similar attacks on actuators have been observed in other robotic applications, such as mobile robotics [142], [143], robotic Cyber-Physical Systems (CPS) [144], [145], and aircraft systems [146].

Security strategies for robot actuators involve several measures to ensure the safety and integrity of the robot's operations. One crucial aspect is the sanitization of control signals that are transmitted to the actuators, which helps protect against adversarial inputs. To achieve this, concepts like dynamic filtering [147] can be applied to filter out anomalous control signals. Moreover, an access control mechanism can also be implemented to ensure only verified programs send control signals to the actuator driver. Another approach is to develop anomaly detection systems to monitor anomalies during robot manipulation tasks. For example, the authors in [148], presented a method for multimodal anomaly detection during robot manipulation. In a separate study, the authors in [149] developed RoboADS, a robot anomaly detection system that is capable of detecting both sensor and actuator misbehaviors. The alerts generated by these detection models can then be analyzed by robot operators to conduct further investigations if necessary. Furthermore, rule-based actions can be implemented in the actuator code to prohibit the robot from executing undesired actions.

Additionally, attacks on robotic actuators possess significant security concerns regarding the kinematics and dynamics factors due to their ability to interfere with the environment. From the kinematic and dynamics standpoint, compromised actuators can generate motions or forces intending to harm their sensors, machinery, and peripherals. For example, in an industrial setting, actuator attacks causing kinematics disruption can destroy the manufacturing pipeline, instruments, and many more. In the work of Zhang et al. [150], the authors demonstrated mitigation strategies against actuator attacks for kinematic disruption in networked industrial systems that can cause a decrease in manufacturing quality. Again, Moaed et al. [151] demonstrated attacks on actuator dynamics and restoration process through attacks on prosthetic arm

actuators, which can disrupt human electromyogram control signals and develop frustration and distrust in the human mind. Furthermore, the casualty from a kinematic or dynamic impact is not limited to the production pipeline or human mind- a carefully triggered kinematics and dynamics attack in the heavy manufacturing (nuclear power plant, chemical gas plant, steel plant, and other) industry can incur catastrophic casualties such as loss of lives. Therefore, the security of actuators requires significant attention when integrated with AI technologies for operations.

Moving forward, we examine the inherent challenges within AI models and their potential impacts on AI-Robotics systems.

3) ATTACK ON AI MODELS

Modern robots contain multiple software components such as object recognition [152], voice commands [153], and facial recognition [154], that are usually built using AI-based pipelines. At a high level, AI pipeline entails data collection, preprocessing, model selection, training, validation, and deployment. These AI pipelines can be used to solve previously intractable problems in robotics using adaptable and data-driven methodologies. However, this same adaptability can also lead to exploitation by attackers [155] because of the inherent complexity and the many stages involved in training, validating, and deploying an AI system. AI components that involve deep learning (i.e. neural networks) can be especially vulnerable due to their “black box” nature, which means that even when the security is compromised, it may be difficult to pinpoint exactly where the problem occurred and how to address it [156]. The nature of this attack is classified as a digital attack, as delineated in our taxonomy in Section III and depicted in Fig 2.

In the following subsections, we explore training and inference attacks on AI models.

a: TRAINING ATTACKS AND SECURITY STRATEGIES

Training attacks involve the deliberate alteration, tainting, and modification of the training dataset. Such an attack intends to manipulate the training process and compromise the performance and behavior of AI models. There are several forms of security attacks on the training process or training data on AI agents of AI-Robotic systems. One form of attack is known as *model poisoning*, where the training data for AI models is tainted through label corruption. Such attacks might lead AI model to function unexpectedly or is unable to recognize the current input data. Figure 5 presents a high-level overview of a training attack. The adversary tampers sensory data obtained from perception sensors such as cameras, IMUs, LiDAR, etc., to create a poisoned dataset, which is then utilized to train an AI model. As a result, the trained AI model is compromised and may exhibit misclassifications, such as inaccurately identifying objects or misjudging distances between objects. There are several well-known model poisoning cases such as

the Microsoft Tay chatbot incident [157] and the VirusTotal poisoning incident [158]. The Tay chatbot incident involved an experimental conversational chatbot released by Microsoft on Twitter, which is a popular social media platform. Tay is designed to work by learning from real-world conversations and improving its language capabilities over time. Unfortunately, malicious internet users exploited this property by coordinating to send messages to it involving racist, misogynistic, and anti-semitic language. As a result, Tay started to repeat these abusive and offensive messages and the experiment had to be suspended. Another example is the VirusTotal poisoning incident, where mutant samples uploaded to the training platform were able to fool an ML model into incorrectly classifying inputs. VirusTotal is a popular platform for analyzing suspected malware files and automatically sharing them with the security community. However, attackers were able to compromise the integrity of this platform by uploading fake samples such that the platform would classify even benign files as malicious. In essence, these model poisoning cases show that an AI model can be compromised if the source of its training data is not validated. Robotic systems that utilize publicly-sourced training data can be vulnerable to these model poisoning attacks. In addition, outsourced training can potentially introduce the risk of malicious backdoors [159].

One strategy to prevent model poisoning attacks is either preventing a model from being poisoned or repairing a network after it is poisoned [160]. Another strategy involves identifying poisoned data as outliers in the input space [161], [162], employing certain input preprocessing techniques [163], or by trimming out irregular samples when computing the loss function [164]. In the event of poisoned model, it may be possible to detect the planted artifacts using generative adversarial networks [165] and repair them using pruning and fine-tuning [159]. Pruning works by removing neurons that are non-essential or dormant for clean inputs, thus reducing the risk that the network will produce unwanted behavior for bad inputs. On the other hand, fine-tuning works by taking a network trained with untrusted sources and then further training the network with trusted inputs. A combination of the two, fine-pruning [159], can even defeat more sophisticated pruning-aware attacks.

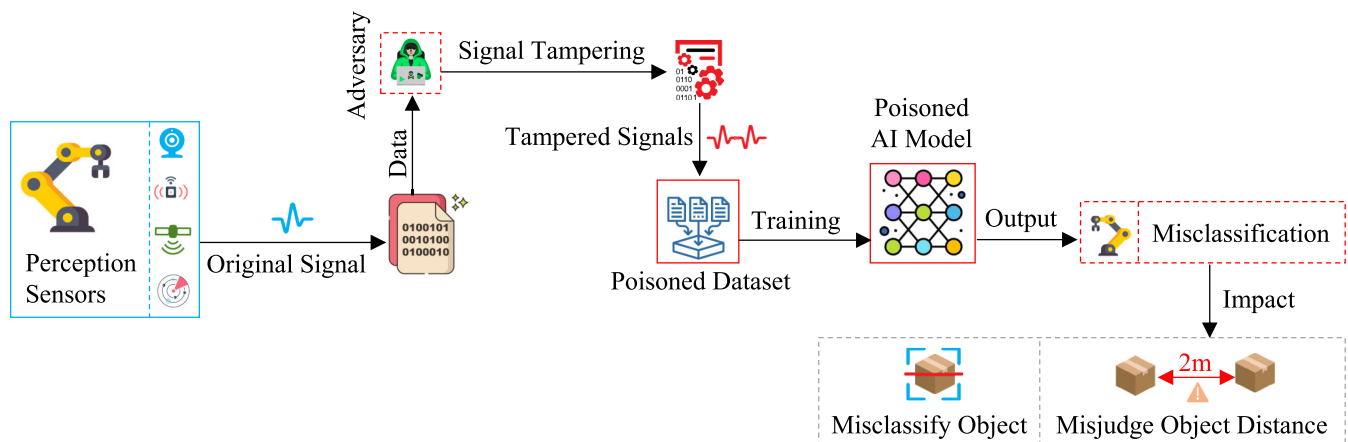
Building upon the exploration of training attacks and associated security strategies, in the next section, we delve into inference attacks and their corresponding security strategies.

b: INFERENCE ATTACKS AND SECURITY STRATEGIES

A model inference attack is a type of privacy attack wherein an adversary attempts to reverse-engineer or infer sensitive information about the training data or individual data points used to train an AI model by querying the model and analyzing its outputs [166]. Construction of model inference attack usually involves generating shadow models (a replica of the target model that mimics its behavior as closely as possible) using shadow training [166]. An example of a model

TABLE 4. Overview of attack patterns in actuators and associated defensive strategies.

Attack Surface	Medium	Attack Summary	Defensive Strategy	Research
Hardware	Backdoors and kill switch	- Installation of hardware-level backdoors or kill switches to bypass control protection mechanisms. - Backdoor programs to remotely control robotic systems, e.g., UAVs.	- Secure design and development. - Utilizing formal verification methods to ensure the robotic system's software and hardware meet specified security requirements.	[132], [133]
Firmware	Insecure network communication	Attackers can exploit communication vulnerabilities of public networks to install malware and obtain full control of robotic system.	- Employing secure communication protocols and encryption mechanisms between robotic components. - Update firmware and patch software using secured connection.	[134]–[136]
Access Controls	Access controls mechanism and authentication systems	Lack of proper access control mechanism allowing anyone to access and control remotely.	Inclusion of robust access control to ensure only verified programs send control signals to the actuator driver.	[138], [139]
Others	Manipulation of surgical functions (surgical robots)	Targeted attacks can bypass the robot's input sensors, including the surgeon's input, and directly impact functioning of actuators.	- Sanitization of control signals. - Dynamic filtering [147] can be applied to filter out anomalous control signals.	[140], [141]

**FIGURE 5. An example of training attack on AI pipeline that causes misclassification of objects and misjudgment of object distance.**

inference attack is depicted in Figure 6. Inference attacks on AI-Robotic systems may include model inversion, model evasion, and model alteration among others.

- *Model Inversion* attacks also known as attribute inference attack [167] occur when an adversary strategically queries the inference API to extract potentially private information embedded within the training data [168]. Research has shown that certain data augmentation techniques can mitigate overfitting and indirectly reduce the risk of membership inference attacks [166] but such methods may incur an accuracy penalty [168].
- *Model Evasion* tactics are used by attackers to prevent the AI model from computing the correct output. In the image domain, adversarial techniques can take the form of simple transformation of the input (cropping, shearing, translation), adding white noise in the background, or constructing adversarial examples by carefully perturbing the input to achieve desired output [169]. In the natural language domain, adversarial

techniques can take the form of a Happy String, wherein the benign input is tacked onto malicious query points, or by using bad characters, words, or sentences that can still appear semantically sound from a human perspective [170]. Certain model evasion strategies can be especially stealthy since they employ optimization techniques to find the smallest perturbations that can still cause a change in the model output [171].

- *Model Alteration* occurs when attackers are able to gain access to the network weights and alter them without the knowledge of the user. These attacks are essentially deployment-stage backdoor attacks where the attacker can cause the network to produce incorrect output for a specific input while producing correct outputs for most other inputs [172], [173]. Such attacks can also cause neural networks to become black-box trojans. Since the behavior of a neural network can significantly change while appearing benign from the outside if only minimal model parameters are changed [172], [174]. These attacks can take the form of network structure

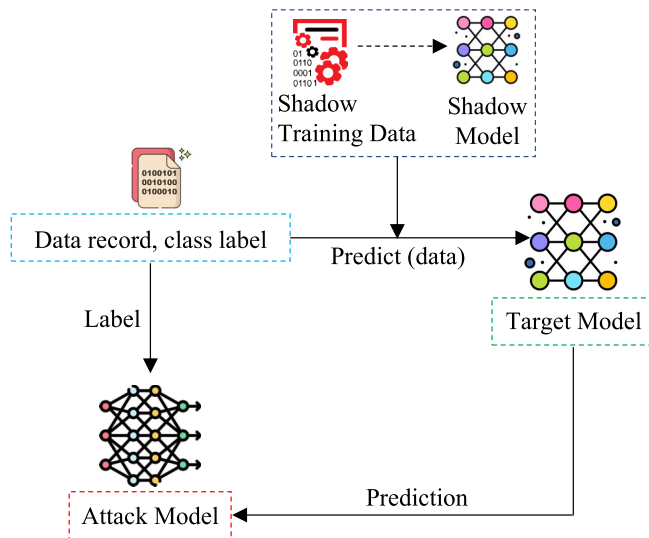


FIGURE 6. An overview of a membership inference attack where the attacker queries the target model utilizing the shadow training set and shadow model to obtain a prediction vector containing probabilities for each class [166].

modification [174], flipping weight bits [172], or subnet replacement [173].

Defense strategies to preserve the privacy of users' data (in case of membership inference attacks), especially images for computer vision problems, is to apply a reconstruction and perturbation network. This method preserves image features while obfuscating the individual's face [175], [176]. Differential Privacy (DP) is another technique that can mitigate privacy issue by computing a theoretical bound on the effect that any particular input has on the output. Employing such strategy may provide guarantees in certain situations that information will not be leaked [177]. Another technique to prevent data leakage is learning with selective forgetting [178], where a class-specific synthetic signal is used to forget arbitrary classes without using the original data. In a similar vein, defense against model evasion techniques that utilize adversarial inputs is to formally check the security properties of a deep neural network with Satisfiability Modulo Theory or symbolic interval analysis [179]. In the case of evasion attacks, two forms of defense can be implemented. The first type focuses on improving an AI model's robustness by introducing methods that make the model's weights and layers more difficult to steal or reverse-engineer [180]. This can be accomplished via online learning and weight readjustment or by introducing bounded perturbations [180]. The second form employs a hybrid approach. This involves integrating AI algorithms to deal with unknown quantities with tried and tested methods from legacy systems, which are designed to tackle known robotics problems [181]. This approach provides an extra degree of verification to ensure comprehensive protection.

C. OTHER RELEVANT ATTACKS

As highlighted in Section III, AI-Robotics systems can also suffer from other common cybersecurity attacks. Based on

our taxonomic Figure 2, some of these attacks include backdoors, malware, ransomware, and attacks on communication networks. In the following paragraphs, we explain these attacks in detail.

A *backdoor attack* is a stealthy technique of circumventing normal authentication methods to gain unauthorized access to a system. Malicious actors can insert a backdoor in robotic systems to remotely control them. In some cases, threat actors such as malicious manufacturers may leave, backdoor, on purpose, in robotic systems to track and monitor the activities of the robot and its operator without the owner's knowledge [10].

Malware, on the other hand, can target robotic systems such as surgical robots. Chung et al. [182], for example, exploited ROS vulnerabilities and developed smart self-learning malware capable of tracking the movements of the robot's arms and triggering the attack payload during a surgical procedure. Similarly, *ransomware attacks* in the context of industrial robots attempt to lock them to extort ransom from the manufacturer. In an effort to augment robot control system security, the authors in [183] presented a customized industrial robot ransomware, which showcased the possibility of industrial robots being susceptible to ransomware attacks.

Another potential candidate of attack is the robotic *communication network*. Adversaries may target vulnerability of existing communication protocols which might lead to jamming, de-authentication, traffic analysis, eavesdropping, false data injection, denial of service, replay, or man-in-the-middle attacks. Jamming and de-authentication attacks are quite common in small UAVs [184]. Traffic analysis and eavesdropping are passive attacks used on the unencrypted channels of UAVs to extract the meaningful and sensitive flight and motion data [13] whereas false data injection and replay attacks involve insertion of incorrect routing information into the caches of UAVs [12], [13].

V. ETHICAL AND LEGAL CONCERNS

Apart from the attack surfaces discussed in preceding section, there are also ethical and legal concerns surrounding robots. In this section, we will explore these aspects in greater detail.

A. ETHICAL CONCERNS

One of the popular topics that have an ethical dimension in robotics domain is *robot ethics* or *roboethics* - a term first coined by Gianmarco Verrugio [185]. It is a combination of ethics (*what's right and wrong*) and robotics (robots and technology). It focuses on comprehending and overseeing the ethical outcomes and aftermaths arising from robotics technology, especially with regard to intelligent and autonomous robots, within our societal context [186]. Roboethics encompasses a diverse range of application areas where ethical considerations play a pivotal role in shaping the development, deployment, and use of robots and autonomous systems.

In following paragraphs, we explain ethical concerns surrounding use of AI-Robotics systems across various domains including healthcare, autonomous vehicles, assistive robots, military, and manufacturing as delineated in Section III and depicted in Fig. 2.

In the realm of healthcare, medical robotic technologies are increasingly utilized for surgeries [4], rehabilitation [5], and caregiving. Ethical questions arise concerning patient privacy, cost of implementation, and informed consent for robotic procedures among others. Consider, for instance, the substantial expense associated with robotic surgery. This naturally prompts us to ask [187]: “Given that there is marginal benefit from using robots, is it ethical to impose financial burden on patients or the medical system?” Other ethical quandaries include patients’ autonomy and raise questions such as how to ensure patients fully understand and consent to robotic procedures? Do they have the right to opt for or against robotic assistance in surgery? Similarly, there are questions about data privacy and security. Questions like, What safeguards are in place to protect patient data collected and transmitted by medical robots? How do we ensure privacy and prevent data breaches?

Medical roboethics guides the development of these systems to ensure patient safety, privacy, and equitable access. Furthermore, the design and development of medical robots must comply with privacy rules, including the Health Insurance Portability and Accountability Act (HIPAA) [188] and the European Union’s General Data Protection Regulation (EU GDPR) [189]. These regulations guide engineers and developers in ensuring the privacy and security of user data (see Subsection VI-A1 for privacy concerns around human robot interaction). By adhering to these policies, developers can mitigate the risk of data breaches, unauthorized access, and misuse of sensitive medical information.

Another critical area of robotic application is Autonomous Vehicles (AV). There are two fundamental ethical concerns surrounding AVs, including the trolley problem and the self-sacrifice dilemma [190]. A visual illustration of these dilemma is presented in Fig 7. The first one refers to a self-driving car deciding whether to continue on its present path, potentially hitting and harming multiple pedestrians, or to swerve and divert its path, potentially killing a single individual [191], [192]. Then, in such a scenario, what is the correct decision? It raises questions about how these decisions should be programmed by AI agents, who bears the ethical responsibility, or how societal values and cultural norms should influence the behavior of these vehicles. The self-sacrifice dilemma, on the other hand, is a variant of the trolley problem that addresses the ethical question of whether an autonomous vehicle should prioritize the safety of its occupants (passengers) over the safety of pedestrians or other individuals outside the vehicle. In this scenario, the self-driving car is confronted with an unavoidable collision and must determine whether to take actions that would minimize harm to its passengers at the expense of greater harm to pedestrians or vice versa [193]. Then again, what is

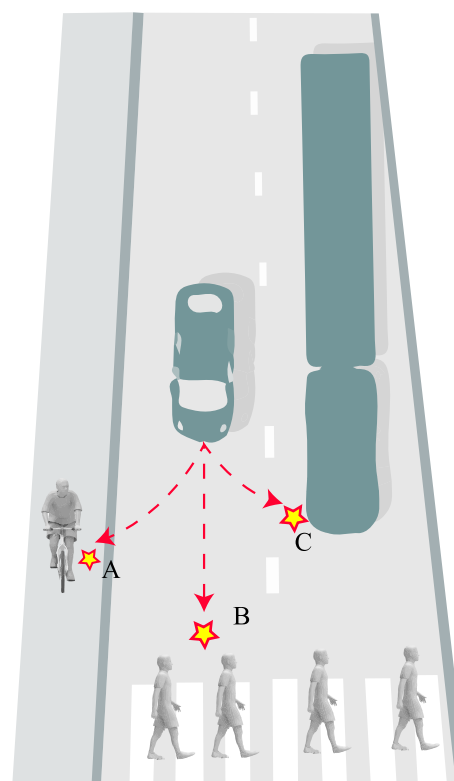


FIGURE 7. A graphical depiction of the ethical dilemma for the AVs in an inevitable crash. Option (A): crash with the cyclist on the right; Option (B): continue straight and crash with two pedestrians; Option (C): self-sacrifice by crashing with the truck on the left. Adapted from [190].

the correct decision? Should the AI agents be programmed in a way that saves more people by sacrificing the driver? How can these decisions be made transparent? This issue underscores the need to consider integrating societal values, legal frameworks, and cultural norms in designing AI agents that can intelligently adapt their behavior in such ethically complex situations.

In a similar vein, assistive robots (for example rehabilitation robots, robotized wheelchairs, walkers, etc.) designed to assist people with disabilities, or the elderly are becoming increasingly prevalent. These robots can improve autonomy and quality of life, but ethical quandaries arise surrounding questions of dependency, risk, cost, benefits, and the potential for social isolation induced by growing reliance on machines for companionship and care. For example, to what extent should individuals using assistive robots maintain control over the actions and decisions of the robot?, who is liable in case of accidents or errors made by assistive robots?, and how can we maintain a balance between the benefits of companionship and the potential risks of emotional dependence [25] on machines?.

The use of sophisticated and automated robotic weapons by the military in conflict has profound ethical concerns. Particularly, lethal autonomous weapons raise concerns about moral accountability for machine actions, and the potential dehumanization of combat. For instance, how can we ensure meaningful human involvement in critical decisions,

especially in life-and-death situations?, how can we program AI agents that distinguish combatants and non-combatants? how can we ensure that targeting decisions made by AI agents aligns with humanitarian principles and laws? This is a critical issue because, during the fog of war, it is difficult enough for humans to effectively distinguish whether a target is legitimate. So, how can we ensure AI-Robotics systems perform better ethical decision-making than human soldiers? [194].

Additionally, in the industries like manufacturing and agriculture, ethical considerations surrounding human-robot collaboration, job displacement, and responsible automation are central to the roboethics discourse.

B. LEGAL CONCERNS

Laws in robotics are influenced by fictional writers such as Isaac Asimov. His three laws of robotics [195], which first appeared in a story called “Runaround”, include the following:

- 1) A robot may not harm a human being, or, through inaction, allow a human being to come to harm.
- 2) A robot must obey the orders given to it by human beings, except where such orders would conflict with the First Law.
- 3) A robot must protect its own existence, as long as such protection does not conflict with the First or Second Law.

While the laws straightforwardly establish robots as subordinate to humans, their implementation can be challenging in diverse real-world scenarios. Nevertheless, robot engineers can view them as guidelines for programming AI agents within robotic systems. Legal and regulatory concerns stem from the application of legal and moral theories pertaining to liability and responsibility [196]. Consider the utilization of surgical robots in hostile military environments for procedures like “remote robotic-assisted telesurgery” or “autonomous robotic surgery”. In the event of an unintended event mid-surgery, where harm is inflicted upon human counterparts due to malfunctioning or attack on robot actuators, this may result in legal consequences. This prompts a discussion regarding which party should be held liable?, such as the system manufacturer, software developer, the autonomous agents involved (this is difficult because as they are not legal persons), or the human operators overseeing the operation of these devices [197].

Another legal dimension within the AI-Robotics systems in security of personal data. To safeguard the personal information various policy framework has been proposed both in the United States (US) known as 2020 California Consumer Privacy Act (CCPA) [27] and 2016 General Data Protection Regulation (GDPR [28]) in the Europe (EU).

Before CCPA, in the US, the Executive Office of the President, National Science and Technology Council, Committee on Technology issued a report, “Preparing for the Future of Artificial Intelligence” (“AI Report”), in October 2016 [198]. Given the importance of AI in the future, the report

recommended “the Department of Transportation should work with industry and researchers on ways to increase sharing of data for safety, research, and other purposes.” Prior to this, the same administration also presented a discussion draft on “Consumer Privacy Bill of Rights Act”. It proposed seven principles including transparency; individual control; respect for context; focused collection and responsible use; security; access and accuracy; and accountability, however, the act was not enacted [199]. To alleviate the concerns of use of personal data by large technology companies CCPA was brought to legislation in 2018. The regulation defines *personal data and information*, enforce requirements for *company data security*, *allow data portability* and *deletion rights*, and *impose fines* on organizations that violate them. This law went to effect in 2020, after undergoing several amendments. The CCPA is based on a consumer protection framework and applies more narrowly to enterprises that meet specified revenue or number of California residents’ personal information collection requirements [200].

Similarly, in the EU, GDPR law was signed in April 2016 and implemented in May 2016. The new GDPR mandates explicit informed consent for data use. Through informed consent, the users are given the right to opt in or out from any potential processing of their data through the “right of explanation.” GDPR also empowers users to track or remove their data through the “right to be forgotten.” To counter the problem of information leakage, the authors in [201] presented two efficient approaches, such as unlearning and amnesiac unlearning. Their approach is successful in removing personal data. In a similar vein, Villaronga et al. [202] examined the issue of AI memory in the context of the “right to be forgotten” and explored the inadequacies of current privacy laws for tackling the complexities and challenges of AI technology. The paper analyzed legal implications for AI as well as the conflict between privacy and transparency under EU privacy law. In a separate study, the authors in [203] proposed an approach that cleanses the weights in a deep neural network to forget information selectively.

In the subsequent section, we delve into challenges and risks that arise in the context of human-robot interaction.

VI. HUMAN-ROBOT INTERACTION (HRI) SECURITY STUDIES

Human-Robot Interaction (HRI) is an emerging field that examines the physical, cognitive, and social interactions between people and robots. While HRI is an important component of robotic autonomy, studies have also identified several key vulnerabilities that can compromise the integrity, privacy, and safety of these interactions. For example, unauthorized access to a collaborative robot (co-bot) in a manufacturing setting could involve potential manipulation of the robot’s behavior. Similarly, privacy is another major concern, as the collection and storage of personal and sensitive information during interactions with social robots used in healthcare settings can result in privacy breaches

or unauthorized use of data. It is important to note that interactions can occur between people and collaborative multi-robot systems (MRS) including robot swarms.

In accordance with the taxonomy introduced in Section III, we conduct a survey of the challenges and concerns in the subsequent subsections. We also recommend countermeasures such as Privacy Enhancing Technologies (PET) that bolsters the integrity, privacy and confidentiality of data exchanged between humans and robots.

A. CHALLENGES AND CONCERNS

1) PRIVACY CONCERNS AROUND ROBOTS

User privacy needs to be considered and protected when designing robotic assistive systems. Privacy concerns might influence human behavior when robots monitor them. In their work [204], the authors analyzed how older adults perceive privacy and their attitudes to engage in privacy-enhancing behavior under three natural settings – a camera, a stationary robot, and a mobile robot. Authors found that more participants engaged in more privacy-enhancing behavior such as self-censoring of speech or covering up the camera when being watched by a camera compared to embodied robots. Their analysis implies that participants are less cautious about their privacy around robots due to their unfamiliarity with them.

Robots endanger people's privacy by directly monitoring them and increasing access to their information. A study in [205], examined 1090 German-speaking older adults verifying the factors that influence their privacy concerns and preferences. The paper observed a significant correlation between the people with practical robotic experience and the interest. The participants interested in human robots showed higher comfort levels around robots than the users who selected machine robots. Most participants did not express interest in sharing their health-related information with anonymous health services. Results indicated that participants prefer not to transmit their location and position to others unless there is an emergency. Furthermore, their level of comfort varied depends on the data collection modalities.

2) SOCIAL ACCEPTANCE TO ROBOT THREATS

People exposed to robots that are capable of rejecting human commands could result in a negative attitude towards robots. In the context of their experimental investigation [206], the authors investigated social acceptance of perceived robot autonomy in case of threats to human identity and threats to jobs. Using the Mechanical Turk crowdsourcing platform, they surveyed 176 US participants showing videos of autonomous robots that could disregard human commands and non-autonomous robots that could only obey human commands. Participants were assessed on their perception of robotic threats to jobs, resources, safety, and human identity. They also evaluated participants' negative attitudes towards robots and support for the robotic research. The survey concluded that the participants viewed autonomous

robots as more realistic threats than those that were not autonomous. Generally, humans are more hesitant to trust especially those they perceive to be deceptive [207], [208]. However, within their study [209] found that robotic failures in their experiments did not negatively affect user attitude when using a human-like robot embodiment. Furthermore, as outlined in their research endeavor [210], researchers confirm that error-prone robots are more believable.

3) INTERNET-CONNECTED DEVICES

Storing and sharing personal information of parents and children is a potential threat in internet-connected robotic toys. Within the scope of empirically evaluating the security and privacy perspective about parent attitudes and child privacy, the authors in [211], made recommendations for toy designers related to the security and privacy of the toys. They focused on identifying parents' privacy expectations and concerns, children's mental models of their privacy, and the ethical perspective of parental controls. They conducted semi-structured interviews with thirty parents and children, following a demonstration and interaction session with two different toys. Authors suggested that robotic toy designers must be transparent and communicate with parents that the toy is recording and delete recordings after a fixed time from the app and the cloud servers. Such studies aid designers in devising internet-connected games and toys without violating privacy rights.

On the other hand, smart homes, more specifically, Robot-integrated Smart Homes (RiSH) [212], are on the rise. These smart homes integrate a home service robot, a home sensor network, a body sensor network, a mobile device, cloud servers, and remote caregivers. RiSH can be used for research in assistive technologies for elderly care. The interconnectivity of these systems through the Internet of Things (IoT), however, exposes them to various cybersecurity attacks and poses a number of security concerns. In their research [213], the authors collected responses from semi-structured interviews with fifteen smart home residents to identify the gap between risks identified by security researchers and the user's security concerns and their mitigation methods. The findings showcased smart home users' lack of security concerns, ranging from not feeling personally targeted or trusting potentially adversarial actors to believing existing mitigation strategies to be sufficient.

4) TRUST

Psychologists define trust as a mental state in humans [214]. In relation to HRI, there are various definitions of trust; for example, Lee et al. [215], define it as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" whereas Hancock et al. [207], defines it as "the reliance by an agent that actions prejudicial to their well-being will not be undertaken by influential others". The key takeaway from these concepts is *whether a robot's activities and behaviors align with human interests* [216].

In their work [207], the authors identified three categories of factors that affect trust in HRI: robot-related factors, task and environmental factors, and human-related factors. The first category pertains to the quality of operation performed by the robot from the perspective of the human operator and includes subtasks [216] such as transparency and feedback in decision-making [217], situational awareness [218], autonomy [219], fault tolerance [220], and engagement [221] to facilitate naturalistic interaction. In contrast, task and environment-related factors, as identified in various other studies, such as the nature of the task [222], risk and human safety [223], and the task site [219], influence trust between robots and humans. The third category, on the other hand, is concerned with human-based factors, such as the general perception about robots [224], expectations of human operators towards automation [225], past experience with robots [226], and operators' understanding of robotic systems [227].

B. PRIVACY ENHANCING TECHNOLOGIES

In the context of AI-Robotics, Privacy Enhancing Technologies (PET) refer to the use of various technological solutions and strategies to protect individuals' privacy when interacting with robotic systems. Their purpose is to protect user identities by offering anonymity, pseudonymity, unobservability, and unlinkability of both users and data subjects [228], [229]. A report by the Federal Reserve Bank of San Francisco [230] categorizes PETs in three categories: *altering data*, *shielding data*, and *systems + architecture*. A graphical depiction of these categories and subcategories is presented in Fig. 8. The first category of PETs are tools that alter data itself. Examples of these include anonymization, pseudonymization, differential privacy, and synthetic data. The second category focuses on hiding and shielding the data. This usually involves the use of cryptographic techniques. Some examples include encryption, homomorphic encryption, and privacy-enhanced hardware. The final category of PETs are new systems and processes for data activities. The report suggested that some of these systems also enable greater transparency and oversight across data activities, including collection, processing, transfer, use, and storage.

Authors in [231]

surveyed PETs for solving personalization paradox. Their approach included much broader PETs as compared to the report by Federal Reserve Bank of San Francisco. Some of these technologies include anti-tracing (fingerprinting), data obfuscation (perturbation, private information retrieval), privacy preservation certification (attribute-based signatures, sanitizable signatures), self-destructing data systems (vanish, neuralyzer), statistical disclosure control (anonymizing DBs), and secure communication (client-service encryption, end-to-end encryption).

PET such as *access control mechanisms* are popular in achieving privacy and data protection in natural settings. Furthermore, the right choice of sensors that adapt to

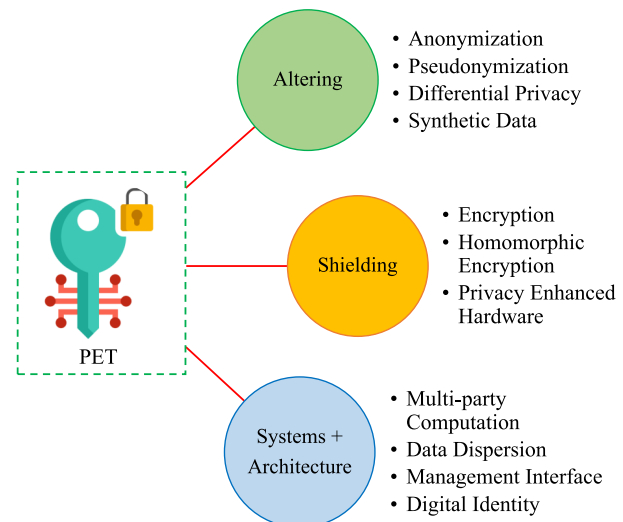


FIGURE 8. A graphical illustration of Privacy Enhancing Technologies (PET). Adapted from [230].

differing privacy expectations is essential in reducing privacy vulnerabilities [232]. To address this, the authors in [233] proposed an app-based user interface to enable users to be in control of their data. The interface is intended to enable transparency and accuracy in data processing by communicating with the software running on the robot. This transparent interface displayed the robot's visual, auditory, position, and communication information to help users identify the possible privacy violations caused by social robots. However, the human feedback evaluation revealed that participants had somewhat mixed attitudes towards this self-learning social robot. Moreover, the work in [234] recommends balancing privacy with utility in robot systems. The qualitative and quantitative evaluations of privacy preferences of end-users in their study have shown concerns about both privacy and physical harm.

In relation to trust in HRI, to enhance trust in robotic applications, trust models must be developed for a specific form of human-robot interaction or a specific type of robotic agent [216]. For example, a trust model for evacuation robots must be different from a trust model for surgical robots because the expected outcomes of these domain-specific robotic systems are distinct.

VII. FUTURE RESEARCH AND DISCUSSION

Previous sections discuss the security risks and challenges faced by AI-Robotics systems. More, specifically, we explored three dimensions of AI-Robotics systems including, *attack surfaces*, *ethical and legal concerns*, and *security of HRI*. As these AI-Robotics systems become more pervasive, their security becomes critical. We believe that the holistic understanding of abovementioned dimensions will effectively strengthen the overall security posture of these systems. Security of input sensors and sensory data from these sensors (e.g., training data, models) becomes essential to ensuring the integrity of these systems. The

data complexity only increases as you consider that an AI-Robotics system could be comprised of many robots (i.e., a system of systems). We see future research focusing on ensuring trustworthy AI-Robotics systems that are capable of withstand various attack scenarios, adhere to the ethical and legal principles of the domain in which they are deployed, and ensure the interactions with these systems remain private.

In the following subsections, we outline seven essential domains that encompass various research areas critical for enhancing the security of AI-Robotics systems. As depicted in Fig 9 these include research on attack surfaces mitigation, securing swarm robotics and knowledge transfer, security of human-robot ecosystems, explainability in ai-robotics, securing verification, validation, and evaluation of ai-robotics systems, and education initiatives to raise robotics security awareness.

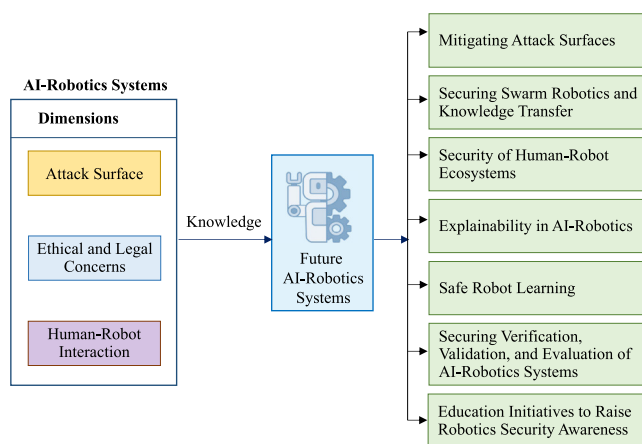


FIGURE 9. A visual representation outlining the future research direction aimed at improving the security of AI-Robotics systems.

A. MITIGATING ATTACK SURFACES

In Section IV, we explored various attack surfaces on AI-Robotics systems. The first component that an adversary may target is the embedded sensors that are responsible to percept the environment. Adversaries can either physically damage sensors or they can manipulate sensors by jamming or spoofing the sensor signals. The future robotic systems should be designed with adversaries' capability in mind. For example, to secure the sensors from a physical attack such as vandalism defenders can apply perimeter security with robust access controls. Similarly, in case of other attack patterns such as jamming and spoofing the defenders may integrate lightweight spoofing and jamming detection algorithms [83], [84]. Another important aspect that future AI-Robotics systems designers and engineers must consider is to have some mechanism for verification and sanity checking of input sensors data. Such process would thwart adversaries attempt to supply adversarial input to AI agents.

To secure the actuators from attacks, the research is leaning towards development of anomaly detection systems. These systems are capable of monitoring the anomalies during robot manipulation task [148]. Such methods can also be extended

to filter out anomalies in sensor data [149]. Furthermore, control signal sanitization mechanism must be integrated in robotics systems of the future to defend against adversarial input to actuators.

Future security strategies to secure the AI agents used by AI-Robotics systems during training phase may include certain preprocessing techniques [163] such as identifying poisoned data as an outlier in the input space [161], [162]. Furthermore, the robotic systems can engineers leverage generative models such as Generative Adversarial Networks to detect the planted artifacts in poisoned model [165]. In case of protecting privacy during inference attack various security strategies can be implemented for future robotic application. Concepts like differential privacy, selective forgetting [178], data obfuscation [175], [176] can be considered.

B. SECURING SWARM ROBOTICS AND KNOWLEDGE TRANSFER

As noted before, robots are becoming more integrated in various aspects of human life, and securing human-robot interactions is going to be vital moving forward. Similarly, securing the interactions between robots in multi-robot systems is another major challenge. Advances in swarm robotics [235] have led to large-scale swarm applications in which teams of robots work together to achieve a common goal. The need to share information across agents has led to new architectures [236] including making use of shared-memory structures called blackboards. Swarms may find themselves inoperative if a constituent is targeted by an adversarial attack. Decentralized security solutions, including secured federated learning [237] and cybersecurity knowledge graphs [238], are becoming more prominent in order to protect swarms from agent-specific targeted attacks.

C. SECURITY OF HUMAN-ROBOT ECOSYSTEMS

In human-robot ecosystems where all agents work independently, a major challenge is preventing conflicts between robotic agents and humans utilizing shared resources [239]. *How are plans prioritized? How is information exchanged? Can negotiations occur?*

In scenarios where AI-Robotics systems assist humans (i.e., HRI applications), *trusting* inferences made by these robotic systems is another challenge [240]. Sensors in robots could become faulty or biased over time and produce inaccurate measurements. *How does a robot re-calibrate a faulty sensor in an autonomous setting? How do you fuse information in a multi-robot system that may or may not have a faulty sensor (or a faulty robot)?* These are important questions that must be addressed in order to *trust* these systems in scenarios with high variability and where human life is at stake.

It is important to note that in human-robot ecosystems, agents can be both physical and digital. In particular, agents can be a digital representation of a physical system (i.e., digital twin [241]). Digital twins are being used to

improve human-robot collaboration in manufacturing [242]. Improving the robotic system through human experience and knowledge is also an important area of research. An emerging concept is the use of a human digital twin (HDT) in an elaborate system-of-systems where human decision making is captured by data-driven approaches [243], [244].

D. EXPLAINABILITY IN AI-ROBOTICS

Explainable AI (XAI) provides a mechanism for securing HRI applications. Roque et al. [245] describe and implement an explainable AI solution that can be used by humans to calibrate their trust in their systems. While most current XAI solutions are geared towards domain experts [246], other approaches [247] target non-experts when explaining robot failures. In order to provide truly comprehensive XAI solutions in AI-Robotics, end-users with no (or very little) AI/ML experience must be considered. In fact, the majority of end-users will not be experts in many robotic applications.

Another emerging research topic is neurorobotics. Since the beginning, AI and robotics have always drawn inspiration from biology. Today, researchers [248] are leveraging neuroethology, the combination of neuroscience and ethology, to study the effect of neural network perturbations in hopes of explaining a robot's behavior.

E. SAFE ROBOT LEARNING

The increased use of data-driven approaches in safety-critical applications has raised the question of safety (and safe guarantees) for such learning algorithms [249]. Brunke et al. [250] define *safe learning control* as the problem of safe decision-making under uncertainties using machine learning. Reinforcement learning (i.e., data-driven) approaches are highly adaptable but do not provide safety guarantees. Conversely, control theory (i.e., model-driven) approaches provide guarantees within known operation conditions but are less adaptable. Current research [251] is investigating the merger of control theory and reinforcement learning to provide generalizable and safe solutions within certain boundaries.

Many of these solutions use physics-based environments during model training. In certain scenarios, these environments will also have to be protected to avoid corruption and unintended disclosure of information (e.g., performance bounds, physical constraints) for any derived model.

F. SECURING VERIFICATION, VALIDATION, AND EVALUATION OF AI-ROBOTICS SYSTEMS

Ensuring safe, secure, and trustworthy AI is an active area of research [252]. This naturally extends to AI-Robotics systems. *How can these systems be verified and validated in large-dimensional and dynamic environments? How do you ensure that these systems are robust to situations different from those encountered during the training/design phase (i.e., new/unseen data)? How do you capture all the decisions made by an AI-Robotics system for traceability?* The future

of AI-Robotics will look at mitigating security risks and data protection.

G. EDUCATION INITIATIVES TO RAISE ROBOTICS SECURITY AWARENESS

Raising robotics security awareness and training is crucial in our increasingly interconnected world, where robots and autonomous systems are becoming more prevalent.

Hands-on hacking exercises on various forms of cyber-attacks on Robotic systems are an effective tool for security education to raise awareness about various forms of cyberattacks to Robotic systems stakeholders. For example, within the scope of their scholarly investigation [253], the authors reviewed the current landscape of hacking exercises, their efficacy, and challenges through a set of pedagogical principles. Moreover, in an effort to promote awareness and bolster the security of AI-Robotics in academic environments, researchers [23] are advocating for the inclusion of robotics curricula that equip students with the knowledge and abilities required for developing resilient and secure robotic applications.

Similarly, researchers also recommend game-based teaching approaches to make learning enjoyable, and such studies indicate that it has positive cognitive and affective effects in training people [254], [255]. In their work [254], the authors designed security requirements educational games to educate stakeholders about cyber security awareness and to empirically evaluate the impact of game-based training. The game included puzzle cards with security-related terms and concepts that gave a deeper understanding of an attack in a real scenario. Along the same lines, the researchers in [255] investigated contrasting security decisions of security experts, managers, and computer scientists from the three demographics to learn the difference in their perception of security. Their game primarily focused on player's decisions in handling physical security, firewalls, antivirus, network monitoring, and intrusion detection.

On the other hand, there are companies and professional associations that provide training and certification programs to educate stakeholders with the necessary skills and knowledge to secure robotic systems. For example, the Association for Advancing Automation (A3) [256] offers certification in robotics systems security to ensure that engineers and integrators have the appropriate cybersecurity knowledge and skills.

VIII. CONCLUSION

Today, AI-Robotics systems are deeply embedded in our everyday lives (e.g., robotic vacuum cleaners, cars with semi-autonomous capabilities, delivery robots). A common AI-Robotics system may use its sensors to collect data about itself and the environment. The physical system, algorithms, and data are all susceptible to security attacks.

In this paper, we provide an overview of attack surfaces and associated defensive strategies relevant to AI-Robotics systems. We introduce a taxonomy of security attacks

on these robotics systems and identified two major categories: physical and digital attacks. In order to highlight the vulnerabilities of AI-Robotics systems, we decompose fundamental robotic AI architectures into primitives (i.e., sense, plan, attack) and stacks (i.e., perception, navigation & planning, and control) providing security risks at each layer. Emerging technologies, including human-robot interaction, are discussed while making note of privacy, ethical, and trustworthiness concerns. Finally, we provide our perceived future research direction to aid researchers interested in this exciting field.

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REFERENCES

- [1] *Robot System Architectures*. [Online]. Available: https://web.stanford.edu/class/cs237b/pdfs/lecture/lecture_14.pdf
- [2] *The World's Most Experienced Driver*. [Online]. Available: <https://waymo.com/>
- [3] F. Sherwani, M. M. Asad, and B. S. K. K. Ibrahim, "Collaborative robots and Industrial Revolution 4.0 (IR 4.0)," in *Proc. Int. Conf. Emerg. Trends Smart Technol. (ICETST)*, Mar. 2020, pp. 1–5.
- [4] S. Leonard, K. L. Wu, Y. Kim, A. Krieger, and P. C. W. Kim, "Smart tissue anastomosis robot (STAR): A vision-guided robotics system for laparoscopic suturing," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 4, pp. 1305–1317, Apr. 2014.
- [5] Q. Ai, Z. Liu, W. Meng, Q. Liu, and S. Q. Xie, "Machine learning in robot assisted upper limb rehabilitation: A focused review," *IEEE Trans. Cognit. Develop. Syst.*, 2021.
- [6] R. R. Shamschiri, C. Weltzien, I. A. Hameed, I. J. Yule, T. E. Grift, S. K. Balasundram, L. Pitonakova, D. Ahmad, and G. Chowdhary, "Research and development in agricultural robotics: A perspective of digital farming," *Int. J. Agricult. Biol. Eng.*, vol. 11, no. 4, pp. 1–14, 2018.
- [7] M. Gupta, M. Abdelsalam, S. Khorsandroo, and S. Mittal, "Security and privacy in smart farming: Challenges and opportunities," *IEEE Access*, vol. 8, pp. 34564–34584, 2020.
- [8] S. Sontowski, M. Gupta, S. S. L. Chukkapalli, M. Abdelsalam, S. Mittal, A. Joshi, and R. Sandhu, "Cyber attacks on smart farming infrastructure," in *Proc. IEEE 6th Int. Conf. Collaboration Internet Comput. (CIC)*, Dec. 2020, pp. 135–143.
- [9] P. Lin, G. Bekey, and K. Abney, "Autonomous military robotics: Risk, ethics, and design," Tech. Rep., 2008.
- [10] J.-P.-A. Yaacoub, H. N. Noura, O. Salman, and A. Chehab, "Robotics cyber security: Vulnerabilities, attacks, countermeasures, and recommendations," *Int. J. Inf. Secur.*, vol. 21, no. 1, pp. 115–158, Feb. 2022.
- [11] S. Jaitly, H. Malhotra, and B. Bhushan, "Security vulnerabilities and countermeasures against jamming attacks in wireless sensor networks: A survey," in *Proc. Int. Conf. Comput., Commun. Electron.*, Jul. 2017, pp. 559–564.
- [12] B. Wu, J. Chen, J. Wu, and M. Cardei, "A survey of attacks and countermeasures in mobile ad hoc networks," in *Wireless Network Security*. Boston, MA, USA: Springer, 2007, pp. 103–135.
- [13] Y. Mekdad, A. Aris, L. Babun, A. E. Fergougui, M. Conti, R. Lazzeretti, and A. S. Uluagac, "A survey on security and privacy issues of UAVs," *Comput. Netw.*, vol. 224, Apr. 2023, Art. no. 109626.
- [14] J. A. Placed, J. Strader, H. Carrillo, N. Atanasov, V. Indelman, L. Carlone, and J. A. Castellanos, "A survey on active simultaneous localization and mapping: State of the art and new frontiers," *IEEE Trans. Robot.*, 2023.
- [15] L. Li, K. Qu, and K.-Y. Lin, "A survey on attack resilient of UAV motion planning," in *Proc. IEEE 16th Int. Conf. Control Autom. (ICCA)*, Oct. 2020, pp. 558–563.
- [16] A. Lamssaggad, N. Benamar, A. S. Hafid, and M. Msahli, "A survey on the current security landscape of intelligent transportation systems," *IEEE Access*, vol. 9, pp. 9180–9208, 2021.
- [17] S. N. Narayanan, S. Mittal, and A. Joshi, "OBD_SecureAlert: An anomaly detection system for vehicles," in *Proc. IEEE Int. Conf. Smart Comput. (SMARTCOMP)*, May 2016, pp. 1–6.
- [18] S. S. L. Chukkapalli, S. Mittal, M. Gupta, M. Abdelsalam, A. Joshi, R. Sandhu, and K. Joshi, "Ontologies and artificial intelligence systems for the cooperative smart farming ecosystem," *IEEE Access*, vol. 8, pp. 164045–164064, 2020.
- [19] (Feb. 2020). *Hackers Can Trick a Tesla Into Accelerating by 50 Miles Per Hour*. Accessed: Aug. 8, 2023. [Online]. Available: <https://www.technologyreview.com/2020/02/19/868188/hackers-can-trick-a-tesla-into-accelerating-by-50-miles-per-hour>
- [20] (2022). *Legendary Hackers Charlie Miller and Chris Valasek Talk Cybersecurity and Autonomous Vehicles At Tc Sessions: Mobility*. Accessed: Sep. 26, 2022. [Online]. Available: <https://tcn.ch/3iN2mRx>
- [21] C. Miller, "Lessons learned from hacking a car," *IEEE Design Test.*, vol. 36, no. 6, pp. 7–9, Dec. 2019.
- [22] (Mar. 2023). *Artificial Intelligence and Cybersecurity Opportunities and Challenges*. Accessed: Jul. 27, 2023. [Online]. Available: <https://www.nitrd.gov/pubs/AI-CS-Tech-Summary-2020.pdf>
- [23] S. Mittal and J. Chen, "AI security threats against pervasive robotic systems: A course for next generation cybersecurity workforce," 2023, *arXiv:2302.07953*.
- [24] E. Bertino, R. Sandhu, B. Thuraisingham, I. Ray, W. Li, M. Gupta, and S. Mittal, "Security and privacy for emerging IoT and CPS domains," in *Proc. 12th ACM Conf. Data Appl. Secur. Privacy*, Apr. 2022, pp. 336–337.
- [25] L. Riek and D. Howard, "A code of ethics for the human–robot interaction profession," *Proc. Robot.*, pp. 1–10, Apr. 2014.
- [26] E. Datteri and G. Tamburrini, "Ethical reflections on health care robotics," *Ethics Robot.*, pp. 35–48, Oct. 2009.
- [27] *California Consumer Privacy Act*. Accessed: Oct. 5, 2023. [Online]. Available: <https://oag.ca.gov/privacy/ccpa>
- [28] *General Data Protection Regulation*. Accessed: Jul. 8, 2023. [Online]. Available: <https://www.techtarget.com/whatis/definition/General-Data-Protection-Regulation-GDPR>
- [29] A. Botta, S. Rotbei, S. Zinno, and G. Ventre, "Cyber security of robots: A comprehensive survey," *Intell. Syst. Appl.*, vol. 18, May 2023, Art. no. 200237.
- [30] V. Dutta and T. Zieli ska, "Cybersecurity of robotic systems: Leading challenges and robotic system design methodology," *Electronics*, vol. 10, no. 22, p. 2850, Nov. 2021.
- [31] S. H. Alsamhi, O. Ma, and M. S. Ansari, "Survey on artificial intelligence based techniques for emerging robotic communication," *Telecommun. Syst.*, vol. 72, no. 3, pp. 483–503, 2019.
- [32] N. Emaminejad and R. Akhavan, "Trustworthy AI and robotics: Implications for the AEC industry," *Autom. Construct.*, vol. 139, Jul. 2022, Art. no. 104298.
- [33] F. Jahan, W. Sun, Q. Niyaz, and M. Alam, "Security modeling of autonomous systems: A survey," *ACM Comput. Surv.*, vol. 52, no. 5, pp. 1–34, Sep. 2020.
- [34] *Flaticon*. Accessed: Aug. 30, 2023. [Online]. Available: <https://www.flaticon.com/>
- [35] R. R. Murphy, *Introduction to AI Robotics*. Cambridge, MA, USA: MIT Press, 2019.
- [36] Z. Kappassov, J.-A. Corrales, and V. Perdureau, "Tactile sensing in dexterous robot hands," *Robot. Auto. Syst.*, vol. 74, pp. 195–220, Dec. 2015.
- [37] P. Dash, M. Karimibiuki, and K. Pattabiraman, "Stealthy attacks against robotic vehicles protected by control-based intrusion detection techniques," *Digit. Threats, Res. Pract.*, vol. 2, no. 1, pp. 1–25, Mar. 2021.
- [38] H. Moravec and A. Elfes, "High resolution maps from wide angle sonar," in *Proc. IEEE Int. Conf. Robot. Autom.*, vol. 2, Mar. 1985, pp. 116–121.

- [39] A. Hornung, K. M. Wurm, M. Bennewitz, C. Stachniss, and W. Burgard, "OctoMap: An efficient probabilistic 3D mapping framework based on octrees," *Auto. Robots*, vol. 34, no. 3, pp. 189–206, Apr. 2013.
- [40] A. I. Karoly, P. Galambos, J. Kuti, and I. J. Rudas, "Deep learning in robotics: Survey on model structures and training strategies," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 51, no. 1, pp. 266–279, Jan. 2021.
- [41] M. Schwarz, H. Schulz, and S. Behnke, "RGB-D object recognition and pose estimation based on pre-trained convolutional neural network features," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2015, pp. 1329–1335.
- [42] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.
- [43] C. Farabet, C. Couprie, L. Najman, and Y. LeCun, "Learning hierarchical features for scene labeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1915–1929, Aug. 2013.
- [44] P. Pinheiro and R. Collobert, "Recurrent convolutional neural networks for scene labeling," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 82–90.
- [45] C. Moore, S. Mitra, N. Pillai, M. Moore, S. Mittal, C. Bethel, and J. Chen, "URA: Uncertainty-aware path planning using image-based aerial-to-ground traversability estimation for off-road environments," 2023, *arXiv:2309.08814*.
- [46] W. Kehl, F. Milletari, F. Tombari, S. Ilıc, and N. Navab, "Deep learning of local RGB-D patches for 3D object detection and 6D pose estimation," in *Proc. 14th Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands: Cham, Switzerland: Springer, 2016, pp. 205–220.
- [47] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
- [48] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, "V2X-VIT: Vehicle-to-everything cooperative perception with vision transformer," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Cham, Switzerland: Springer, 2022, pp. 107–124.
- [49] C. Fu, W. Peng, S. Li, J. Ye, and Z. Cao, "Local perception-aware transformer for aerial tracking," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2022, pp. 12122–12129.
- [50] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Trans. Syst. Sci. Cybern.*, vol. SCS-4, no. 2, pp. 100–107, Jul. 1968.
- [51] S. LaValle, "Rapidly-exploring random trees: A new tool for path planning," Dept. Comput. Sci., Iowa State Univ., Ames, Iowa, Tech. Rep. 9811, 1998.
- [52] J. Reeds and L. Shepp, "Optimal paths for a car that goes both forwards and backwards," *Pac. J. Math.*, vol. 145, no. 2, pp. 367–393, 1990.
- [53] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robot. Autom. Mag.*, vol. 4, no. 1, pp. 23–33, Mar. 1997.
- [54] L. Kästner, X. Zhao, T. Buiyan, J. Li, Z. Shen, J. Lambrecht, and C. Marx, "Connecting deep-reinforcement-learning-based obstacle avoidance with conventional global planners using waypoint generators," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2021, pp. 1213–1220.
- [55] N. Pérez-Higueras, F. Caballero, and L. Merino, "Learning human-aware path planning with fully convolutional networks," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 5897–5902.
- [56] G. Chen, L. Pan, Y. Chen, P. Xu, Z. Wang, P. Wu, J. Ji, and X. Chen, "Robot navigation with map-based deep reinforcement learning," in *Proc. IEEE Int. Conf. Netw., Sens. Control (ICNSC)*, Oct. 2020, pp. 1–6.
- [57] A. R. Puthussery, K. P. Haradi, B. A. Erol, P. Benavidez, P. Rad, and M. Jamshidi, "A deep vision landmark framework for robot navigation," in *Proc. 12th Syst. Syst. Eng. Conf. (SoSE)*, Jun. 2017, pp. 1–6.
- [58] C. Richter and N. Roy, *Safe Visual Navigation via Deep Learning and Novelty Detection*. Cambridge, MA, USA: MIT Press, 2017.
- [59] P. Regier, L. Gesing, and M. Bennewitz, "Deep reinforcement learning for navigation in cluttered environments," in *Proc. Intl. Conf. Mach. Learn. Appl. (CMLA)*, 2020, pp. 1–12.
- [60] P. Chen, J. Pei, W. Lu, and M. Li, "A deep reinforcement learning based method for real-time path planning and dynamic obstacle avoidance," *Neurocomputing*, vol. 497, pp. 64–75, Aug. 2022.
- [61] D. Morrison, P. Corke, and J. Leitner, "Closing the loop for robotic grasping: A real-time, generative grasp synthesis approach," 2018, *arXiv:1804.05172*.
- [62] J. A. Shaw, *The PID Control Algorithm*. Boston, MA, USA: Process Control Solutions, 2003.
- [63] L. B. Prasad, B. Tyagi, and H. O. Gupta, "Optimal control of nonlinear inverted pendulum system using PID controller and LQR: Performance analysis without and with disturbance input," *Int. J. Autom. Comput.*, vol. 11, no. 6, pp. 661–670, Dec. 2014.
- [64] J. Kober, J. A. Bagnell, and J. Peters, "Reinforcement learning in robotics: A survey," *Int. J. Robot. Res.*, vol. 32, no. 11, pp. 1238–1274, Sep. 2013.
- [65] D. Kalashnikov, A. Irpan, P. Pastor, J. Ibarz, A. Herzog, E. Jang, D. Quillen, E. Holly, M. Kalakrishnan, V. Vanhoucke, and S. Levine, "Scalable deep reinforcement learning for vision-based robotic manipulation," in *Proc. Int. Conf. Robot Learn.*, 2018, pp. 651–673.
- [66] S. Levine, N. Wagener, and P. Abbeel, "Learning contact-rich manipulation skills with guided policy search," 2015, *arXiv:1501.05611*.
- [67] K. Eykholt, I. Evtimov, E. Fernandes, B. Li, A. Rahmati, C. Xiao, A. Prakash, T. Kohno, and D. Song, "Robust physical-world attacks on deep learning visual classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1625–1634.
- [68] R. Ross, V. Pillitteri, K. Dempsey, M. Riddle, and G. Guisanie, *Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations*. Gaithersburg, MD, USA: Nat. Inst. Standards Technol., 2019.
- [69] S. Hollerer, C. Fischer, B. Brenner, M. Papa, S. Schlund, W. Kastner, J. Babini, and T. Zseby, "Cobot attack: A security assessment exemplified by a specific collaborative robot," *Proc. Manuf.*, vol. 54, pp. 191–196, Jan. 2021.
- [70] S. O. Oruma, M. Sánchez-Gordón, R. Colomo-Palacios, V. Gkioulos, and J. K. Hansen, "A systematic review on social robots in public spaces: Threat landscape and attack surface," *Computers*, vol. 11, no. 12, p. 181, Dec. 2022.
- [71] S. Bhunia, M. S. Hsiao, M. Banga, and S. Narasimhan, "Hardware Trojan attacks: Threat analysis and countermeasures," *Proc. IEEE*, vol. 102, no. 8, pp. 1229–1247, Aug. 2014.
- [72] Y. Deng, T. Zhang, G. Lou, X. Zheng, J. Jin, and Q. L. Han, "Deep learning-based autonomous driving systems: A survey of attacks and defenses," *IEEE Trans. Ind. Informat.*, vol. 17, no. 12, pp. 7897–7912, Dec. 2021.
- [73] C. Yan, W. Xu, and J. Liu, "Can you trust autonomous vehicles: Contactless attacks against sensors of self-driving vehicle," *Defcon*, vol. 24, no. 8, p. 109, 2016.
- [74] H. Shin, D. Kim, Y. Kwon, and Y. Kim, "Illusion and dazzle: Adversarial optical channel exploits against LiDARs for automotive applications," in *Proc. 19th Int. Conf. Cryptograph. Hardw. Embedded Syst.*, Taipei, Taiwan, Cham, Switzerland: Springer, 2017, pp. 445–467.
- [75] B. S. Lim, S. L. Keoh, and V. L. L. Thing, "Autonomous vehicle ultrasonic sensor vulnerability and impact assessment," in *Proc. IEEE 4th World Forum Internet Things (WF-IoT)*, Feb. 2018, pp. 231–236.
- [76] G. Kar, H. Mustafa, Y. Wang, Y. Chen, W. Xu, M. Gruteser, and T. Vu, "Detection of on-road vehicles emanating GPS interference," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2014, pp. 621–632.
- [77] Y. Cao, C. Xiao, B. Cyr, Y. Zhou, W. Park, S. Rampazzi, Q. A. Chen, K. Fu, and Z. M. Mao, "Adversarial sensor attack on LiDAR-based perception in autonomous driving," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2019, pp. 2267–2281.
- [78] Z. Jin, X. Ji, Y. Cheng, B. Yang, C. Yan, and W. Xu, "PLA-LiDAR: Physical laser attacks against LiDAR-based 3D object detection in autonomous vehicle," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2023, pp. 710–727.
- [79] Y. Cao, N. Wang, C. Xiao, D. Yang, J. Fang, R. Yang, Q. A. Chen, M. Liu, and B. Li, "Invisible for both camera and LiDAR: Security of multi-sensor fusion based perception in autonomous driving under physical-world attacks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2021, pp. 176–194.
- [80] Y. Zhu, C. Miao, T. Zheng, F. Hajiaghajani, L. Su, and C. Qiao, "Can we use arbitrary objects to attack LiDAR perception in autonomous driving?" in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2021, pp. 1945–1960.

- [81] D. F. Kune, J. Backes, S. S. Clark, D. Kramer, M. Reynolds, K. Fu, Y. Kim, and W. Xu, "Ghost Talk: Mitigating EMI signal injection attacks against analog sensors," in *Proc. IEEE Symp. Secur. Privacy*, May 2013, pp. 145–159.
- [82] W. Xu, C. Yan, W. Jia, X. Ji, and J. Liu, "Analyzing and enhancing the security of ultrasonic sensors for autonomous vehicles," *IEEE Internet Things J.*, vol. 5, no. 6, pp. 5015–5029, Dec. 2018.
- [83] S. Rivera, S. Lagraa, A. K. Iannillo, and R. State, "Auto-encoding robot state against sensor spoofing attacks," in *Proc. IEEE Int. Symp. Softw. Rel. Eng. Workshops (ISSREW)*, Oct. 2019, pp. 252–257.
- [84] P. Kapoor, A. Vora, and K. Kang, "Detecting and mitigating spoofing attack against an automotive radar," in *Proc. IEEE 88th Veh. Technol. Conf. (VTC-Fall)*, Aug. 2018, pp. 1–6.
- [85] K. C. Zeng, S. Liu, Y. Shu, D. Wang, H. Li, Y. Dou, G. Wang, and Y. Yang, "All your GPS are belong to us: Towards stealthy manipulation of road navigation systems," in *Proc. 27th USENIX Secur. Symp.*, 2018, pp. 1527–1544.
- [86] X. Ji, Y. Cheng, Y. Zhang, K. Wang, C. Yan, W. Xu, and K. Fu, "Poltergeist: Acoustic adversarial machine learning against cameras and computer vision," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2021, pp. 160–175.
- [87] G. Bradski, "The OpenCV library," *Dr. Dobbs's J., Softw. Tools Prof. Programmer*, vol. 25, no. 11, pp. 120–123, Nov. 2000.
- [88] M. Abadi et al., "TensorFlow: Large-scale machine learning on heterogeneous distributed systems," 2016, *arXiv:1603.04467*.
- [89] Q.-Y. Zhou, J. Park, and V. Koltun, "Open3D: A modern library for 3D data processing," 2018, *arXiv:1801.09847*.
- [90] S. Nashimoto, D. Suzuki, T. Sugawara, and K. Sakiyama, "Sensor CON-fusion: Defeating Kalman filter in signal injection attack," in *Proc. Asia Conf. Comput. Commun. Secur.*, May 2018, pp. 511–524.
- [91] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.
- [92] CVE-2018-5268 Detail. [Online]. Available: <https://nvd.nist.gov/vuln/detail/CVE-2018-5268>
- [93] CVE-2021-37650 Detail. [Online]. Available: <https://nvd.nist.gov/vuln/detail/CVE-2021-37650>
- [94] J. Petit, B. Stottelaar, M. Feiri, and F. Kargl, "Remote attacks on automated vehicles sensors: Experiments on camera and LiDAR," *Black Hat Eur.*, vol. 11, p. 995, Nov. 2015.
- [95] Y. Xu, X. Han, G. Deng, J. Li, Y. Liu, and T. Zhang, "SoK: Rethinking sensor spoofing attacks against robotic vehicles from a systematic view," 2022, *arXiv:2205.04662*.
- [96] J. H. Ryu, M. Irfan, and A. Reyaz, "A review on sensor network issues and robotics," *J. Sensors*, vol. 2015, pp. 1–14, Aug. 2015.
- [97] A. Bhardwaj, V. Avasthi, and S. Goundar, "Cyber security attacks on robotic platforms," *Netw. Secur.*, vol. 2019, no. 10, pp. 13–19, Oct. 2019.
- [98] G. Bianchin, Y.-C. Liu, and F. Pasqualetti, "Secure navigation of robots in adversarial environments," *IEEE Control Syst. Lett.*, vol. 4, no. 1, pp. 1–6, Jan. 2020.
- [99] C. H. Yang, J. Qi, P.-Y. Chen, Y. Ouyang, I. D. Hung, C.-H. Lee, and X. Ma, "Enhanced adversarial strategically-timed attacks against deep reinforcement learning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 3407–3411.
- [100] V. Dey, V. Pudi, A. Chattopadhyay, and Y. Elovici, "Security vulnerabilities of unmanned aerial vehicles and countermeasures: An experimental study," in *Proc. 31st Int. Conf. VLSI Design 17th Int. Conf. Embedded Syst. (VLSID)*, Jan. 2018, pp. 398–403.
- [101] D. He, H. Liu, S. Chan, and M. Guizani, "How to govern the non-cooperative amateur drones?" *IEEE Netw.*, vol. 33, no. 3, pp. 184–189, May 2019.
- [102] D. He, Y. Qiao, S. Chen, X. Du, W. Chen, S. Zhu, and M. Guizani, "A friendly and low-cost technique for capturing non-cooperative civilian unmanned aerial vehicles," *IEEE Netw.*, vol. 33, no. 2, pp. 146–151, Mar. 2019.
- [103] H. Sathaye, M. Strohmeier, V. Lenders, and A. Ranganathan, "An experimental study of GPS spoofing and takeover attacks on UAVs," in *Proc. 31st USENIX Secur. Symp.*, 2022, pp. 3503–3520.
- [104] J. Shen, J. Y. Won, Z. Chen, and Q. A. Chen, "Drift with devil: Security of multi-sensor fusion based localization in high-level autonomous driving under GPS spoofing," in *Proc. 29th USENIX Secur. Symp.*, 2020, pp. 931–948.
- [105] Y. Li, C. Wen, F. Juefei-Xu, and C. Feng, "Fooling LiDAR perception via adversarial trajectory perturbation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7898–7907.
- [106] K. Tang, J. Shen, and Q. A. Chen, "Fooling perception via location: A case of region-of-interest attacks on traffic light detection in autonomous driving," in *Proc. 3rd Int. Workshop Automot. Auto. Vehicle Secur.*, 2021, pp. 1–4.
- [107] Z. Jin, X. Ji, Y. Cheng, B. Yang, C. Yan, and W. Xu, "PLA-LiDAR: Physical laser attacks against LiDAR-based 3D object detection in autonomous vehicle," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2023, pp. 1822–1839.
- [108] J. Sun, Y. Cao, Q. A. Chen, and Z. M. Mao, "Towards robust LiDAR-based perception in autonomous driving: General black-box adversarial sensor attack and countermeasures," in *Proc. 29th USENIX Secur. Symp.*, 2020, pp. 877–894.
- [109] Y. Son, H. Shin, D. Kim, Y. Park, J. Noh, K. Choi, J. Choi, and Y. Kim, "Rocking drones with intentional sound noise on gyroscopic sensors," in *Proc. 24th USENIX Secur. Symp.*, 2015, pp. 881–896.
- [110] T. Trippel, O. Weisse, W. Xu, P. Honeyman, and K. Fu, "WALNUT: Waging doubt on the integrity of MEMS accelerometers with acoustic injection attacks," in *Proc. IEEE Eur. Symp. Secur. Privacy (EuroSP)*, Apr. 2017, pp. 3–18.
- [111] Y. Tu, Z. Lin, I. Lee, and X. Hei, "Injected and delivered: Fabricating implicit control over actuation systems by spoofing inertial sensors," in *Proc. 27th USENIX Secur. Symp.*, 2018, pp. 1545–1562.
- [112] S. Vemprala and A. Kapoor, "Adversarial attacks on optimization based planners," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2021, pp. 9943–9949.
- [113] D. Rempe, J. Philion, L. J. Guibas, S. Fidler, and O. Litany, "Generating useful accident-prone driving scenarios via a learned traffic prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jul. 2022, pp. 17305–17315.
- [114] A. B. Bhandarkar, S. K. Jayaweera, and S. A. Lane, "Adversarial Sybil attacks against deep RL based drone trajectory planning," in *Proc. IEEE Mil. Commun. Conf. (MILCOM)*, Nov. 2022, pp. 1–6.
- [115] T. Hickling, N. Aouf, and P. Spencer, "Robust adversarial attacks detection based on explainable deep reinforcement learning for UAV guidance and planning," 2022, *arXiv:2206.02670*.
- [116] X. Peng and D. Xu, "Intelligent online path planning for UAVs in adversarial environments," *Int. J. Adv. Robotic Syst.*, vol. 9, no. 1, p. 3, Mar. 2012.
- [117] W. Wang, Y. Yao, X. Liu, X. Li, P. Hao, and T. Zhu, "I can see the light: Attacks on autonomous vehicles using invisible lights," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2021, pp. 1930–1944.
- [118] L. Zhou, V. Tzoumas, G. J. Pappas, and P. Tokekar, "Resilient active target tracking with multiple robots," *IEEE Robot. Autom. Lett.*, vol. 4, no. 1, pp. 129–136, Jan. 2019.
- [119] D. Saldaña, A. Prorok, S. Sundaram, M. F. M. Campos, and V. Kumar, "Resilient consensus for time-varying networks of dynamic agents," in *Proc. Amer. Control Conf. (ACC)*, May 2017, pp. 252–258.
- [120] R. K. Ramachandran, J. A. Preiss, and G. S. Sukhatme, "Resilience by reconfiguration: Exploiting heterogeneity in robot teams," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Nov. 2019, pp. 6518–6525.
- [121] J. Song and S. Gupta, "CARE: Cooperative autonomy for resilience and efficiency of robot teams for complete coverage of unknown environments under robot failures," *Auto. Robots*, vol. 44, nos. 3–4, pp. 647–671, Mar. 2020.
- [122] L. Zhou and P. Tokekar, "Multi-robot coordination and planning in uncertain and adversarial environments," *Current Robot. Rep.*, vol. 2, no. 2, pp. 147–157, Apr. 2021.
- [123] H. Park and S. Hutchinson, "Robust rendezvous for multi-robot system with random node failures: An optimization approach," *Auto. Robots*, vol. 42, no. 8, pp. 1807–1818, Dec. 2018.
- [124] F. Yang and N. Chakraborty, "Algorithm for optimal chance constrained linear assignment," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2017, pp. 801–808.
- [125] L. Zhou and P. Tokekar, "An approximation algorithm for risk-averse submodular optimization," in *Proc. Int. Workshop Algorithmic Found. Robot. Cham, Switzerland: Springer*, 2018, pp. 144–159.
- [126] E. Tolstaya, F. Gama, J. Paulos, G. Pappas, V. Kumar, and A. Ribeiro, "Learning decentralized controllers for robot swarms with graph neural networks," in *Proc. Conf. Robot Learn.*, 2020, pp. 671–682.

- [127] Q. Li, F. Gama, A. Ribeiro, and A. Prorok, "Graph neural networks for decentralized multi-robot path planning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2020, pp. 11785–11792.
- [128] A. Li, J. Wang, and N. Zhang, "Chronos: Timing interference as a new attack vector on autonomous cyber-physical systems," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2021, pp. 2426–2428.
- [129] X. Wang and M. C. Gursay, "Resilient UAV path planning for data collection under adversarial attacks," in *Proc. IEEE Int. Conf. Commun.*, May 2022, pp. 625–630.
- [130] X. Wang and M. C. Gursay, "Resilient path planning for UAVs in data collection under adversarial attacks," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2766–2779, 2023.
- [131] D. Herath and D. St-Onge, *Foundations of Robotics: A Multidisciplinary Approach With Python and ROS*. Berlin, Germany: Springer, 2022.
- [132] X. Wang, T. Mal-Sarkar, A. Krishna, S. Narasimhan, and S. Bhunia, "Software exploitable hardware Trojans in embedded processor," in *Proc. IEEE Int. Symp. Defect Fault Tolerance VLSI Nanotechnol. Syst. (DFT)*, Oct. 2012, pp. 55–58.
- [133] R. Sasi, "Maldrone the first backdoor for drones," *Rahul Sasi's Blog*, 2015.
- [134] V. M. Vilches, L. A. Kirschgens, A. B. Calvo, A. H. Cordero, R. I. Pisón, D. M. Vilches, A. M. Rosas, G. O. Mendia, L. U. S. Juan, I. Z. Ugarte, E. Gil-Uriarte, E. Tews, and A. Peter, "Introducing the robot security framework (RSF), a standardized methodology to perform security assessments in robotics," 2018, *arXiv:1806.04042*.
- [135] L. A. Kirschgens, I. Z. Ugarte, E. G. Uriarte, A. M. Rosas, and V. M. Vilches, "Robot hazards: From safety to security," 2018, *arXiv:1806.06681*.
- [136] M. Pogliani, D. Quarta, M. Polino, M. Vittone, F. Maggi, and S. Zanero, "Security of controlled manufacturing systems in the connected factory: The case of industrial robots," *J. Comput. Virol. Hacking Techn.*, vol. 15, no. 3, pp. 161–175, Sep. 2019.
- [137] R. Douceur, "The Sybil attack," in *Proc. 1st Int. Workshop Peer-to-Peer Syst. Cham, Switzerland: Springer*, 2002, pp. 251–260.
- [138] G. Lacava, A. Marotta, F. Martinelli, A. Saracino, A. L. Marra, E. Gil-Uriarte, and V. M. Vilches, "Cybersecurity issues in robotics," *J. Wireless Mobile Netw., Ubiquitous Comput. Dependable Appl.*, vol. 12, no. 3, pp. 1–28, 2021.
- [139] D. Quarta, M. Pogliani, M. Polino, F. Maggi, A. M. Zanchettin, and S. Zanero, "An experimental security analysis of an industrial robot controller," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 268–286.
- [140] T. Bonaci, J. Herron, T. Yusuf, J. Yan, T. Kohno, and H. J. Chizeck, "To make a robot secure: An experimental analysis of cyber security threats against teleoperated surgical robots," 2015, *arXiv:1504.04339*.
- [141] H. Alemzadeh, D. Chen, X. Li, T. Kesavadas, Z. T. Kalbarczyk, and R. K. Iyer, "Targeted attacks on teleoperated surgical robots: Dynamic model-based detection and mitigation," in *Proc. 46th Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. (DSN)*, Jun. 2016, pp. 395–406.
- [142] P. Guo, H. Kim, N. Virani, J. Xu, M. Zhu, and P. Liu, "Exploiting physical dynamics to detect actuator and sensor attacks in mobile robots," 2017, *arXiv:1708.01834*.
- [143] Y. Gao, G. Sun, J. Liu, Y. Shi, and L. Wu, "State estimation and self-triggered control of CPSs against joint sensor and actuator attacks," *Automatica*, vol. 113, Mar. 2020, Art. no. 108687.
- [144] J. Giraldo, S. H. Kafash, J. Ruths, and A. A. Cardenas, "DARIA: Designing actuators to resist arbitrary attacks against cyber-physical systems," in *Proc. IEEE Eur. Symp. Secur. Privacy (EuroS&P)*, Sep. 2020, pp. 339–353.
- [145] X. Huang and J. Dong, "Reliable control policy of cyber-physical systems against a class of frequency-constrained sensor and actuator attacks," *IEEE Trans. Cybern.*, vol. 48, no. 12, pp. 3432–3439, Dec. 2018.
- [146] H. He, W. Qi, and Y. Kao, "HMM-based adaptive attack-resilient control for Markov jump system and application to an aircraft model," *Appl. Math. Comput.*, vol. 392, Mar. 2021, Art. no. 125668.
- [147] C. Escudero, C. Murguía, P. Massioni, and E. Zamañ, "Enforcing safety under actuator injection attacks through input filtering," in *Proc. Eur. Control Conf. (ECC)*, Jul. 2022, pp. 1521–1528.
- [148] D. Park, Z. Erickson, T. Bhattacharjee, and C. C. Kemp, "Multimodal execution monitoring for anomaly detection during robot manipulation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2016, pp. 407–414.
- [149] P. Guo, H. Kim, N. Virani, J. Xu, M. Zhu, and P. Liu, "RoboADS: Anomaly detection against sensor and actuator misbehaviors in mobile robots," in *Proc. 48th Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. (DSN)*, Jun. 2018, pp. 574–585.
- [150] Y. Zhang and S. Li, "Kinematic control of serial manipulators under false data injection attack," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 4, pp. 1009–1019, Apr. 2023.
- [151] M. A. Abd, I. Gonzalez, C. Ades, M. Nojournian, and E. D. Engeberg, "Simulated robotic device malfunctions resembling malicious cyber-attacks impact human perception of trust, satisfaction, and frustration," *Int. J. Adv. Robotic Syst.*, vol. 16, no. 5, Sep. 2019, Art. no. 172988141987496.
- [152] B. Browatzki, V. Tikhonoff, G. Metta, H. H. Bülthoff, and C. Wallraven, "Active object recognition on a humanoid robot," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2012, pp. 2021–2028.
- [153] F. Alonso-Martín and M. A. Salichs, "Integration of a voice recognition system in a social robot," *Cybern. Syst.*, vol. 42, no. 4, pp. 215–245, May 2011.
- [154] C. Cruz, L. E. Sucar, and E. F. Morales, "Real-time face recognition for human-robot interaction," in *Proc. 8th IEEE Int. Conf. Autom. Face Gesture Recognit.*, Sep. 2008, pp. 1–6.
- [155] M. Barreno, B. Nelson, A. D. Joseph, and J. D. Tygar, "The security of machine learning," *Mach. Learn.*, vol. 81, no. 2, pp. 121–148, 2010.
- [156] D. Castellevecchi, "Can we open the black box of AI?" *Nature*, vol. 538, no. 7623, pp. 20–23, Oct. 2016.
- [157] (2016). In 2016, Microsoft's Racist Chatbot Revealed the Dangers of Online Conversation. Accessed: Aug. 23, 2021. [Online]. Available: <https://spectrum.ieee.org/in-2016-microsofts-racist-chatbot-revealed-the-dangers-of-online-conversation>
- [158] *How Data Poisoning Attacks Can Corrupt Machine Learning Models*. Accessed: Aug. 23, 2021. [Online]. Available: <https://www.linkedin.com/pulse/how-data-poisoning-attacks-can-corrupt-machine-learning-misra>
- [159] K. Liu, B. Dolan-Gavitt, and S. Garg, "Fine-pruning: Defending against backdooring attacks on deep neural networks," in *Proc. RAID*, 2018, pp. 273–294.
- [160] M. Goldblum, D. Tsipras, C. Xie, X. Chen, A. Schwarzschild, D. Song, A. Madry, B. Li, and T. Goldstein, "Dataset security for machine learning: Data poisoning, backdoor attacks, and defenses," 2020, *arXiv:2012.10544*.
- [161] M. Charikar, J. Steinhardt, and G. Valiant, "Learning from untrusted data," in *Proc. 49th Annu. ACM SIGACT Symp. Theory Comput.* New York, NY, USA: Association for Computing Machinery, Jun. 2017, pp. 47–60.
- [162] G. Severi, J. Meyer, S. Coull, and A. Oprea, "Explanation-guided backdoor poisoning attacks against malware classifiers," in *Proc. 30th USENIX Secur. Symp.* Berkeley, CA, USA: USENIX Association, 2021, pp. 1487–1504.
- [163] Y. Liu, Y. Xie, and A. Srivastava, "Neural Trojans," in *Proc. IEEE Int. Conf. Comput. Design (ICCD)*. Los Alamitos, MA, USA: IEEE Computer Society, Nov. 2017, pp. 45–48.
- [164] M. Jagielski, A. Oprea, B. Biggio, C. Liu, C. N. Rotaru, and B. Li, "Manipulating machine learning: Poisoning attacks and countermeasures for regression learning," in *Proc. IEEE Symp. Security Privacy*, San Francisco, CA, USA, 2018, pp. 19–35.
- [165] L. Zhu, R. Ning, C. Wang, C. Xin, and H. Wu, "GangSweep: Sweep out neural backdoors by GAN," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 3173–3181.
- [166] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 3–18.
- [167] M. Fredrikson, S. Jha, and T. Ristenpart, "Model inversion attacks that exploit confidence information and basic countermeasures," in *Proc. 22nd ACM SIGSAC Conf. Comput. Commun. Secur.*, Oct. 2015, pp. 1322–1333.
- [168] Y. Kaya and T. Dumitras, "When does data augmentation help with membership inference attacks?" in *Proc. ICML*, 2021, pp. 5345–5355.
- [169] G. R. Machado, E. Silva, and R. R. Goldschmidt, "Adversarial machine learning in image classification: A survey toward the defender's perspective," *ACM Comput. Surv.*, vol. 55, no. 1, pp. 1–38, Nov. 2021.
- [170] X. Chen, A. Salem, D. Chen, M. Backes, S. Ma, Q. Shen, Z. Wu, and Y. Zhang, "BadNL: Backdoor attacks against NLP models with semantic-preserving improvements," 2020, *arXiv:2006.01043*.

- [171] A. Kashyap, S. M. Iqbal, K. Pattabiraman, and M. Seltzer, "ReLUSyn: Synthesizing stealthy attacks for deep neural network based cyber-physical systems," 2021, *arXiv:2105.10393*.
- [172] J. Bai, B. Wu, Y. Zhang, Y. Li, Z. Li, and S.-T. Xia, "Targeted attack against deep neural networks via flipping limited weight bits," 2021, *arXiv:2102.10496*.
- [173] X. Qi, J. Zhu, C. Xie, and Y. Yang, "Subnet replacement: Deployment-stage backdoor attack against deep neural networks in gray-box setting," 2021, *arXiv:2107.07240*.
- [174] J. Pan, "Blackbox trojanising of deep learning models: Using non-intrusive network structure and binary alterations," in *Proc. IEEE REGION 10th Conf. (TENCON)*, Nov. 2020, pp. 891–896.
- [175] Y. Wen, L. Song, B. Liu, M. Ding, and R. Xie, "IdentityDP: Differential private identification protection for face images," 2021, *arXiv:2103.01745*.
- [176] M. Maximov, I. Elezi, and L. Leal-Taixé, "CIAGAN: Conditional identity anonymization generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5446–5455.
- [177] S. Zanella-Béguelin, L. Wutschitz, S. Tople, V. Rühle, A. Pavard, O. Ohrimenko, B. Köpf, and M. Brockschmidt, "Analyzing information leakage of updates to natural language models," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Oct. 2020, pp. 1–13.
- [178] T. Shibata, G. Irie, D. Ikami, and Y. Mitsuzumi, "Learning with selective forgetting," in *Proc. 30th Int. Joint Conf. Artif. Intell.*, Aug. 2021, pp. 989–996.
- [179] S. Wang, K. Pei, J. Whitehouse, J. Yang, and S. Jana, "Formal security analysis of neural networks using symbolic intervals," in *Proc. USENIX Secur. Symp.*, 2018, pp. 1599–1614.
- [180] T. Orekondy, B. Schiele, and M. Fritz, "Prediction poisoning: Towards defenses against DNN model stealing attacks," 2019, *arXiv:1906.10908*.
- [181] A. Ashkenazy and S. Zini, "Cylance, I kill you," Tech. Rep., 2019.
- [182] K. Chung, X. Li, P. Tang, Z. Zhu, Z. T. Kalbarczyk, R. K. Iyer, and T. Kesavadas, "Smart malware that uses leaked control data of robotic applications: The case of Raven-II surgical robots," in *Proc. 22nd Int. Symp. Res. Attacks, Intrusions Defenses*, 2019, pp. 337–351.
- [183] V. Mayoral-Vilches, U. A. Carbajo, and E. Gil-Urriarte, "Industrial robot ransomware: Akerbeltz," in *Proc. 4th IEEE Int. Conf. Robot. Comput. (IRC)*, Nov. 2020, pp. 432–435.
- [184] C. L. Krishna and R. R. Murphy, "A review on cybersecurity vulnerabilities for unmanned aerial vehicles," in *Proc. IEEE Int. Symp. Safety Security Rescue Robot. (SSRR)*, Oct. 2017, pp. 194–199.
- [185] G. Veruggio, "The birth of Roboethics," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Apr. 2005, pp. 1–4.
- [186] (2023). *Robot Ethics*. Accessed: Aug. 20, 2023. [Online]. Available: <https://www.ieee-ras.org/robot-ethics>
- [187] S. G. Tzafestas, *Roboethics: A Navigating Overview*. Berlin, Germany: Springer, 2016.
- [188] A. Act, "Health insurance portability and accountability act of 1996," *Public Law*, vol. 104, p. 191, Aug. 1996.
- [189] *Regulation (eu) 2016/679 of the European Parliament and of the Council*, document 679, Protection Regulation, 2016.
- [190] H. Wang, A. Khajepour, D. Cao, and T. Liu, "Ethical decision making in autonomous vehicles: Challenges and research progress," *IEEE Intell. Transp. Syst. Mag.*, vol. 14, no. 1, pp. 6–17, Jan. 2022.
- [191] N. J. Goodall, "Can you program ethics into a self-driving car?" *IEEE Spectr.*, vol. 53, no. 6, pp. 28–58, Jun. 2016.
- [192] B. Goodman and S. Flaxman, "European union regulations on algorithmic decision-making and a 'right to explanation,'" *AI Mag.*, vol. 38, no. 3, pp. 50–57, 2017.
- [193] P. Lin, "The ethics of saving lives with autonomous cars are far murkier than you think," *Wired Opinion*, vol. 30, Jul. 2013.
- [194] E. A. McDaniel, "Robot wars: Legal and ethical dilemmas of using unmanned robotic systems in 21st century warfare and beyond," M.S. thesis, U.S. Army Command Gen. Staff College, Leavenworth, KS, USA, 2008.
- [195] I. Asimov, "Three laws of robotics," *Asimov, I. Runaround*, vol. 2, 1941.
- [196] P. M. Asaro, "The liability problem for autonomous artificial agents," in *Proc. AAAI Spring Symp. Ser.*, 2016, pp. 190–194.
- [197] S. O'Sullivan, N. Nevejans, C. Allen, A. Blyth, S. Leonard, U. Pagallo, K. Holzinger, A. Holzinger, M. I. Sajid, and H. Ashrafian, "Legal, regulatory, and ethical frameworks for development of standards in artificial intelligence (AI) and autonomous robotic surgery," *Int. J. Med. Robot. Comput. Assist. Surgery*, vol. 15, no. 1, p. e1968, Feb. 2019.
- [198] *Preparing for the Future of Artificial Intelligence*. Accessed: Sep. 30, 2023. [Online]. Available: https://obamawhitehouse.archives.gov/sites/default/files/whitehouse_files/microsites/ostp/NSTC/preparing_for_the_future_of_ai.pdf
- [199] K. Ishii, "Comparative legal study on privacy and personal data protection for robots equipped with artificial intelligence: Looking at functional and technological aspects," *AI Soc.*, vol. 34, no. 3, pp. 509–533, Sep. 2019.
- [200] R. Y. Wong, A. Chong, and R. C. Aspegren, "Privacy legislation as business risks: How GDPR and CCPA are represented in technology companies' investment risk disclosures," *Proc. ACM Human Comput. Interact.*, vol. 7, no. 1, pp. 1–26, Apr. 2023.
- [201] L. Graves, V. Nagisetty, and V. Ganesh, "Amnesiac machine learning," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 13, pp. 11516–11524.
- [202] E. F. Villaronga, P. Kieseberg, and T. Li, "Humans forget, machines remember: Artificial intelligence and the right to be forgotten," *Comput. Law Secur. Rev.*, vol. 34, no. 2, pp. 304–313, Apr. 2018.
- [203] A. Golatkar, A. Achille, and S. Soatto, "Eternal sunshine of the spotless net: Selective forgetting in deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9301–9309.
- [204] K. Caine, S. Šabanovic, and M. Carter, "The effect of monitoring by cameras and robots on the privacy enhancing behaviors of older adults," in *Proc. 7th Annu. ACM/IEEE Int. Conf. Human Robot Interact.* New York, NY, USA: Association for Computing Machinery, Mar. 2012, pp. 343–350.
- [205] D. Reinhardt, M. Khurana, and L. H. Acosta, "'I still need my privacy': Exploring the level of comfort and privacy preferences of German-speaking older adults in the case of mobile assistant robots," *Pervasive Mobile Comput.*, vol. 74, Jul. 2021, Art. no. 101397.
- [206] J. Zlotowski, K. Yogeewaran, and C. Bartneck, "Can we control it? Autonomous robots threaten human identity, uniqueness, safety, and resources," *Int. J. Hum. Comput. Stud.*, vol. 100, pp. 48–54, Apr. 2017.
- [207] P. A. Hancock, D. R. Billings, and K. E. Schaefer, "Can you trust your robot?" *Ergonom. Design*, vol. 19, no. 3, pp. 24–29, Jul. 2011.
- [208] A. R. Wagner and R. C. Arkin, "Acting deceptively: Providing robots with the capacity for deception," *Int. J. Social Robot.*, vol. 3, no. 1, pp. 5–26, Jan. 2011.
- [209] D. Kontogiorgos, S. van Waveren, O. Wallberg, A. Pereira, I. Leite, and J. Gustafson, "Embodiment effects in interactions with failing robots," in *Proc. CHI Conf. Human Factors Comput. Syst.*, Apr. 2020, pp. 1–14.
- [210] N. Mirmig, G. Stollnberger, M. Miksch, S. Stadler, M. Giuliani, and M. Tscheligi, "To err is robot: How humans assess and act toward an erroneous social robot," *Frontiers Robot. AI*, vol. 4, no. 21, p. 1, May 2017.
- [211] E. McReynolds, S. Hubbard, T. Lau, A. Saraf, M. Cakmak, and F. Roesner, "Toys that listen: A study of parents, children, and Internet-connected toys," in *Proc. CHI Conf. Human Factors Comput. Syst.* New York, NY, USA: Association for Computing Machinery, May 2017, pp. 5197–5207.
- [212] H. M. Do, M. Pham, W. Sheng, D. Yang, and M. Liu, "RiSH: A robot-integrated smart home for elderly care," *Robot. Auto. Syst.*, vol. 101, pp. 74–92, Mar. 2018.
- [213] E. Zeng, S. Mare, and F. Roesner, "End user security and privacy concerns with smart homes," in *Proc. 13th Symp. Usable Privacy Security (SOUPS)*. Santa Clara, CA, USA: USENIX Association, Jul. 2017, pp. 65–80.
- [214] H. Atoyan, J.-R. Duquet, and J.-M. Robert, "Trust in new decision aid systems," in *Proc. 18th Int. Conf. Assoc. Francophone d'Interaction Homme Mach.*, 2006, pp. 115–122.
- [215] J. D. Lee and K. A. See, "Trust in automation: Designing for appropriate reliance," *Human Factors, J. Human Factors Ergonom. Soc.*, vol. 46, no. 1, pp. 50–80, 2004.
- [216] Z. R. Khavas, S. R. Ahmadzadeh, and P. Robinette, "Modeling trust in human–robot interaction: A survey," in *Proc. Int. Conf. Social Robot.* Cham, Switzerland: Springer, 2020, pp. 529–541.
- [217] M. Salem, G. Lakatos, F. Amirabdollahian, and K. Dautenhahn, "Would you trust a (Faulty) robot? Effects of error, task type and personality on human–robot cooperation and trust," in *Proc. 10th ACM/IEEE Int. Conf. Human Robot Interact. (HRI)*, Mar. 2015, pp. 141–148.
- [218] M. W. Boyce, J. Y. C. Chen, A. R. Selkowitz, and S. G. Lakhmani, "Effects of agent transparency on operator trust," in *Proc. 10th Annu. ACM/IEEE Int. Conf. Human Robot Interact. Extended Abstr.*, Mar. 2015, pp. 179–180.

- [219] K. Lazányi and G. Marácz, “Dispositional trust—Do we trust autonomous cars?” in *Proc. IEEE 15th Int. Symp. Intell. Syst. Informat. (SISY)*, Sep. 2017, pp. 000135–000140.
- [220] M. Desai, M. Medvedev, M. Vázquez, S. McSheehy, S. Gadea-Omelchenko, C. Bruggeman, A. Steinfeld, and H. Yanco, “Effects of changing reliability on trust of robot systems,” in *Proc. 7th ACM/IEEE Int. Conf. Human Robot Interact. (HRI)*, Mar. 2012, pp. 73–80.
- [221] P. Robinette, A. R. Wagner, and A. M. Howard, “Building and maintaining trust between humans and guidance robots in an emergency,” in *Proc. AAAI Spring Symp.*, 2013, pp. 78–83.
- [222] B. Tay, Y. Jung, and T. Park, “When stereotypes meet robots: The double-edge sword of robot gender and personality in human–robot interaction,” *Comput. Hum. Behav.*, vol. 38, pp. 75–84, Sep. 2014.
- [223] P. Robinette, W. Li, R. Allen, A. M. Howard, and A. R. Wagner, “Overtrust of robots in emergency evacuation scenarios,” in *Proc. 11th ACM/IEEE Int. Conf. Human Robot Interact. (HRI)*, Mar. 2016, pp. 101–108.
- [224] M. Obaid, E. B. Sandoval, J. Zlotowski, E. Moltchanova, C. A. Basedow, and C. Bartneck, “Stop! That is close enough. How body postures influence human–robot proximity,” in *Proc. 25th IEEE Int. Symp. Robot Human Interact. Commun. (RO-MAN)*, Aug. 2016, pp. 354–361.
- [225] X. J. Yang, V. V. Unhelkar, K. Li, and J. A. Shah, “Evaluating effects of user experience and system transparency on trust in automation,” in *Proc. 12th ACM/IEEE Int. Conf. Human Robot Interact. (HRI)*, Mar. 2017, pp. 408–416.
- [226] M. L. Walters, M. A. Oskoei, D. S. Syrdal, and K. Dautenhahn, “A long-term human–robot proxemic study,” in *Proc. RO-MAN*, Jul. 2011, pp. 137–142.
- [227] S. Ososky, D. Schuster, E. Phillips, and F. G. Jentsch, “Building appropriate trust in human–robot teams,” in *Proc. AAAI Spring Symp. Ser.*, 2013, pp. 60–65.
- [228] S. Fischer-Hübner, *IT-Security and Privacy: Design and Use of Privacy-Enhancing Security Mechanisms*. Berlin, Germany: Springer, 2001.
- [229] A. Pfitzmann and M. Hansen, *Anonymity, Unlinkability, Unobservability, Pseudonymity, and Identity Management—A Consolidated Proposal for Terminology*, document Version v0.28, 2005.
- [230] *Privacy Enhancing Technologies*. Accessed: Sep. 30, 2023. [Online]. Available: https://www.frbsf.org/banking/wp-content/uploads/sites/5/Privacy-Enhancing-Technologies_FINAL_V2_TOC-Update.pdf
- [231] N. Kaaniche, M. Laurent, and S. Belguith, “Privacy enhancing technologies for solving the privacy-personalization paradox: Taxonomy and survey,” *J. Netw. Comput. Appl.*, vol. 171, Dec. 2020, Art. no. 102807.
- [232] S. Eick and A. I. Antón, “Enhancing privacy in robotics via judicious sensor selection,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 7156–7165.
- [233] B. Horstmann, N. Diekmann, H. Buschmeier, and T. Hassan, “Towards designing privacy-compliant social robots for use in private households: A use case based identification of privacy implications and potential technical measures for mitigation,” in *Proc. 29th IEEE Int. Conf. Robot Human Interact. Commun. (RO-MAN)*, Aug. 2020, pp. 869–876.
- [234] D. J. Butler, J. Huang, F. Roesner, and M. Cakmak, “The privacy-utility tradeoff for remotely teleoperated robots,” in *Proc. 10th ACM/IEEE Int. Conf. Human Robot Interact. (HRI)*, Mar. 2015, pp. 27–34.
- [235] M. Dorigo, G. Theraulaz, and V. Trianni, “Reflections on the future of swarm robotics,” *Sci. Robot.*, vol. 5, no. 49, Dec. 2020, Art. no. eabe4385.
- [236] S. S. O. Venkata, R. Parasuraman, and R. Pidaparti, “KT-BT: A framework for knowledge transfer through behavior trees in multirobot systems,” *IEEE Trans. Robot.*, vol. 39, no. 5, pp. 4114–4130, Oct. 2023.
- [237] C. Shan, H. Jiao, and J. Fu, “Towards representation identical privacy-preserving graph neural network via split learning,” 2021, *arXiv:2107.05917*.
- [238] S. Mitra, A. Piplai, S. Mittal, and A. Joshi, “Combating fake cyber threat intelligence using provenance in cybersecurity knowledge graphs,” in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2021, pp. 3316–3323.
- [239] T. Chakraborti, Y. Zhang, D. E. Smith, and S. Kambhampati, “Planning with resource conflicts in human–robot cohabitation,” in *Proc. 15th Int. Conf. Auton. Agents Multiagent Syst.*, May 2016, pp. 1069–1077.
- [240] B. C. Kok and H. Soh, “Trust in robots: Challenges and opportunities,” *Current Robot. Rep.*, vol. 1, no. 4, pp. 297–309, Dec. 2020.
- [241] A. Fuller, Z. Fan, C. Day, and C. Barlow, “Digital twin: Enabling technologies, challenges and open research,” *IEEE Access*, vol. 8, pp. 108952–108971, 2020.
- [242] A. Bilberg and A. A. Malik, “Digital twin driven human–robot collaborative assembly,” *CIRP Ann.*, vol. 68, no. 1, pp. 499–502, 2019.
- [243] B. Wang, H. Zhou, G. Yang, X. Li, and H. Yang, “Human digital twin (HDT) driven human-cyber-physical systems: Key technologies and applications,” *Chin. J. Mech. Eng.*, vol. 35, no. 1, p. 11, Dec. 2022.
- [244] S. Neupane, I. A. Fernandez, W. Patterson, S. Mittal, M. Parmar, and S. Rahimi, “TwinExplainer: Explaining predictions of an automotive digital twin,” 2023, *arXiv:2302.00152*.
- [245] A. Roque and S. K. Damodaran, “Explainable AI for security of human-interactive robots,” *Int. J. Human Comput. Interact.*, vol. 38, nos. 18–20, pp. 1789–1807, Dec. 2022.
- [246] S. Neupane, J. Ables, W. Anderson, S. Mittal, S. Rahimi, I. Banicescu, and M. Seale, “Explainable intrusion detection systems (X-IDS): A survey of current methods, challenges, and opportunities,” *IEEE Access*, vol. 10, pp. 112392–112415, 2022.
- [247] D. Das, S. Banerjee, and S. Chernova, “Explainable AI for robot failures: Generating explanations that improve user assistance in fault recovery,” in *Proc. 16th ACM/IEEE Int. Conf. Human Robot Interact. (HRI)*. New York, NY, USA: Association for Computing Machinery, Mar. 2021, pp. 351–360.
- [248] K. Chen, T. Hwu, H. J. Kashyap, J. L. Krichmar, K. Stewart, J. Xing, and X. Zou, “Neurobots as a means toward neuroethology and explainable AI,” *Frontiers Neurobot.*, vol. 14, Oct. 2020, Art. no. 570308.
- [249] N. Fulton and A. Platzter, “Verifiably safe off-model reinforcement learning,” in *Tools and Algorithms for the Construction and Analysis of Systems, T. Vojnar and L. Zhang, Eds.* Cham, Switzerland: Springer, 2019, pp. 413–430.
- [250] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, “Safe learning in robotics: From learning-based control to safe reinforcement learning,” *Annu. Rev. Control, Robot., Auto. Syst.*, vol. 5, no. 1, pp. 411–444, May 2022.
- [251] F. Berkenkamp, M. Turchetta, A. Schoellig, and A. Krause, “Safe model-based reinforcement learning with stability guarantees,” in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
- [252] S. Thiebes, S. Lins, and A. Sunyaev, “Trustworthy artificial intelligence,” *Electron. Markets*, vol. 31, no. 2, pp. 447–464, Jun. 2021.
- [253] D. Votipka, E. Zhang, and M. L. Mazurek, “HackEd: A pedagogical analysis of online vulnerability discovery exercises,” in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2021, pp. 1268–1285.
- [254] A. Yasin, L. Liu, T. Li, J. Wang, and D. Zowghi, “Design and preliminary evaluation of a cyber security requirements education game (SREG),” *Inf. Softw. Technol.*, vol. 95, pp. 179–200, Mar. 2018.
- [255] S. Frey, A. Rashid, P. Anthonysamy, M. Pinto-Albuquerque, and S. A. Naqvi, “The good, the bad and the ugly: A study of security decisions in a cyber-physical systems game,” *IEEE Trans. Softw. Eng.*, vol. 45, no. 5, pp. 521–536, May 2019.
- [256] *Association for Advancing Automation*. Accessed: Oct. 4, 2023. [Online]. Available: <https://www.automate.org/a3-content/who-we-are>



SUBASH NEUPANE received the Bachelor of Computer Engineering degree from Kathmandu University, Nepal, in 2011, the M.S. degree in information technology (professional computing) from the Swinburne University of Technology, Melbourne, Australia, in 2016, and the M.S. degree in information systems and security management from Tuskegee University, Alabama, USA, in 2020. He is currently pursuing the Ph.D. degree in systems and security with Mississippi State University. He was a Telecom Network Engineer with Tandem Corporation, Melbourne, before moving to the USA. His current research interests include systems and security, machine learning, and blockchain.



SHASWATA MITRA received the B.Tech. degree in computer science and engineering from the West Bengal University of Technology, India. He is currently pursuing the Ph.D. degree in artificial intelligence with Mississippi State University (MSU). He is also with the PATENT Laboratory, Department of Computer Science and Engineering, MSU. Before moving to the USA, he was a Full-Stack Developer with Tata Consultancy Services, India. His current research

interests include the application of machine learning to the field of cybersecurity and robotics.



IVAN A. FERNANDEZ received the B.S. degree in aerospace and ocean engineering from Virginia Tech, in 2010, and the M.S. degree in computer science from Georgia Tech, in 2018. He is currently pursuing the Ph.D. degree in computer science with Mississippi State University. He has been working in the defense industry, since 2011, developing advanced algorithms and providing AI/ML solutions. His current research interests

include security of cyber-physical systems, verification and validation (V&V) of AI-based systems, and adversarial machine learning.



SWAYAMJIT SAHA received the B.Tech. degree in computer science and engineering from the West Bengal University of Technology, India. He is currently pursuing the Ph.D. degree in artificial intelligence with Mississippi State University (MSU). Before coming to USA, he was a Database Administrator with DXC Technology, India. His current research interests include security of machine learning algorithms in the domains of cybersecurity and robotics.



SUDIP MITTAL (Member, IEEE) received the Ph.D. degree in computer science from the University of Maryland Baltimore County, in 2019. He is currently an Assistant Professor with the Department of Computer Science and Engineering, Mississippi State University. His primary research interests include cybersecurity and artificial intelligence. His goal is to develop the next generation of cyber defense systems that help protect various organizations and people.

At Mississippi State, he leads the Secure and Trustworthy Cyberspace (SECRETS) Laboratory and has published over 80 journals and conference papers in leading cybersecurity and AI venues. He has received funding from the NSF, USAF, USACE, and various other Department of Defense programs. He also serves as a program committee member or the program chair for leading AI and cybersecurity conferences and workshops. His work has been cited in the LA times, Business Insider, WIRED, and the Cyberwire. He is a member of the ACM.



JINGDAO CHEN (Member, IEEE) received the B.S. degree in electrical engineering from Washington University in St. Louis, in 2015, and the master's degree in computer science and the Ph.D. degree in robotics from the Georgia Institute of Technology, in 2019 and 2021, respectively. He is currently an Assistant Professor with the Department of Computer Science and Engineering, Mississippi State University. His research

interests include AI, computer vision, robotics, and exploring data-driven deep learning techniques for scene understanding to enable mobile robots to autonomously navigate and explore unknown environments.



NISHA PILLAI (Member, IEEE) received the Ph.D. degree in computer science from the University of Maryland Baltimore County, in 2021. She is currently with the Department of Computer Science and Engineering, Mississippi State University, as a Research Scientist. Her research interests include the applications of artificial intelligence in robotics, biology, healthcare, and food safety. Her research has been published

in some of the most prestigious journals and conference proceedings in the fields of artificial intelligence and bioinformatics. Her professional experience has included serving on the senior program committee for AAAI 2023, the Board of Directors for MCBIOS (2023–2025), the Co-Chair for MCBIOS (2022–2023), and an Editorial Board Member for *Frontiers in Medical Technology*, *Frontiers in Plant Science*, and *Frontiers in Genetics*.



SHAHRAM RAHIMI (Member, IEEE) is currently a Professor and the Head of the Department of Computer Science and Engineering, Mississippi State University. Prior to that, he led the Department of Computer Science, Southern Illinois University, for five years. He is also a recognized Leader in the area of artificial and computational intelligence with over 220 peer-reviewed publications and a few patents or pending patents in this area. He has served as the Editor-in-Chief for two leading computational intelligence journals and sits on the editorial board of several others. He is a member of IEEE New Standards Committee in Computational Intelligence and provides advice to staff and administration at federal government on predictive analytics for foreign policy. He was a recipient of 2016 Illinois Rising Star Award from ISBA, selected among 100's of other highly qualified candidates. His intelligent algorithm for patient flow optimization and hospital staffing is currently

used in over 1000 emergency departments across the nation and was named top 10 AI technology for healthcare, in 2018, by *HealthTech Magazine*. He has secured over \$20M of federal and industry funding as a PI or a Co-PI in the last 20 years. He has also organized 15 conferences and workshops in the areas of computational intelligence and multi-agent systems over the past two decades.

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